

BH3FR — formula report

213 inline formulas, 325 display equations, 0 tables, 0 unrecovered image regions.

Ink comparison not available for this document — no residual measurement has been run for it. MathPix confidence is shown; the residual column is absent.

Display equations

| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Page | Conf.                                  | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Scan image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| <div><div>BH3FR_</div><div>EQ0138 (60) <b>[conf 0.001]</b></div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 116  | <div><div></div><div>0.001</div></div> | $\begin{aligned} & \& \left[ \begin{array}{c} \widehat{\kappa} \\ \lambda \end{array} \right]_{-} = \hat{0}, \quad \left[ \begin{array}{c} \kappa \\ -\mu \nu \end{array} \right] ; () \left[ \begin{array}{c} \widehat{\kappa} \\ -+ \\ \mu \nu \end{array} \right] \times ; () \partial \Omega < \infty, \quad sp \left( {}^2 Q_{(\mu \nu)}^{(\kappa \lambda)} \times \left[ \begin{array}{c} \widehat{\kappa} \\ -+ \\ \mu \nu \end{array} \right] \right) ; () \left( sp \left( {}^2 \bar{Q}_{(\mu \nu)}^{(\kappa \lambda)} \times \left[ \begin{array}{c} \widehat{\kappa} \\ -+ \\ \mu \nu \end{array} \right] \right) \right)^\times ; () \partial \Omega < \infty, \\ & \Lambda_{(\mu \nu)}^{(\kappa \lambda)} = \left( \Lambda_{(\mu \nu)}^{(\kappa \lambda)} \right)^\times \end{aligned}$ | $\begin{aligned} & \left[ \begin{array}{c} \widehat{\kappa} \\ -+ \\ \mu \nu \end{array} \right]_{-} = \hat{0}, \quad \left[ \begin{array}{c} \kappa \\ -\mu \nu \end{array} \right] ; () \left[ \begin{array}{c} \widehat{\kappa} \\ -+ \\ \mu \nu \end{array} \right] \times ; () \partial \Omega < \infty, \quad sp \left( {}^2 Q_{(\mu \nu)}^{(\kappa \lambda)} \times \left[ \begin{array}{c} \widehat{\kappa} \\ -+ \\ \mu \nu \end{array} \right] \right) ; () \left( sp \left( {}^2 \bar{Q}_{(\mu \nu)}^{(\kappa \lambda)} \times \left[ \begin{array}{c} \widehat{\kappa} \\ -+ \\ \mu \nu \end{array} \right] \right) \right)^\times ; () \partial \Omega < \infty, \\ & \Lambda_{(\mu \nu)}^{(\kappa \lambda)} = \left( \Lambda_{(\mu \nu)}^{(\kappa \lambda)} \right)^\times \end{aligned}$ | $\begin{aligned} & \left[ \begin{array}{c} \widehat{\kappa} \\ -+ \\ \mu \nu \end{array} \right]_{-} = \hat{0}, \quad \left[ \begin{array}{c} \kappa \\ -\mu \nu \end{array} \right] ; () \left[ \begin{array}{c} \widehat{\kappa} \\ -+ \\ \mu \nu \end{array} \right] \times ; () \partial \Omega < \infty, \quad sp \left( {}^2 Q_{(\mu \nu)}^{(\kappa \lambda)} \times \left[ \begin{array}{c} \widehat{\kappa} \\ -+ \\ \mu \nu \end{array} \right] \right) ; () \left( sp \left( {}^2 \bar{Q}_{(\mu \nu)}^{(\kappa \lambda)} \times \left[ \begin{array}{c} \widehat{\kappa} \\ -+ \\ \mu \nu \end{array} \right] \right) \right)^\times ; () \partial \Omega < \infty, \\ & A_{(\mu \nu)}^{(\kappa \lambda)} = \left( A_{(\mu \nu)}^{(\kappa \lambda)} \right)^\times \end{aligned}$ |
| <div><b>Conf.</b> MathPix confidence: <div><div></div><div>≥0.9</div><div>0.5–0.9</div><div>&lt;0.5</div><div>— no value</div></div></div> <div><b>LaTeX source</b> MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ</div> <div><b>Residual</b> not shown for this document — no residual measurements are paired with these rows.</div> <div><b>[refined: basis]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</div> |      |                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

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| BH3FR_EQ0158 <b>[conf 0.004]</b> | 120  | ■ 0.004 | $\begin{aligned} & \& K_{14}^{++}(2,\gamma)=2\,d,4\,d,5\,d,1\,s,3\,s,4\,s,6\,s,1\,e \\ & \& K_{15}^{++}(3,\gamma)=1\,d,4\,d,5\,d,1\,s,2\,s,4\,s,5\,s,1\,e \\ & \& K_{16}^{++}(\alpha,\beta)=1\,d,2\,d,3\,d,6\,d,7\,s,8\,s,9\,s \\ & \& K_{17}^{++}(\alpha,\gamma)=1\,d,2\,d,3\,d,5\,d,4\,s,5\,s,6\,s \\ & \& K_{18}^{++}(\beta,\gamma)=1\,d,2\,d,3\,d,4\,d,1\,s,2\,s,3\,s \\ & K_1^{- -}\left(\frac{1}{2}\right)=3\,d,2\,d,5\,s,8\,s \\ & C_1^{-} =2\,d,1\,s,4\,s,7\,s \\ & K_2^{-}\left(\frac{1}{3}\right)=3\,d,3\,s,6\,s,9\,s \\ & C_2^{-} =1\,d,1\,s,4\,s,7\,s \\ & K_3^{-}\left(\frac{2}{3}\right)=2\,d,3\,s,6\,s,9\,s \\ & C_3^{-} =1\,d,2\,s,5\,s,8\,s \\ & K_4^{-}\left(\frac{1}{\alpha,\alpha^*}\right)=4\,d,2\,s,3\,s,1\,e,2\,e \\ & C_4^{-} =1\,d,2\,d,4\,s,7\,s \\ & K_5^{-}\left(\frac{2}{\alpha,\alpha^*}\right)=4\,d,1\,s,3\,s,1\,e,2\,e \\ & C_5^{-} =1\,d,3\,d,5\,s,8\,s \\ & K_6^{-}\left(\frac{3}{\alpha,\alpha^*}\right)=4\,d,1\,s,2\,s,1\,e,2\,e \\ & C_6^{-} =2\,d,3\,d,6\,s,9\,s \\ & K_7^{-}\left(\frac{1}{\beta,\beta^*}\right)=5\,d,5\,s,6\,s,1\,e,3\,e \\ & C_7^{-} =1\,d,2\,d,1\,s,7\,s \\ & K_8^{-}\left(\frac{2}{\beta,\beta^*}\right)=5\,d,4\,s,6\,s,1\,e,3\,e \\ & C_8^{-} =1\,d,3\,d,2\,s,8\,s \\ & K_9^{-}\left(\frac{3}{\beta,\beta^*}\right)=5\,d,4\,s,5\,s,1\,e,3\,e \\ & C_9^{-} =2\,d,3\,d,3\,s,9\,s \\ & K_{10}^{-}\left(\frac{1}{\gamma,\gamma^*}\right)=6\,d,8\,s,9\,s,2\,e,3\,e \\ & C_{10}^{-} =1\,d,2\,d,1\,s,4\,s \end{aligned}$ | $\begin{aligned} & K_{14}^{++}(2,\gamma)=2\,d,4\,d,5\,d,1\,s,3\,s,4\,s,6\,s,1\,e \\ & K_{15}^{++}(3,\gamma)=1\,d,4\,d,5\,d,1\,s,2\,s,4\,s,5\,s,1\,e \\ & K_{16}^{++}(\alpha,\beta)=1\,d,2\,d,3\,d,6\,d,7\,s,8\,s,9\,s \\ & K_{17}^{++}(\alpha,\gamma)=1\,d,2\,d,3\,d,5\,d,4\,s,5\,s,6\,s \\ & K_{18}^{++}(\beta,\gamma)=1\,d,2\,d,3\,d,4\,d,1\,s,2\,s,3\,s \\ & K_1^{- -}\left(\frac{1}{2}\right)=3\,d,2\,d,5\,s,8\,s \\ & C_1^{-} =2\,d,1\,s,4\,s,7\,s \\ & K_2^{-}\left(\frac{1}{3}\right)=3\,d,3\,s,6\,s,9\,s \\ & C_2^{-} =1\,d,1\,s,4\,s,7\,s \\ & K_3^{-}\left(\frac{2}{3}\right)=2\,d,3\,s,6\,s,9\,s \\ & C_3^{-} =1\,d,2\,s,5\,s,8\,s \\ & K_4^{-}\left(\frac{1}{\alpha,\alpha^*}\right)=4\,d,2\,s,3\,s,1\,e,2\,e \\ & C_4^{-} \\ & K_5^{-}\left(\frac{2}{\alpha,\alpha^*}\right)=1\,d,2\,d,4\,s,7\,s \\ & C_5^{-} \\ & K_6^{-}\left(\frac{3}{\alpha,\alpha^*}\right)=4\,d,1\,s,3\,s,1\,e,2\,e \\ & C_6^{-} \\ & K_7^{-}\left(\frac{1}{\beta,\beta^*}\right)=2\,d,3\,d,6\,s,9\,s \\ & C_7^{-} \\ & K_8^{-}\left(\frac{2}{\beta,\beta^*}\right)=1\,d,2\,d,1\,s,7\,s,4\,s,6\,s,1\,e,3\,e \\ & C_8^{-} \\ & K_9^{-}\left(\frac{3}{\beta,\beta^*}\right) \\ & C_9^{-} \\ & K_{10}^{-}\left(\frac{1}{\gamma,\gamma^*}\right)=6\,d,3\,d,2\,s,8\,s \\ & C_{10}^{-} =2\,d,3\,d,3\,s,9\,s,1\,e,3\,e \\ & =1\,d,2\,d,1\,s,4\,s \end{aligned}$ | $\begin{aligned} & K_{14}^{++}(2,\gamma)=2\,d,4\,d,5\,d,1\,s,3\,s,4\,s,6\,s,1\,e \\ & K_{15}^{++}(3,\gamma)=1\,d,4\,d,5\,d,1\,s,2\,s,4\,s,5\,s,1\,e \\ & K_{16}^{++}(\alpha,\beta)=1\,d,2\,d,3\,d,6\,d,7\,s,8\,s,9\,s \\ & K_{17}^{++}(\alpha,\gamma)=1\,d,2\,d,3\,d,5\,d,4\,s,5\,s,6\,s \\ & K_{18}^{++}(\beta,\gamma)=1\,d,2\,d,3\,d,4\,d,1\,s,2\,s,3\,s \\ & K_1^{-}\left(\frac{1}{2}\right)=3\,d,2\,d,5\,s,8\,s \\ & C_1^{-} =2\,d,1\,s,4\,s,7\,s \\ & K_2^{-}\left(\frac{1}{3}\right)=3\,d,3\,s,6\,s,9\,s \\ & C_2^{-} =1\,d,1\,s,4\,s,7\,s \\ & K_3^{-}\left(\frac{2}{3}\right)=2\,d,3\,s,6\,s,9\,s \\ & C_3^{-} =1\,d,2\,s,5\,s,8\,s \\ & K_4^{-}\left(\frac{1}{\alpha,\alpha^*}\right)=4\,d,2\,s,3\,s,1\,e,2\,e \\ & C_4^{-} =1\,d,2\,d,4\,s,7\,s \\ & K_5^{-}\left(\frac{2}{\alpha,\alpha^*}\right)=4\,d,1\,s,3\,s,1\,e,2\,e \\ & C_5^{-} =1\,d,3\,d,5\,s,8\,s \\ & K_6^{-}\left(\frac{3}{\alpha,\alpha^*}\right)=4\,d,1\,s,2\,s,1\,e,2\,e \\ & C_6^{-} =2\,d,3\,d,6\,s,9\,s \\ & K_7^{-}\left(\frac{1}{\beta,\beta^*}\right)=5\,d,5\,s,6\,s,1\,e,3\,e \\ & C_7^{-} =1\,d,2\,d,1\,s,7\,s \\ & K_8^{-}\left(\frac{2}{\beta,\beta^*}\right)=5\,d,4\,s,6\,s,1\,e,3\,e \\ & C_8^{-} =1\,d,3\,d,2\,s,8\,s \\ & K_9^{-}\left(\frac{3}{\beta,\beta^*}\right)=5\,d,4\,s,5\,s,1\,e,3\,e \\ & C_9^{-} =2\,d,3\,d,3\,s,9\,s \\ & K_{10}^{-}\left(\frac{1}{\gamma,\gamma^*}\right)=6\,d,8\,s,9\,s,2\,e,3\,e \\ & C_{10}^{-} =1\,d,2\,d,1\,s,4\,s \end{aligned}$ |

| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Page | Conf.  | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Scan image |
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| BH3FR_<br>EQ0142 (62) <b>[conf 0.005]</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 117  | ■0.005 | $\begin{aligned} & \left[\begin{array}{c} i \\ k\ l \end{array}\right] (\mu\lambda) + Q_{(\mu\nu)+}^{(\kappa\lambda)i} \left[\begin{array}{c} m \\ k \end{array}\right] \left(\begin{array}{c} \kappa\lambda \\ (\mu\nu)+ \end{array}\right) = \\ & = A_{(\mu\nu)kl}^{(\kappa\lambda)\frac{i}{kl}} \exp\left(\underline{\lambda}_{kl}\mu - \underline{c}_s \boldsymbol{S} \left[\begin{array}{c} s \\ k\ l \end{array}\right]; ()\right) \partial_\mu \end{aligned}$                                                                                                                                                                                                                                                                                               | $\left[\begin{array}{c} i \\ k\ l \end{array}\right]_{(\mu\nu)+}^{(\kappa\lambda)} + Q_{(\mu\nu)m}^{(\kappa\lambda)\underline{l}} \left[\begin{array}{c} m \\ k\ l \end{array}\right]_{(\mu\nu)+}^{(\kappa\lambda)} =$ $= A_{(\mu\nu)kl}^{(\kappa\lambda)\underline{l}} \exp\left(\underline{\lambda}_{kl}\mu - \underline{c}_s \boldsymbol{S} \left[\begin{array}{c} s \\ k\ l \end{array}\right]; ()\right) \delta\mu$                                                                                                                                                                                                                                                                                                                                                                                                                             |            |
| BH3FR_<br>EQ0141 (61a) <b>[conf 0.006]</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 117  | ■0.006 | $\begin{aligned} & L_{(\mu\nu)+}^{(\kappa\lambda)} = \left(K_{(\mu\nu)+}^{(\kappa\lambda)} - \bar{\lambda}_{(\mu\nu)}^{(\kappa\lambda)} \times ()\right); \left(1 + sp^2 \bar{Q}_{(\mu\nu)}^{(\kappa\lambda)} \times ()\right) + D_{(\mu\nu)}^{(\kappa\lambda)}, \\ & \Lambda_{(\mu\nu)+}^{(\kappa\lambda)} = K_{(\mu\nu)+}^{(\kappa\lambda)}; \left(1 + sp^2 \bar{Q}_{(\mu\nu)}^{(\kappa\lambda)} \times ()\right) + D_{(\mu\nu)}^{(\kappa\lambda)}, \\ & K_{(\mu\nu)+m}^{(\kappa\lambda)} = \left(\begin{array}{c} \kappa\lambda + \\ \mu\nu \end{array}\right)_{(-)l}^{(+,-)}; \delta(l, m) - \left(\begin{array}{c} \kappa\lambda \\ \mu^+ \end{array}\right)_{(-)m}^+_{(-,-)}^{(+\lambda)} \end{aligned}$                                                     | $L_{(\mu\nu)+}^{(\kappa\lambda)} = \left(K_{(\mu\nu)+}^{(\kappa\lambda)} - \bar{\lambda}_{(\mu\nu)}^{(\kappa\lambda)} \times ()\right); \left(1 + sp^2 \bar{Q}_{(\mu\nu)}^{(\kappa\lambda)} \times ()\right) + D_{(\mu\nu)}^{(\kappa\lambda)},$ $\Lambda_{(\mu\nu)+}^{(\kappa\lambda)} = K_{(\mu\nu)+}^{(\kappa\lambda)}; \left(1 + sp^2 \bar{Q}_{(\mu\nu)}^{(\kappa\lambda)} \times ()\right) + D_{(\mu\nu)}^{(\kappa\lambda)},$ $K_{(\mu\nu)+m}^{(\kappa\lambda)} = \left(\begin{array}{c} \kappa\lambda + \\ \mu\nu \end{array}\right)_{(-)l}^{(+,-)}; \delta(l, m) - \left(\begin{array}{c} \kappa\lambda \\ \mu^+ \end{array}\right)_{(-)m}^+_{(-,-)}^{(+\lambda)}$                                                                                                                                                                             |            |
| BH3FR_EQ0315 <b>[conf 0.014]</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 146  | ■0.014 | $\begin{aligned} & \left[\begin{array}{c} \widehat{cd} \\ -at \\ ab \end{array}\right] \neq \left[\begin{array}{c} \widehat{cd} \\ +\overline{ab} \end{array}\right], \quad \left[\begin{array}{c} \widehat{cd} \\ -ab \end{array}\right] \neq \left[\begin{array}{c} \widehat{cd} \\ -a \end{array}\right] \times, \\ & {}^2\bar{\gamma}_{(ab)} \neq {}^2\bar{\gamma}_{(ab)}^\times, \quad \left[\begin{array}{c} \widehat{cd} \\ -ab \end{array}\right] = \left[\begin{array}{c} \widehat{cd} \\ -ab \end{array}\right]_+ + \left[\begin{array}{c} \widehat{cd} \\ -ab \end{array}\right]_-, \\ & \left[\begin{array}{c} \widehat{cd} \\ -ab \end{array}\right]_\pm = \pm \left[\begin{array}{c} \widehat{cd} \\ -a \end{array}\right]_\pm^\times \end{aligned}$ | $\left[\begin{array}{c} \widehat{cd} \\ -\overline{ab} \end{array}\right] \neq \left[\begin{array}{c} \widehat{cd} \\ +\overline{ab} \end{array}\right], \quad {}^2\bar{c} \neq {}^2\bar{d}, \quad \left[\begin{array}{c} \widehat{cd} \\ -\overline{ab} \end{array}\right] \neq \left[\begin{array}{c} \widehat{cd} \\ -\overline{ab} \end{array}\right]^\times,$ ${}^2\bar{\gamma}_{(ab)} \neq {}^2\bar{\gamma}_{(ab)}^\times, \quad \left[\begin{array}{c} \widehat{cd} \\ -\overline{ab} \end{array}\right] = \left[\begin{array}{c} \widehat{cd} \\ -\overline{ab} \end{array}\right]_+ + \left[\begin{array}{c} \widehat{cd} \\ -\overline{ab} \end{array}\right]_-,$ $\left[\begin{array}{c} \widehat{cd} \\ -\overline{ab} \end{array}\right]_\pm = \pm \left[\begin{array}{c} \widehat{cd} \\ -\overline{ab} \end{array}\right]_\pm^\times$ |            |
| <b>Conf.</b> MathPix confidence: <span>■</span> ≥0.9 <span>■</span> 0.5-0.9 <span>■</span> <0.5 — no value<br><b>LaTeX source</b> MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ<br><b>Residual</b> not shown for this document — no residual measurements are paired with these rows.<br><b>[refined: <i>basis</i>]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here. |      |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |            |

| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Scan image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| BH3FR_ EQ8310 (M24a) <b>I</b> conf <b>0.015</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 145  | ■ 0.015 | $\begin{aligned} & \{ \}^2 \bar{\gamma}_{(c\,d)} = \\ & \operatorname{sp}\left(\{ \}^2 \bar{c} \times \{ \}^2 \bar{d}\right), \quad \gamma^i(c\,d)[p\,k\,l]_{(a\,b)} = \\ & \left[\begin{array}{c} i \\ k \quad l \end{array}\right]_{(ab)}^{\widehat{cd}} \\ & \left[\begin{array}{c} i \\ k \quad l \end{array}\right]_{(ab)}^{\widehat{cd}} = \left[\begin{array}{c} \widehat{cd} \\ -a \end{array}\right] \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | ${}^2\bar{\gamma}_{(cd)} = \operatorname{sp}\left({}^2\bar{c} \times {}^2\bar{d}\right), \quad \gamma^i(c\,d)[p\,k\,l]_{(ab)} = \left[\begin{array}{c} i \\ k \quad l \end{array}\right]_{(ab)}^{(cd)},$<br>${}^{[3]}\left[\left[\begin{array}{c} i \\ k \quad l \end{array}\right]_{(ab)}^{(cd)}\right]_N = \left[\begin{array}{c} \widehat{cd} \\ -a \end{array}\right]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | ${}^2\bar{\gamma}_{(cd)} = \operatorname{sp}\left({}^2\bar{c} \times {}^2\bar{d}\right), \quad \gamma^i_{(cd)}\left[\begin{array}{c} p\,k\,l \end{array}\right]_{(ab)} = \left[\begin{array}{c} i \\ k \quad l \end{array}\right]_{(ab)}^{(cd)},$<br>${}^{[3]}\left[\left[\begin{array}{c} i \\ k \quad l \end{array}\right]_{(ab)}^{(cd)}\right]_N = \left[\begin{array}{c} \widehat{cd} \\ -a \end{array}\right]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| BH3FR_ EQ8317 <b>I</b> conf <b>0.021</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 146  | ■ 0.021 | $\begin{array}{l} \left[\begin{array}{c} \beta_{\pm} \\ \alpha \end{array}\right]_{(\pm)}^{(s_1)(s_2)} = \left(\left(\begin{array}{c} \beta_{\pm} \\ \alpha \end{array}\right)_{(\pm)}^{(s_1)(s_2)}\right)_{P,Q}, \quad \bigcap = \left(\left(\begin{array}{c} \beta_{\pm} \\ \alpha \end{array}\right)\right)_{V,W} \\ \left[\begin{array}{c} \beta_{\pm} \\ \alpha \end{array}\right]_{(\pm)}^{(s_1)(s_2)} = \left(\left(\begin{array}{c} \beta_{\pm} \\ \alpha \end{array}\right)_{(\pm)}^{(s_1)(s_2)}\right)_{P,Q}, \quad \bigcap = \left(\left(\begin{array}{c} \beta_{\pm} \\ \alpha \end{array}\right)\right)_{V,W} \\ \left[\begin{array}{c} \beta_{\pm} \\ \alpha \end{array}\right]_{(\pm)}^{(s_1)(s_2)} = \left(\left(\begin{array}{c} \beta_{\pm} \\ \alpha \end{array}\right)_{(\pm)}^{(s_1)(s_2)}\right)_{P,Q}, \quad \bigcap = \left(\left(\begin{array}{c} \beta_{\pm} \\ \alpha \end{array}\right)\right)_{V,W} \end{array}$ | $\left(\begin{array}{c} \beta_{\pm} \\ \alpha \end{array}\right)_{(\pm)}^{(s_1)(s_2)} = \left(\left(\begin{array}{c} \beta_{\pm} \\ \alpha \end{array}\right)_{(\pm)}^{(s_1)(s_2)}\right)_{P,Q}, \quad \bigcap = \left(\left(\begin{array}{c} \beta_{\pm} \\ \alpha \end{array}\right)\right)_{V,W}$<br>${}^2\bar{\gamma}_{(\mu\nu)} = {}^2\bar{E}, \quad (\mu, \nu) \neq 1, \quad {}^2\bar{\gamma}_{(11)} = {}^2\bar{\gamma} \neq {}^2\bar{E},$<br>${}^2\bar{\gamma} = \operatorname{sp}\left({}^2\bar{\kappa} \times {}^2\bar{\kappa}\right) \neq {}^2\bar{\gamma}^{\times}, \quad \left[\begin{array}{c} \widehat{cd} \\ -a \end{array}\right] = [\widehat{\kappa}],$<br>$\left(\begin{array}{c} \beta_{\pm} \\ \alpha \end{array}\right)_{(\pm)}^{(s_1)(s_2)} = \left(\left(\begin{array}{c} \beta_{\pm} \\ \alpha \end{array}\right)_{(\pm)}^{(s_1)(s_2)}\right)_{P,Q}, \quad \bigcap = \left(\left(\begin{array}{c} \beta_{\pm} \\ \alpha \end{array}\right)\right)_{V,W}$ | $\left(\begin{array}{c} \beta_{\pm} \\ \alpha \end{array}\right)_{(\pm)}^{(s_1)(s_2)} = \left(\left(\begin{array}{c} \beta_{\pm} \\ \alpha \end{array}\right)_{(\pm)}^{(s_1)(s_2)}\right)_{P,Q}, \quad \bigcap = \left(\left(\begin{array}{c} \beta_{\pm} \\ \alpha \end{array}\right)\right)_{V,W}$<br>${}^2\bar{\gamma}_{(\mu\nu)} = {}^2\bar{E}, \quad (\mu, \nu) \neq 1, \quad {}^2\bar{\gamma}_{(11)} = {}^2\bar{\gamma} \neq {}^2\bar{E},$<br>${}^2\bar{\gamma} = \operatorname{sp}\left({}^2\bar{\kappa} \times {}^2\bar{\kappa}\right) \neq {}^2\bar{\gamma}^{\times}, \quad \left[\begin{array}{c} \widehat{cd} \\ -a \end{array}\right] = [\widehat{\kappa}],$<br>$\left(\begin{array}{c} \beta_{\pm} \\ \alpha \end{array}\right)_{(\pm)}^{(s_1)(s_2)} = \left(\left(\begin{array}{c} \beta_{\pm} \\ \alpha \end{array}\right)_{(\pm)}^{(s_1)(s_2)}\right)_{P,Q}, \quad \bigcap = \left(\left(\begin{array}{c} \beta_{\pm} \\ \alpha \end{array}\right)\right)_{V,W}$ |
| BH3FR_ EQ8309 (M24) <b>I</b> conf <b>0.027</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 145  | ■ 0.027 | $\begin{aligned} & \{ \}^2 \bar{\gamma}_{(a\,b)} = \\ & \operatorname{sp}\left(\{ \}^2 \bar{a} \times \{ \}^2 \bar{b}\right), \quad 2[p\,k\,l]_{(ab)} = \frac{1}{\alpha_k} \partial_k \gamma_{(ab)p\,l} + \\ & + \frac{1}{\alpha_l} \partial_l \gamma_{(ab)k\,p} - \frac{1}{\alpha_p} \partial_p \gamma_{(ab)k\,l}, \quad [3] \left[\left[\begin{array}{c} p\,k\,l \\ (ab) \end{array}\right]_N\right] = \left[\begin{array}{c} \widehat{ab} \end{array}\right] \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                | ${}^2\bar{\gamma}_{(ab)} = \operatorname{sp}\left({}^2\bar{a} \times {}^2\bar{b}\right), \quad 2[p\,k\,l]_{(ab)} = \frac{1}{\alpha_k} \partial_k \gamma_{(ab)p\,l} +$<br>$+ \frac{1}{\alpha_l} \partial_l \gamma_{(ab)k\,p} - \frac{1}{\alpha_p} \partial_p \gamma_{(ab)k\,l}, \quad [3] \left[\left[\begin{array}{c} p\,k\,l \\ (ab) \end{array}\right]_N\right] = \left[\begin{array}{c} \widehat{ab} \end{array}\right]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | ${}^2\bar{\gamma}_{(ab)} = \operatorname{sp}\left({}^2\bar{a} \times {}^2\bar{b}\right), \quad 2\left[\begin{array}{c} p\,k\,l \end{array}\right]_{(ab)} = \frac{1}{\alpha_k} \delta_k \gamma_{(ab)p\,l} +$<br>$+ \frac{1}{\alpha_l} \delta_l \gamma_{(ab)k\,p} - \frac{1}{\alpha_p} \delta_p \gamma_{(ab)k\,l}, \quad {}^{[3]}\left[\left[\begin{array}{c} p\,k\,l \end{array}\right]_{(ab)}\right]_N = \left[\begin{array}{c} \widehat{ab} \end{array}\right]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Conf.</b> MathPix confidence: ■ $\geq 0.9$ ■ $0.5\text{--}0.9$ ■ $< 0.5$ — no value<br><b>LaTeX source</b> MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ<br><b>Residual</b> not shown for this document — no residual measurements are paired with these rows.<br><b>[refined: <i>basis</i>]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here. |      |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |







| Identifier                              | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Scan image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------------|------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BH3FR_ EQ0147 (66) <b>[conf 0.075]</b>  | 117  | ■ 0.075 | \begin{aligned} & \partial \gamma_{+kl}^{\mu\nu} = 0, \\ & \psi_{(\mu\nu)kl}^{(\kappa\lambda)} = 0, \\ & \partial^2 \gamma_{+kl}^{\mu\nu} = 0, \\ & \partial q_{(\mu\nu)kl}^{(\kappa\lambda)} = 0, \\ & \partial \psi_{(\mu\nu)kl}^{(\kappa\lambda)} = 0, \\ & q_{(\mu\nu)kl}^{(\kappa\lambda)} = C_{(\mu\nu)kl}^{(\kappa\lambda)} \sum_{s=1}^q \left( \sum_{i=1}^q \left( \gamma_{(\kappa\lambda)}^i \frac{S}{\lambda} + Q_{(\mu\nu)m}^{(\kappa\lambda)} \gamma_{(\kappa\lambda)}^m \frac{S}{\lambda} \right) \right) - 1 \end{aligned} | $\partial \gamma_{+kl}^{(\mu\nu)} = 0, \quad \psi_{(\mu\nu)kl}^{(\kappa\lambda)} = 0, \quad \partial^2 \gamma_{+kl}^{(\mu\nu)} = 0,$ $\partial q_{(\mu\nu)kl}^{(\kappa\lambda)} = 0, \quad \partial \psi_{(\mu\nu)kl}^{(\kappa\lambda)} = 0,$ $q_{(\mu\nu)kl}^{(\kappa\lambda)} = C_{(\mu\nu)kl}^{(\kappa\lambda)} \sum_{s=1}^q \left( \sum_{i=1}^q \left( \gamma_{(\kappa\lambda)}^i \frac{S}{\lambda} + Q_{(\mu\nu)m}^{(\kappa\lambda)} \gamma_{(\kappa\lambda)}^m \frac{S}{\lambda} \right) \right) - 1$ | $\delta \gamma_{+kl}^{(\mu\nu)} = 0, \quad \psi_{(\mu\nu)kl}^{(\kappa\lambda)} = 0, \quad \delta^2 \gamma_{+kl}^{(\mu\nu)} = 0,$ $\delta q_{(\mu\nu)kl}^{(\kappa\lambda)} = 0, \quad \delta \psi_{(\mu\nu)kl}^{(\kappa\lambda)} = 0,$ $q_{(\mu\nu)kl}^{(\kappa\lambda)} = C_{(\mu\nu)kl}^{(\kappa\lambda)} \sum_{s=1}^q \left( \sum_{i=1}^q \left( \gamma_{(\kappa\lambda)}^i \frac{S}{\lambda} + Q_{(\mu\nu)m}^{(\kappa\lambda)} \gamma_{(\kappa\lambda)}^m \frac{S}{\lambda} \right) \right) - 1$ |
| BH3FR_EQ0303 <b>[conf 0.086]</b>        | 144  | ■ 0.086 | \begin{aligned} & \bar{\psi} = \boldsymbol{S}^2 \bar{\kappa}; \delta \bar{x}, \quad {}^2 \bar{\gamma} = sp({}^2 \bar{\kappa} \times {}^2 \bar{\kappa}), \quad {}^2 \bar{\gamma}_+ \neq {}^2 \bar{0}, \\ & \bar{\psi}; n = \bar{\xi}, \quad \bar{x} = \sum_{k=1}^N \bar{e}_k C^k \end{aligned}                                                                                                                                                                                                                                            | $\bar{\psi} = \boldsymbol{S}^2 \bar{\kappa}; \delta \bar{x}, \quad {}^2 \bar{\gamma} = sp({}^2 \bar{\kappa} \times {}^2 \bar{\kappa}), \quad {}^2 \bar{\gamma},$ $\bar{\psi}; n = \bar{\xi}, \quad \bar{x} = \sum_{k=1}^N \bar{e}_k C^k$                                                                                                                                                                                                                                                                    | $\bar{\psi} = \boldsymbol{S}^2 \bar{\kappa}; \delta \bar{x}, \quad {}^2 \bar{\gamma} = sp({}^2 \bar{\kappa} \times {}^2 \bar{\kappa}), \quad {}^2 \bar{\gamma}_+ \neq {}^2 \bar{0},$ $\bar{\psi}; n = \bar{\xi}, \quad \bar{x} = \sum_{k=1}^N \bar{e}_k C^k$                                                                                                                                                                                                                                        |
| BH3FR_ EQ0126 (54a) <b>[conf 0.090]</b> | 114  | ■ 0.090 | \begin{aligned} & \bar{\psi} = \boldsymbol{S}^2 \bar{\kappa}; \delta \bar{x}, \quad {}^2 \bar{\gamma} = sp({}^2 \bar{\kappa} \times {}^2 \bar{\kappa}), \quad {}^2 \bar{\gamma}, \\ & \bar{\psi}; n = \bar{\xi}, \quad \bar{x} = \sum_{k=1}^N \bar{e}_k C^k \end{aligned}                                                                                                                                                                                                                                                                | $\bar{\psi} = \boldsymbol{S}^2 \bar{\kappa}; \delta \bar{x}, \quad {}^2 \bar{\gamma} = sp({}^2 \bar{\kappa} \times {}^2 \bar{\kappa}), \quad {}^2 \bar{\gamma},$ $\bar{\psi}; n = \bar{\xi}, \quad \bar{x} = \sum_{k=1}^N \bar{e}_k C^k$                                                                                                                                                                                                                                                                    | $\bar{\psi} = \boldsymbol{S}^2 \bar{\kappa}; \delta \bar{x}, \quad {}^2 \bar{\gamma} = sp({}^2 \bar{\kappa} \times {}^2 \bar{\kappa}), \quad {}^2 \bar{\gamma}_+ \neq {}^2 \bar{0},$ $\bar{\psi}; n = \bar{\xi}, \quad \bar{x} = \sum_{k=1}^N \bar{e}_k C^k$                                                                                                                                                                                                                                        |
| BH3FR_ EQ0137 (59) <b>[conf 0.094]</b>  | 116  | ■ 0.094 | \bar{\lambda} \hat{\Pi} = \sum_{\varepsilon} \bar{\lambda}_{(\mu\nu)}^{(\kappa\lambda)} \left[ \begin{array}{c} \widehat{\kappa\lambda} \\ - \circ + \\ \mu\nu \end{array} \right], \quad \left[ \begin{array}{c} \widehat{\kappa\lambda} \\ - \circ + \\ \mu\nu \end{array} \right] = \left( 1 + sp^2 \bar{Q}_{(\mu\nu)}^{(\kappa\lambda)} \times () \right); \left[ \begin{array}{c} \widehat{\kappa\lambda} \\ - + \\ \mu\nu \end{array} \right]                                                                                      | $\bar{\lambda} \hat{\Pi} = \sum_{\varepsilon} \bar{\lambda}_{(\mu\nu)}^{(\kappa\lambda)} \left[ \begin{array}{c} \widehat{\kappa\lambda} \\ - \circ + \\ \mu\nu \end{array} \right], \quad \left[ \begin{array}{c} \widehat{\kappa\lambda} \\ - \circ + \\ \mu\nu \end{array} \right] = \left( 1 + sp^2 \bar{Q}_{(\mu\nu)}^{(\kappa\lambda)} \times () \right); \left[ \begin{array}{c} \widehat{\kappa\lambda} \\ - + \\ \mu\nu \end{array} \right]$                                                       | $\bar{\lambda} \hat{\Pi} = \sum_{\varepsilon} \bar{\lambda}_{(\mu\nu)}^{(\kappa\lambda)} \left[ \begin{array}{c} \widehat{\kappa\lambda} \\ - + \\ \mu\nu \end{array} \right], \quad \left[ \begin{array}{c} \widehat{\kappa\lambda} \\ - + \\ \mu\nu \end{array} \right] = \left( 1 + sp^2 \bar{Q}_{(\mu\nu)}^{(\kappa\lambda)} \times () \right); \left[ \begin{array}{c} \widehat{\kappa\lambda} \\ - + \\ \mu\nu \end{array} \right]$                                                           |

**Conf.** MathPix confidence: ■ ≥0.9   ■ 0.5–0.9   ■ <0.5   — no value

**LaTeX source** MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

**Residual** not shown for this document — no residual measurements are paired with these rows.

**[refined: *basis*]** beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Page | Conf.  | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Scan image |
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| BH3FR_ E00136 (58a) <b>[conf 0.099]</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 116  | ■0.099 | $\begin{aligned} & + \delta(\underline{k}, s); () \sum_{\varepsilon} Q_{(\alpha\beta)t}^{(\gamma\delta)\underline{s}} \begin{bmatrix} t \\ k \ m \end{bmatrix}_{(\alpha\beta)}^{(\gamma\delta)} - \\ & - \delta(\underline{k} \underline{l}, sm); () \sum_{\varepsilon} Q_{(\alpha\beta)t}^{(\gamma\delta)\underline{s}} \begin{bmatrix} t \\ k' m \end{bmatrix}_{(\alpha\beta)}^{(\gamma\delta)} + \\ & + Q_{(\mu\nu)t}^{(\kappa\lambda)\underline{i}} \delta(\underline{i} \underline{k}, ts); () \sum_{\varepsilon} \begin{bmatrix} s \\ km \end{bmatrix}_{(\alpha\beta)}^{(\gamma\delta)} (1 - \delta) - \\ & - Q_{(\mu\nu)t}^{(\kappa\lambda)\underline{i}} \delta(\underline{i} \underline{k} \underline{l}, tsm); () \sum_{\varepsilon} \begin{bmatrix} s \\ kl \end{bmatrix}_{(\alpha\beta)}^{(\gamma\delta)} (1 - \delta) \end{aligned}$ $\delta = \delta_{\mu\alpha} \delta_{\nu\beta} \delta_{\kappa\gamma} \delta_{\lambda\delta}, \quad \delta(\underline{p} \dots \underline{q}, m \dots s); A_{p \dots q} = A_{m \dots s}$ | $+ \delta(\underline{k}, s); () \sum_{\varepsilon} Q_{(\alpha\beta)t}^{(\gamma\delta)\underline{s}} \begin{bmatrix} t \\ k \ m \end{bmatrix}_{(\alpha\beta)}^{(\gamma\delta)} -$ $- \delta(\underline{k} \underline{l}, sm); () \sum_{\varepsilon} Q_{(\alpha\beta)t}^{(\gamma\delta)\underline{s}} \begin{bmatrix} t \\ k \ m \end{bmatrix}_{(\alpha\beta)}^{(\gamma\delta)} +$ $+ Q_{(\mu\nu)t}^{(\kappa\lambda)\underline{i}} \delta(\underline{i} \underline{k}, ts); () \sum_{\varepsilon} \begin{bmatrix} s \\ k \ m \end{bmatrix}_{(\alpha\beta)}^{(\gamma\delta)} (1 - \delta) -$ $- Q_{(\mu\nu)t}^{(\kappa\lambda)\underline{i}} \delta(\underline{i} \underline{k} \underline{l}, tsm); () \sum_{\varepsilon} \begin{bmatrix} s \\ k \ l \end{bmatrix}_{(\alpha\beta)}^{(\gamma\delta)} (1 - \delta),$ $\delta = \delta_{\mu\alpha} \delta_{\nu\beta} \delta_{\kappa\gamma} \delta_{\lambda\delta}, \quad \delta(\underline{p} \dots \underline{q}, m \dots s); A_{p \dots q} = A_{m \dots s}$ |            |
| BH3FR_ E00312 (M25a)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 145  | ■0.124 | $\partial_{\underline{p}}^2 x^{\underline{i}} + \partial_{\underline{p}} x^{\underline{k}} \partial_{\underline{p}} x' \begin{bmatrix} i \\ kl \end{bmatrix}_{(ab)(C')}^{(cd)}; n = 0$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $\partial_{\underline{p}}^2 x^{\underline{i}} + \partial_{\underline{p}} x^{\underline{k}} \partial_{\underline{p}} x' \begin{bmatrix} i \\ kl \end{bmatrix}_{(ab)(C')}^{(cd)}; n = 0$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |            |
| BH3FR_ E00140 (61)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 116  | ■0.125 | $L_{(\mu\nu)+}^{(\kappa\lambda)}; \left[ \begin{array}{c} \widehat{\kappa\lambda} \\ -\mu\nu \end{array} \right]_+ = {}^4\overline{0}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $L_{(\mu\nu)+}^{(\kappa\lambda)}; \left[ \begin{array}{c} \widehat{\kappa\lambda} \\ -\mu\nu \end{array} \right]_+ = {}^4\overline{0}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |            |
| BH3FR_ E00271                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 139  | ■0.133 | $\begin{aligned} & \partial \sum_j u_j(n) = \sum_j \partial u_j, \quad \partial C = 0, \quad \partial(Cu) = C \partial u, \\ & C = \text{const}(n), \quad \partial(uv) = u \partial v + v \partial u - \partial u \partial v, \\ & \partial \left( \frac{u}{v} \right) = \frac{1}{v} \left  \begin{array}{cc} \partial u \partial v & \\ u & v \end{array} \right  \cdot \left  \begin{array}{cc} v \partial v & \\ 1 & 1 \end{array} \right ^{-1} \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | $\partial \sum_j u_j(n) = \sum_j \partial u_j, \quad \partial C = 0, \quad \partial(Cu) = C \partial u,$ $C = \text{const}(n), \quad \partial(uv) = u \partial v + v \partial u - \partial u \partial v,$ $\partial \left( \frac{u}{v} \right) = \frac{1}{v} \left  \begin{array}{cc} \partial u \partial v & \\ u & v \end{array} \right  \cdot \left  \begin{array}{cc} v \partial v & \\ 1 & 1 \end{array} \right ^{-1}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |            |
| <b>Conf.</b> MathPix confidence: <span style="color: green;">■</span> ≥0.9 <span style="color: orange;">■</span> 0.5–0.9 <span style="color: red;">■</span> <0.5 — no value<br><b>LaTeX source</b> MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ<br><b>Residual</b> not shown for this document — no residual measurements are paired with these rows.<br><b>[refined: <i>basis</i>]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here. |      |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |

| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                          | Rendered                                                                                                                                                                                                                                                                                                                                                                                      | Scan image                                                                                                                                                                                                                                                                                                                                                                                     |
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| BH3FR_EQ0130 (56a)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 115  | ■ 0.138 | $\left[\begin{array}{c} \widehat{\kappa \lambda} \\ -t \\ \mu \nu \end{array}\right] = \left[\begin{array}{c} \widehat{\kappa \lambda} \\ -t \\ \mu \nu \end{array}\right]_+ = \left[\begin{array}{c} \widehat{\kappa \lambda} \\ -t \\ \mu \nu \end{array}\right] x$                                                                                                                                                 | $\left[\begin{array}{c} \widehat{\kappa \lambda} \\ -t \\ \mu \nu \end{array}\right] = \left[\begin{array}{c} \widehat{\kappa \lambda} \\ -t \\ \mu \nu \end{array}\right]_+ = \left[\begin{array}{c} \widehat{\kappa \lambda} \\ -t \\ \mu \nu \end{array}\right] x$                                                                                                                         | $\left[\begin{array}{c} \widehat{\kappa \lambda} \\ -+ \\ \mu \nu \end{array}\right] = \left[\begin{array}{c} \widehat{\kappa \lambda} \\ -+ \\ \mu \nu \end{array}\right]_+ = \left[\begin{array}{c} \widehat{\kappa \lambda} \\ -+ \\ \mu \nu \end{array}\right] x$                                                                                                                          |
| BH3FR_EQ0191 (87)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 126  | ■ 0.151 | $\{ \}_x^p (\mu)_p^q \sim S \frac{\bar{\xi}_\mu}{w} \sum_\varepsilon s p \Lambda_{(\kappa \lambda)}^{(\mu \nu)}; \left[\begin{array}{c} \widehat{\mu \nu} \\ -\frac{1}{\kappa \lambda} \end{array}\right]_+; () \prod_{k=1}^3 \partial ()_k$                                                                                                                                                                          | $p(\mu)_\pm^q \sim S \frac{\bar{\xi}_\mu}{w} \sum_\varepsilon s p \Lambda_{(\kappa \lambda)}^{(\mu \nu)}; \left[\begin{array}{c} \widehat{\mu \nu} \\ -\frac{1}{\kappa \lambda} \end{array}\right]_+; () \prod_{k=1}^3 \partial ()_k$                                                                                                                                                         | $p(\mu)_\pm^q \sim S \frac{\bar{\xi}_\mu}{w} \sum_\varepsilon s p \Lambda_{(\kappa \lambda)}^{(\mu \nu)}; \left[\begin{array}{c} \widehat{\mu \nu} \\ -\frac{1}{\kappa \lambda} \end{array}\right]_+; () \prod_{k=1}^3 \partial ()_k$                                                                                                                                                          |
| BH3FR_EQ0311 (M25)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 145  | ■ 0.153 | $\begin{aligned} & \partial_p^2 n^i + \frac{\alpha_k \alpha_l}{\alpha_i} \partial_p n^k \partial_p n^l \left[\begin{array}{c} i \\ k \\ l \end{array}\right]_{(ab)}^{(cd)}; n=0, \quad n^i = n^i(p), \\ & \partial_p^2 \xi^i = 0, \quad \left[\begin{array}{c} \widehat{cd} \\ a \end{array}\right]_{(\xi)} = \hat{0}, \quad \left[\begin{array}{c} \widehat{cd} \\ ab \end{array}\right] \neq \hat{0} \end{aligned}$ | $\partial_p^2 n^i + \frac{\alpha_k \alpha_l}{\alpha_i} \partial_p n^k \partial_p n^l \left[\begin{array}{c} i \\ k \\ l \end{array}\right]_{(ab)}^{(cd)}; n=0, \quad n^i = n^i(p),$ $\partial_p^2 \xi^i = 0, \quad \left[\begin{array}{c} \widehat{cd} \\ a \end{array}\right]_{(\xi)} = \hat{0}, \quad \left[\begin{array}{c} \widehat{cd} \\ ab \end{array}\right] \neq \hat{0}$            | $\partial_p^2 n^i + \frac{\alpha_k \alpha_l}{\alpha_i} \partial_p n^k \partial_p n^l \left[\begin{array}{c} i \\ k \\ l \end{array}\right]_{(ab)}^{(cd)}; n=0, \quad n^i = n^i(p),$ $\partial_p^2 \xi^i = 0, \quad \left[\begin{array}{c} \widehat{cd} \\ a \end{array}\right]_{(\xi)} = \hat{0}, \quad \left[\begin{array}{c} \widehat{cd} \\ ab \end{array}\right] \neq \hat{0}$             |
| BH3FR_EQ0065 (21c)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 107  | ■ 0.168 | $\begin{aligned} & \underline{\gamma}_i = A_i \exp \left( \sum_{k,l=1}^q c_{kl}^{(i)} S \underline{\psi}_{kl}; () \partial \underline{\mu} \right), \quad \hat{S}^2 \bar{\gamma}_q \hat{S}^\times = {}^2 \bar{\gamma}_-, \\ & \underline{\beta}_k = \sum_{k=1}^q \alpha_i \beta_{ik} \end{aligned}$                                                                                                                   | $\underline{\gamma}_{ii} = A_i \exp \left( \sum_{k,l=1}^q c_{kl}^{(i)} S \underline{\psi}_{kl}; () \partial \underline{\mu} \right), \quad \hat{S}^2 \bar{\gamma}_q \hat{S}^\times = {}^2 \bar{\gamma}_-$ $\hat{S} \hat{S}^\times = \hat{E}, \quad \hat{\beta} \hat{\beta}^\times = \hat{E}, \quad \underline{\mu} = \sum_{k=1}^q \beta_k ()_k,$ $\beta_k = \sum_{k=1}^q \alpha_i \beta_{ik}$ | $\underline{\gamma}_{ii} = A_i \exp \left( \sum_{k,l=1}^q c_{kl}^{(i)} S \underline{\psi}_{kl}; () \partial \underline{\mu} \right), \quad \hat{S}^2 \bar{\gamma}_q \hat{S}^\times = {}^2 \bar{\gamma}_-,$ $\hat{S} \hat{S}^\times = \hat{E}, \quad \hat{\beta} \hat{\beta}^\times = \hat{E}, \quad \underline{\mu} = \sum_{k=1}^q \beta_k ()_k,$ $\beta_k = \sum_{k=1}^q \alpha_i \beta_{ik}$ |
| BH3FR_EQ0132 (58)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 115  | ■ 0.169 | $L_{(\mu \nu)}^{(\kappa \lambda)}; \left[\begin{array}{c} \widehat{\kappa \lambda} \\ -+ \\ \mu \nu \end{array}\right] = {}^4 \overline{0}, \quad \left[\begin{array}{c} \widehat{\kappa \lambda} \\ -+ \\ \mu \nu \end{array}\right] = \left[\begin{array}{c} \widehat{\kappa \lambda} \\ -+ \\ \mu \nu \end{array}\right]_+$                                                                                        | <i>(not rendered)</i>                                                                                                                                                                                                                                                                                                                                                                         | $L_{(\mu \nu)}^{(\kappa \lambda)}; \left[\begin{array}{c} \widehat{\kappa \lambda} \\ -+ \\ \mu \nu \end{array}\right] = {}^4 \overline{0}, \quad \left[\begin{array}{c} \widehat{\kappa \lambda} \\ -+ \\ \mu \nu \end{array}\right] = \left[\begin{array}{c} \widehat{\kappa \lambda} \\ -+ \\ \mu \nu \end{array}\right]_+$                                                                 |
| BH3FR_EQ0307 (M23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 145  | ■ 0.170 | $N = \bar{K} \bar{\partial} \bar{n}, \quad \bar{n} = \sum_{k=1}^N \bar{e}_k Z(k); n$                                                                                                                                                                                                                                                                                                                                  | $N = \bar{K} \bar{\partial} \bar{n}, \quad \bar{n} = \sum_{k=1}^N \bar{e}_k Z(k); n$                                                                                                                                                                                                                                                                                                          | $\underline{N} = S \bar{K} \bar{\partial} \bar{n}, \quad \bar{n} = \sum_{k=1}^N \bar{e}_k Z(k); n$                                                                                                                                                                                                                                                                                             |
| <b>Conf.</b> MathPix confidence: ■ $\geq 0.9$ ■ 0.5–0.9 ■ $< 0.5$ — no value<br><b>LaTeX source</b> MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ<br><b>Residual</b> not shown for this document — no residual measurements are paired with these rows.<br><b>[refined: <i>basis</i>]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here. |      |         |                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                |

| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                 | Rendered | Scan image                                                                                                                                                                                                                                                                                                                                                                                                 |
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| BH3FR_EQ0084 (31)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 109  | ■ 0.179 | $\begin{aligned} & m_q \approx \frac{h}{\beta c} - \sqrt[3]{4\alpha_{q/\varepsilon}}, \quad \beta = \sqrt{\tau\sqrt{\pi}}, \quad \varepsilon\tau = E, \\ & \alpha_q = 1 + 4\left(\frac{q}{\pi}\right)^4 \end{aligned}$                                                                                                                                                                                                                       |          | $m_q \approx \frac{h}{\beta c} - \sqrt[3]{4\alpha_{q/\varepsilon}}, \quad \beta = \sqrt{\tau\sqrt{\pi}}, \quad \varepsilon\tau = E,$ $\alpha_q = 1 + 4\left(\frac{q}{\pi}\right)^4$                                                                                                                                                                                                                        |
| BH3FR_EQ0018 (5c)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 101  | ■ 0.182 | $\begin{aligned} & \hat{C}(R_6) = \begin{pmatrix} \cos\psi & 0 & 0 & \sin\psi & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ -\sin\psi & 0 & 0 & \cos\psi & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad u\beta = v, \quad i\beta = \operatorname{tg}\psi, \\ & \psi = -\psi^* \end{aligned}$                                                                                                                                               |          | $\hat{C}(R_6) = \begin{pmatrix} \cos\psi & 0 & 0 & \sin\psi & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ -\sin\psi & 0 & 0 & \cos\psi & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}, \quad u\beta = v, \quad i\beta = \operatorname{tg}\psi,$ $\psi = -\psi^*$                                                                                                                                                |
| BH3FR_EQ00300                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 143  | ■ 0.222 | $\begin{array}{l} \hat{s} = \operatorname{ROT}_N \hat{\phi}, \quad (\hat{s}; n)_{n=1} = \hat{\tau}, \quad \hat{s} = ({}^2\bar{s}_{\alpha\beta})_p, \\ \hat{\phi} = (\bar{\phi}_{\alpha\beta})_p, \quad {}^2\bar{s}_{\alpha\beta}; n = \boldsymbol{S} \boldsymbol{S} \boldsymbol{X} \bar{\xi}_{\alpha} \times \partial \bar{\xi}_{\beta} \end{array}$                                                                                         |          | $\hat{s} = \operatorname{ROT}_N \hat{\phi}, \quad (\hat{s}; n)_{n=1} = \hat{\tau}, \quad \hat{s} = ({}^2\bar{s}_{\alpha\beta})_p,$ $\hat{\phi} = (\bar{\phi}_{\alpha\beta})_p, \quad {}^2\bar{s}_{\alpha\beta}; n = \boldsymbol{S} \boldsymbol{S} \partial \bar{\xi}_{\alpha} \times \bar{\partial} \bar{\xi}_{\beta}$                                                                                     |
| BH3FR_EQ00323 (M31)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 148  | ■ 0.229 | $\begin{aligned} & \varsigma_{klm}^i = \frac{1}{\alpha_l} l \begin{bmatrix} i \\ k \ l \end{bmatrix} - \frac{1}{\alpha_m} \partial_m \begin{bmatrix} i \\ k \ l \end{bmatrix} + \begin{bmatrix} i \\ s \ l \end{bmatrix}; () \begin{bmatrix} s \\ k \ m \end{bmatrix} - \begin{bmatrix} i \\ s \ l \end{bmatrix}; () \begin{bmatrix} s \\ k \ l \end{bmatrix} \\ & \lim_{\tau \rightarrow 0} {}^4\bar{\zeta}; n = {}^4\bar{R} \end{aligned}$ |          | $\varsigma_{klm}^i = \frac{1}{\alpha_l} l \begin{bmatrix} i \\ k \ m \end{bmatrix} - \frac{1}{\alpha_m} \partial_m \begin{bmatrix} i \\ k \ l \end{bmatrix} + \begin{bmatrix} i \\ s \ l \end{bmatrix}; () \begin{bmatrix} s \\ k \ m \end{bmatrix} - \begin{bmatrix} i \\ s \ m \end{bmatrix}; () \begin{bmatrix} s \\ k \ l \end{bmatrix},$ $\lim_{\tau \rightarrow 0} {}^4\bar{\zeta}; n = {}^4\bar{R}$ |
| BH3FR_EQ00131 (57)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 115  | ■ 0.241 | $\begin{aligned} & \widehat{[\ ]} = \sum_{\varepsilon} \left( \begin{bmatrix} \widehat{\kappa\lambda} \\ -+ \\ \mu\nu \end{bmatrix}_+ + sp^2 Q_{(\mu\nu)}^{(\kappa\lambda)}; () \times \begin{bmatrix} \widehat{\kappa\lambda} \\ -+ \\ \mu\nu \end{bmatrix}_+ \right), \\ & \varepsilon \equiv (\kappa, \lambda, \mu, \nu) \end{aligned}$                                                                                                   |          | $\widehat{[\ ]} = \sum_{\varepsilon} \left( \begin{bmatrix} \widehat{\kappa\lambda} \\ -+ \\ \mu\nu \end{bmatrix}_+ + sp^2 Q_{(\mu\nu)}^{(\kappa\lambda)}; () \times \begin{bmatrix} \widehat{\kappa\lambda} \\ -+ \\ \mu\nu \end{bmatrix}_+ \right),$ $\varepsilon \equiv (\kappa, \lambda, \mu, \nu)$                                                                                                    |
| <p><b>Conf.</b> MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ &lt;0.5 — no value</p> <p><b>LaTeX source</b> MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ</p> <p><b>Residual</b> not shown for this document — no residual measurements are paired with these rows.</p> <p><b>[refined: <i>basis</i>]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</p> |      |         |                                                                                                                                                                                                                                                                                                                                                                                                                                              |          |                                                                                                                                                                                                                                                                                                                                                                                                            |



| Identifier         | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Scan image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| BH3FR_EQ0155 (70)  | 119  | ■ 0.274 | $\begin{array}{l} K_1^{-}=1,4,5 \quad . \quad \& K_2^{-}=2,4,6 \quad . \\ \& K_3^{-}=5,3,6 \quad . \quad \backslash \backslash K_1^{-}=1,4 \quad . \quad \& K_2^{-}=1,5 \quad . \\ \& K_3^{-}=2,4 \quad . \quad \backslash \backslash K_4^{-}=2,6 \quad . \quad \& K_5^{-}=3,5 \quad . \quad \& K_6^{-}=3,6, \quad \backslash \backslash (1 \ldots 6)_{-}=(6 \ldots 1)_{+} \quad \& \end{array}$                                                                                                                                                                                                                                                                                                                              | $\begin{array}{lll} K_1^{-}=1,4,5. & K_2^{-}=2,4,6. & K_3^{-}=5,3,6. \\ K_1^{-}=1,4. & K_2^{-}=1,5. & K_3^{-}=2,4. \\ K_4^{-}=2,6. & K_5^{-}=3,5. & K_6^{-}=3,6, \\ (1 \ldots 6)_{-}=(6 \ldots 1)_{+} \end{array}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | $\begin{array}{lll} K_1^{-}=1,4,5. & K_2^{-}=2,4,6. & K_3^{-}=5,3,6. \\ K_1^{-}=1,4. & K_2^{-}=1,5. & K_3^{-}=2,4. \\ K_4^{-}=2,6. & K_5^{-}=3,5. & K_6^{-}=3,6, \\ (1 \ldots 6)_{-}=(6 \ldots 1)_{+} \end{array}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| BH3FR_EQ0186 (85a) | 125  | ■ 0.289 | $\begin{array}{l} \bar{F}_{pq}(\mu) \perp \bar{\sigma}_{pq} = \omega_{pq} \bar{s}_{pq}, \quad \bar{s}_{pq} = \bar{s}_{pq}^{(o)} \cdot s_{pq}, \\ \left  \bar{s}_{pq}^{(o)} \right  = 1, \quad \bar{s}_p \parallel \bar{s}_q, \quad \varphi = \pi/2 \end{array}$                                                                                                                                                                                                                                                                                                                                                                                                                                                               | $\begin{array}{lll} \bar{F}_{pq}(\mu) \perp \bar{\sigma}_{pq} = \omega_{pq} \bar{s}_{pq}, & \bar{s}_{pq} = \bar{s}_{pq}^{(o)} \cdot s_{pq}, \\ \left  \bar{s}_{pq}^{(o)} \right  = 1, & \bar{s}_p \parallel \bar{s}_q, & \varphi = \pi/2 \end{array}$                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $\begin{array}{lll} \bar{F}_{pq}(\mu) \perp \bar{\sigma}_{pq} = \omega_{pq} \bar{s}_{pq}, & \bar{s}_{pq} = \bar{s}_{pq}^{(o)} \cdot s_{pq}, \\ \left  \bar{s}_{pq}^{(o)} \right  = 1, & \bar{s}_p \parallel \bar{s}_q, & \varphi = \pi/2 \end{array}$                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| BH3FR_EQ0156 (70a) | 119  | ■ 0.290 | $\begin{array}{l} \& Q_1(\mathrm{I}h \rightarrow \mathrm{I}a) = Q_1(K_i^{++} \rightarrow K_i^{--})_1^3, \\ \& Q_1^{(o)} = Q_1(1,2,3 \rightarrow 6,5,4), \\ \& Q_2^{+}(\mathrm{I}h \rightarrow \mathrm{II}a) = Q_2(1,2,3 \rightarrow K_{3,5}^{-} K_{1,6}^{-} K_{2,4}^{-}), \\ \& Q_2^{-}(\mathrm{II}h \rightarrow \mathrm{I}a) = Q_2((1,6)(2,5)(3,4) \rightarrow (4,5)(4,6)(5,6)), \\ \& Q_3(\mathrm{II}h \rightarrow \mathrm{II}a) = Q_3(K_k^{+} \rightarrow K_k^{-})_1^6, \\ \& Q_4^{+}(\mathrm{I}h \rightarrow \mathrm{II}h) = Q_4(1,2,3 \rightarrow K_{5,3}^{+}, K_{6,1}^{+} K_{4,2}^{+}), \\ \& Q_4^{-}(\mathrm{II}a \rightarrow \mathrm{I}a) = Q_4(K_{2,4}^{-}, K_{1,6}^{-}, K_{3,5}^{-} \rightarrow 4,5,6) \end{array}$ | $\begin{array}{l} Q_1(\mathrm{I}h \rightarrow \mathrm{I}a) = Q_1(K_i^{++} \rightarrow K_i^{--})_1^3, \\ Q_1^{(o)} = Q_1(1,2,3 \rightarrow 6,5,4), \\ Q_2^{+}(\mathrm{I}h \rightarrow \mathrm{II}a) = Q_2(1,2,3 \rightarrow K_{3,5}^{-} K_{1,6}^{-} K_{2,4}^{-}), \\ Q_2^{-}(\mathrm{II}h \rightarrow \mathrm{I}a) = Q_2((1,6)(2,5)(3,4) \rightarrow (4,5)(4,6)(5,6)), \\ Q_3(\mathrm{II}h \rightarrow \mathrm{II}a) = Q_3(K_k^{+} \rightarrow K_k^{-})_1^6, \\ Q_4^{+}(\mathrm{I}h \rightarrow \mathrm{II}h) = Q_4(1,2,3 \rightarrow K_{5,3}^{+}, K_{6,1}^{+} K_{4,2}^{+}), \\ Q_4^{-}(\mathrm{II}a \rightarrow \mathrm{I}a) = Q_4(K_{2,4}^{-}, K_{1,6}^{-}, K_{3,5}^{-} \rightarrow 4,5,6) \end{array}$ | $\begin{array}{l} Q_1(\mathrm{I}h \rightarrow \mathrm{I}a) = Q_1(K_i^{++} \rightarrow K_i^{--})_1^3, \\ Q_1^{(o)} = Q_1(1,2,3 \rightarrow 6,5,4), \\ Q_2^{+}(\mathrm{I}h \rightarrow \mathrm{II}a) = Q_2(1,2,3 \rightarrow K_{3,5}^{-} K_{1,6}^{-} K_{2,4}^{-}), \\ Q_2^{-}(\mathrm{II}h \rightarrow \mathrm{I}a) = Q_2((1,6)(2,5)(3,4) \rightarrow (4,5)(4,6)(5,6)), \\ Q_3(\mathrm{II}h \rightarrow \mathrm{II}a) = Q_3(K_k^{+} \rightarrow K_k^{-})_1^6, \\ Q_4^{+}(\mathrm{I}h \rightarrow \mathrm{II}h) = Q_4(1,2,3 \rightarrow K_{5,3}^{+}, K_{6,1}^{+} K_{4,2}^{+}), \\ Q_4^{-}(\mathrm{II}a \rightarrow \mathrm{I}a) = Q_4(K_{2,4}^{-}, K_{1,6}^{-}, K_{3,5}^{-} \rightarrow 4,5,6) \end{array}$ |
| BH3FR_EQ0314       | 146  | ■ 0.297 | $\begin{array}{l} \& \partial_l \varphi = \alpha_l \left[ \begin{array}{c} k \\ k \end{array} \right]_{(\kappa)}^{(\kappa)}, \quad \ln \sqrt{ g } = \varphi; n, \quad {}^2\bar{g} = {}^2\bar{\gamma}; n, \\ \& {}^2\bar{\gamma} = \mathrm{sp} \left( {}^2\bar{\kappa} \times {}^2\bar{\kappa} \right) = {}^2\bar{\gamma} \times \end{array}$                                                                                                                                                                                                                                                                                                                                                                                  | $\begin{array}{l} \partial_l \varphi = \alpha_l \left[ \begin{array}{c} k \\ k \end{array} \right]_{(\kappa)}^{(\kappa)}, \quad \ln \sqrt{ g } = \varphi; n, \quad {}^2\bar{g} = {}^2\bar{\gamma}; n, \\ {}^2\bar{\gamma} = \mathrm{sp} \left( {}^2\bar{\kappa} \times {}^2\bar{\kappa} \right) = {}^2\bar{\gamma} \times \end{array}$                                                                                                                                                                                                                                                                                                                                                                   | $\begin{array}{l} \partial_l \varphi = \alpha_l \left[ \begin{array}{c} k \\ k \end{array} \right]_{(\kappa)}^{(\kappa)}, \quad \ln \sqrt{ g } = \varphi; n, \quad {}^2\bar{g} = {}^2\bar{\gamma}; n, \\ {}^2\bar{\gamma} = \mathrm{sp} \left( {}^2\bar{\kappa} \times {}^2\bar{\kappa} \right) = {}^2\bar{\gamma} \times \end{array}$                                                                                                                                                                                                                                                                                                                                                                   |

**Conf.** MathPix confidence: ■ ≥0.9    ■ 0.5–0.9    ■ <0.5    — no value

**LaTeX source** MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

**Residual** not shown for this document — no residual measurements are paired with these rows.

**[refined: *basis*]** beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Scan image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| BH3FR_EQ0320 (M29)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 147  | ■ 0.303 | $\begin{aligned} & \frac{1}{\alpha_m} \frac{\partial}{\partial p} \frac{1}{p} (\kappa)_l; p - \frac{1}{\alpha_l} \frac{\partial}{\partial l} \frac{1}{p} (\kappa)_m; p = \\ & = \frac{1}{\alpha_l} \frac{\partial}{\partial l} \left[ \begin{matrix} s & s \\ m & s \end{matrix} \right]_+; n - \frac{1}{\alpha_m} \frac{\partial}{\partial m} \left[ \begin{matrix} s & \\ l & s \end{matrix} \right]_+; n, \quad \left[ \begin{matrix} i \\ k \end{matrix} \right]_{(\kappa)}^{(\kappa)} = \left[ \begin{matrix} i \\ k & l \end{matrix} \right] \end{aligned}$ | $\frac{1}{\alpha_m} \frac{\partial}{\partial p} \frac{1}{p} (\kappa)_l; p - \frac{1}{\alpha_l} \frac{\partial}{\partial l} \frac{1}{p} (\kappa)_m; p =$ $= \frac{1}{\alpha_l} \frac{\partial}{\partial l} \left[ \begin{matrix} s & s \\ m & s \end{matrix} \right]_+; n - \frac{1}{\alpha_m} \frac{\partial}{\partial m} \left[ \begin{matrix} s & \\ l & s \end{matrix} \right]_+; n, \quad \left[ \begin{matrix} i \\ k \end{matrix} \right]_{(\kappa)}^{(\kappa)} = \left[ \begin{matrix} i \\ k & l \end{matrix} \right]$ | $\frac{1}{\alpha_m} \frac{\partial}{\partial p} \frac{1}{p} (\kappa)_l; p - \frac{1}{\alpha_l} \frac{\partial}{\partial l} \frac{1}{p} (\kappa)_m; p =$ $= \frac{1}{\alpha_l} \frac{\partial}{\partial l} \left[ \begin{matrix} s & s \\ m & s \end{matrix} \right]_+; n - \frac{1}{\alpha_m} \frac{\partial}{\partial m} \left[ \begin{matrix} s & \\ l & s \end{matrix} \right]_+; n, \quad \left[ \begin{matrix} i \\ k \end{matrix} \right]_{(\kappa)}^{(\kappa)} = \left[ \begin{matrix} i \\ k & l \end{matrix} \right]$ |
| BH3FR_EQ0185 (85)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 125  | ■ 0.307 | $\begin{aligned} & (-p) \equiv F_p(\mu, N^\pm) = N^+(p) \binom{\kappa \lambda}{v \mu} N^-(p), \\ & (-q) \equiv F_q(\mu, N^\pm) = N^+(q) \binom{\kappa' \lambda'}{v' \mu} N^-(q), \\ & (\kappa \lambda v) \neq (\kappa' \lambda' v'), \quad F_p(N^\pm(p), \mu) = F_q(N^\pm(q), \mu), \\ & \mu \stackrel{\wedge}{=} {}^2 \bar{\kappa}_{(\mu)} \end{aligned}$                                                                                                                                                                                                        | $(-p) \equiv F_p(\mu, N^\pm) = N^+(p) \binom{\kappa \lambda}{v} N^-(p)$ $(-q) \equiv F_q(\mu, N^\pm) = N^+(q) \binom{\kappa' \lambda'}{v' \mu} N^-(q)$ $(\kappa \lambda v) \neq (\kappa' \lambda' v'), \quad F_p(N^\pm(p), \mu) = F_q(N^\pm(q), \mu),$ $\mu \stackrel{\wedge}{=} {}^2 \bar{\kappa}_{(\mu)}$                                                                                                                                                                                                                    | $(-p) \equiv F_p(\mu, N^\pm) = N^+(p) \binom{\kappa \lambda}{v \mu} N^-(p),$ $(-q) \equiv F_q(\mu, N^\pm) = N^+(q) \binom{\kappa' \lambda'}{v' \mu} N^-(q),$ $(\kappa \lambda v) \neq (\kappa' \lambda' v'), \quad F_p(N^\pm(p), \mu) = F_q(N^\pm(q), \mu),$ $\mu \stackrel{\wedge}{=} {}^2 \bar{\kappa}_{(\mu)}$                                                                                                                                                                                                              |
| <div>Conf. MathPix confidence: <span>■ ≥0.9</span> <span>■ 0.5-0.9</span> <span>■ &lt;0.5</span> — no value</div> <div>LaTeX source MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ</div> <div>Residual not shown for this document — no residual measurements are paired with these rows.</div> <div><b>[refined: <i>basis</i>]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</div> |      |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Scan image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| BH3FR_EQ0241 (109a)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 133  | ■ 0.314 | $\begin{aligned} & \& w_{\{1\}+1-k=(1-Q)\left[A_{\{11\}}-P\left(A_{\{12\}}+\frac{\kappa q}{\eta_{qk}}A_{\{13\}}\right)-\binom{P}{2}\left(A_{\{14\}}-\frac{q}{\eta_{qk}}A_{\{15\}}\right)\right]+\kappa Q\eta_{qk}A_{\{16\}}, \quad w_2-2+k=(q-1)A_{\{21\}}+ \\ & + (1-P)A_{\{22\}}+\binom{P}{2}\left(A_{\{23\}}-q_x\eta_{qk}(1+A_{\{24\}}(1+q_x))^{-1}A_{\{25\}}\right)+ \\ & + \kappa\left(A_{\{26\}}+q\eta_{qk}^2A_{\{31\}}\right)+\binom{P}{3}\left[q^3(3-q)^{-1}(\varepsilon q_x-(-1)^q)A_{\{33\}}+ \right. \\ & + \frac{e(P-Q)\eta_{qk}}{(8-q(q-1)A_{\{66\}})\eta^2}\left(1-\frac{q}{\eta_{qk}}(2- \right. \\ & \left. -q)A_{\{34\}}^{1-\varepsilon q_x}A_{\{35\}}\right)\eta^{(q-1)q/4}-A_{\{36\}}\left]+\binom{Q}{3}\eta_{qk}A_{\{32\}} \end{aligned}$ | $w_1+1-k=(1-Q)\left[A_{11}-P\left(A_{12}+\frac{\kappa q}{\eta_{qk}}A_{13}\right)-\binom{P}{2}\left(A_{14}-\frac{q}{\eta_{qk}}A_{15}\right)\right]+\kappa Q\eta_{qk}A_{16}, \quad w_2-2+k=(q-1)A_{21}+ \\ + (1-P)A_{22}+\binom{P}{2}\left(A_{23}-q_x\eta_{qk}(1+A_{24}(1+q_x))^{-1}A_{25}\right)+ \\ + \kappa\left(A_{26}+q\eta_{qk}^2A_{31}\right)+\binom{P}{3}\left[q^3(3-q)^{-1}(\varepsilon q_x-(-1)^q)A_{33}+ \right. \\ + \frac{e(P-Q)\eta_{qk}}{(8-q(q-1)A_{66})\eta^2}\left(1-\frac{q}{\eta_{qk}}(2- \right. \\ \left. -q)A_{34}^{1-\varepsilon q_x}A_{35}\right)\eta^{(q-1)q/4}-A_{36}\left]+\binom{Q}{3}\eta_{qk}A_{32}$ | $w_1+1-k=(1-Q)\left[A_{11}-P\left(A_{12}+\frac{\kappa q}{\eta_{qk}}A_{13}\right)-\binom{P}{2}\left(A_{14}-\frac{q}{\eta_{qk}}A_{15}\right)\right]+\kappa Q\eta_{qk}A_{16}, \quad w_2-2+k=(q-1)A_{21}+ \\ + (1-P)A_{22}+\binom{P}{2}\left(A_{23}-q_x\eta_{qk}(1+A_{24}(1+q_x))^{-1}A_{25}\right)+ \\ + \kappa\left(A_{26}+q\eta_{qk}^2A_{31}\right)+\binom{P}{3}\left[q^3(3-q)^{-1}(\varepsilon q_x-(-1)^q)A_{33}+ \right. \\ + \frac{e(P-Q)\eta_{qk}}{(8-q(q-1)A_{66})\eta^2}\left(1-\frac{q}{\eta_{qk}}(2- \right. \\ \left. -q)A_{34}^{1-\varepsilon q_x}A_{35}\right)\eta^{(q-1)q/4}-A_{36}\left]+\binom{Q}{3}\eta_{qk}A_{32}$ |
| BH3FR_EQ0059 (20)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 105  | ■ 0.315 | $\begin{aligned} & \& \left(E-\psi_{kl}\right)^{\Lambda_{kl}+1} \cdot \psi_{kl}^{\Lambda_{kl}-1}=2^{-2 \Lambda_{kl}} \cdot C_{kl} e^{-\lambda_{kl} \mu}, \\ & \psi_{kl}=\frac{b_i^{(kl)}}{\lambda(k l)}\left[\begin{array}{c} i \\ k l \end{array}\right], \quad \mu=\sum_{m=1}^q \alpha_m()_m, \quad C_{kl}=\text {const} \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                     | $(E-\psi_{kl})^{\Lambda_{kl}+1} \cdot \psi_{kl}^{\Lambda_{kl}-1}=2^{-2 \Lambda_{kl}} \cdot C_{kl} e^{-\lambda_{kl} \mu},$ $\psi_{kl}=\frac{b_i^{(kl)}}{\lambda(k l)}\left[\begin{array}{c} i \\ k l \end{array}\right], \quad \mu=\sum_{m=1}^q \alpha_m()_m, \quad C_{kl}=\text {const}$                                                                                                                                                                                                                                                                                                                                          | $(E-\psi_{kl})^{A_{kl}+1} \cdot \psi_{kl}^{A_{kl}-1}=2^{-2 A_{kl}} \cdot C_{kl} e^{-\lambda_{kl} \mu},$ $\psi_{kl}=\frac{b_i^{(kl)}}{\lambda(k l)}\left[\begin{array}{c} i \\ k l \end{array}\right], \quad \mu=\sum_{m=1}^q \alpha_m()_m, \quad C_{kl}=\text {const}$                                                                                                                                                                                                                                                                                                                                                            |
| BH3FR_EQ0299                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 143  | ■ 0.322 | $\begin{array}{l} C_{\{k\}}=\kappa_{\{k\}} \sqrt[p]{\tau}()_{\{k\}}, \quad \& x_{\{k\}}=C_{\{k\}} ; n, \quad \& C_{\{k\}}=X_{\{k\}}, \\ \& x_{\{k\}}=C_{\{k\}} ; n, \quad \& C_{\{k\}}=X_{\{k\}}, \quad \& \{ \}^{\{2\}} \bar{\gamma} \\ ; n=\text {text } \{ \text {const } \}, \quad \& C_{\{k\}} \neq X_{\{k\}}, \quad \& \{ \}^{\{2\}} \bar{\gamma} \end{array}$                                                                                                                                                                                                                                                                                                                                                                                          | $C_k=\kappa_k \sqrt[p]{\tau}()_k, \quad x_k=C_k ; n, \quad C_k=X_k, \\ {}^{2\overline{\gamma}} ; n=\text{const}, \quad C_k \neq X_k, \quad {}^{2\overline{\gamma}} ; n={^2\overline{g}}$                                                                                                                                                                                                                                                                                                                                                                                                                                          | $C_k=\kappa_k \sqrt[p]{\tau}()_k, \quad x_k=C_k ; n, \quad C_k=X_k, \\ {}^{2\overline{\gamma}} ; n=\text{const}, \quad C_k \neq X_k, \quad {}^{2\overline{\gamma}} ; n={^2\overline{g}}$                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| BH3FR_EQ0051 (17)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 104  | ■ 0.333 | $\begin{array}{l} \Gamma ; \psi=\lambda \psi, \quad \varphi=\psi ; n, \quad S_{\Omega}\left(\varphi^* \Gamma ; \varphi-\varphi(\Gamma ; \varphi)^*\right) \delta \Omega=0, \\ S_{\Omega} \varphi \varphi^* \partial \Omega<\infty, \quad \lambda=\lambda^* \end{array}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | $\Gamma ; \psi=\lambda \psi, \quad \varphi=\psi ; n, \quad S_{\Omega}\left(\varphi^* \Gamma ; \varphi-\varphi(\Gamma ; \varphi)^*\right) \delta \Omega=0, \\ S_{\Omega} \varphi \varphi^* \partial \Omega<\infty, \quad \lambda=\lambda^*$                                                                                                                                                                                                                                                                                                                                                                                        | $\Gamma ; \psi=\lambda \psi, \quad \varphi=\psi ; n, \quad S_{\Omega}\left(\varphi^* \Gamma ; \varphi-\varphi(\Gamma ; \varphi)^*\right) \delta \Omega=0, \\ S_{\Omega} \varphi \varphi^* \delta \Omega<\infty, \quad \lambda=\lambda^*$                                                                                                                                                                                                                                                                                                                                                                                          |
| <p><b>Conf.</b> MathPix confidence: ■ <math>\geq 0.9</math> ■ <math>0.5-0.9</math> ■ <math>&lt; 0.5</math> — no value</p> <p><b>LaTeX source</b> MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ</p> <p><b>Residual</b> not shown for this document — no residual measurements are paired with these rows.</p> <p><b>[refined: <i>basis</i>]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</p> |      |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

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| BH3FR_EQ0295        | 143  | ■ 0.337 | $\begin{array}{l} \hat{\kappa}=\left(\kappa^i-\kappa^k\right)_N, \text { \& } \hat{\kappa} _N \neq 0, \text { \& } 2_{\{\bar{\gamma}\}} \\ \gamma_{\pm i k}=\frac{1}{2}\left(\gamma_i \times \gamma_k\right)_{\pm}, \quad \kappa^i-\kappa^k\left(\gamma_i \times \gamma_k\right)_{+}-2 \alpha \frac{()}{()}=0 \end{array}$                                          | $\hat{\kappa}=\left(\kappa^i-\kappa^k\right)_N, \quad \hat{\kappa} _N \neq 0, \quad 2_{\bar{\gamma}}=\bar{\gamma} \times \bar{\gamma},$ $=\frac{1}{2}\left(\gamma_i \times \gamma_k\right)_{\pm}, \quad \kappa^i-\kappa^k\left(\gamma_i \times \gamma_k\right)_{+}-2 \alpha \frac{()}{()}=0$                                          | $\hat{\kappa}=\left(\kappa^i \kappa^k\right)_N, \quad \hat{\kappa} _N \neq 0, \quad {}^2 \bar{\gamma}=\bar{\gamma} \times \bar{\gamma}, \quad \gamma_{\pm i k}=\frac{1}{2}\left(\gamma_i \times \gamma_k\right)_{\pm}, \quad \kappa^i \kappa^k\left(\gamma_i \times \gamma_k\right)_{+}-2 \alpha \frac{()}{()}=0$                     |
| BH3FR_EQ0272        | 139  | ■ 0.340 | $\begin{array}{l} \partial \varphi=0, \quad n=n_{\mathrm{ext}}, \quad \varphi=\varphi_{\mathrm{ext}}, \quad \varphi_{\mathrm{ext}}=\varphi_{\max } \\ \partial^2 \varphi<0, \quad \varphi_{\mathrm{ext}}=\varphi_{\min }, \quad \partial^2 \varphi>0, \quad \varphi_{\mathrm{ext}}=\varphi_w \\ \partial^2 \varphi=0 \end{array}$                                   | $\partial \varphi=0, \quad n=n_{\mathrm{ext}}, \quad \varphi=\varphi_{\mathrm{ext}}, \quad \varphi_{\mathrm{ext}}=\varphi_{\max }$ $\partial^2 \varphi<0, \quad \varphi_{\mathrm{ext}}=\varphi_{\min }, \quad \partial^2 \varphi>0, \quad \varphi_{\mathrm{ext}}=\varphi_w$ $\partial^2 \varphi=0$                                    | $\delta \varphi=0, \quad n=n_{\mathrm{ext}}, \quad \varphi=\varphi_{\mathrm{ext}}, \quad \varphi_{\mathrm{ext}}=\varphi_{\max },$ $\delta^2 \varphi<0, \quad \varphi_{\mathrm{ext}}=\varphi_{\min }, \quad \delta^2 \varphi>0, \quad \varphi_{\mathrm{ext}}=\varphi_w,$ $\delta^2 \varphi=0$                                          |
| BH3FR_EQ0108 (45)   | 112  | ■ 0.340 | $\begin{array}{l} \text {\& } z \approx B s(1-B s)^{-1}>0, \quad \text {\& } c B=\sqrt{\pi e \gamma \sigma}, \quad s \gg \varrho, \\ \gamma m^3 \varrho=h^2, \quad z \lambda \sim \delta \lambda \end{array}$                                                                                                                                                       | $z \approx B s(1-B s)^{-1}>0, \quad c B=\sqrt{\pi e \gamma \sigma}, \quad s \geqslant \rho,$ $\gamma m^3 \varrho=h^2, \quad z \lambda \sim \delta \lambda$                                                                                                                                                                            | $z \approx B s(1-B s)^{-1}>0, \quad c B=\sqrt{\pi e \gamma \sigma}, \quad s \gg \varrho,$ $\gamma m^3 \varrho=h^2, \quad z \lambda \sim \delta \lambda$                                                                                                                                                                               |
| BH3FR_EQ0273        | 139  | ■ 0.362 | $\begin{array}{l} n_{\boldsymbol{S}}=0, \quad \boldsymbol{S} \varphi \partial n=\varphi\left(n_1\right), \\ n_1+1 \end{array}$                                                                                                                                                                                                                                      | $\begin{array}{l} n_{\boldsymbol{S}}=0, \quad \boldsymbol{S} \varphi \partial n=\varphi\left(n_1\right), \\ n_1+1 \end{array}$                                                                                                                                                                                                        | $\begin{array}{l} n_{\boldsymbol{S}}=0, \quad \boldsymbol{S} \varphi \partial n=\varphi\left(n_1\right), \\ n_1+1 \end{array}$                                                                                                                                                                                                        |
| BH3FR_EQ0286 (M11c) | 141  | ■ 0.373 | $\begin{array}{l} \bar{C}_i=\bar{e}_i C_i, \quad C_i ; n=\varphi_i\left(n_k\right)_1^L, \quad {}^m \bar{C}=[m]\left[\prod_{k=1}^m C_{i_k}\right] \\ {}^m \bar{T}={ }^m \bar{C} ; n, \quad 0 \leqslant m \leqslant L \end{array}$                                                                                                                                    | $\bar{C}_i=\bar{e}_i C_i, \quad C_i ; n=\varphi_i\left(n_k\right)_1^L, \quad {}^m \bar{C}=[m]\left[\prod_{k=1}^m C_{i_k}\right]$ ${}^m \bar{T}={ }^m \bar{C} ; n, \quad 0 \leqslant m \leqslant L$                                                                                                                                    | $\bar{C}_i=\bar{e}_i C_i, \quad C_i ; n=\varphi_i\left(n_k\right)_1^L, \quad {}^m \bar{C}=[m]\left[\prod_{k=1}^m C_{i_k}\right],$ ${}^m \bar{T}={ }^m \bar{C} ; n, \quad 0 \leqslant m \leqslant L$                                                                                                                                   |
| BH3FR_EQ0276        | 139  | ■ 0.376 | $\begin{array}{l} \boldsymbol{S} \Sigma=\Sigma \boldsymbol{S}, \quad \boldsymbol{S} \partial \varphi=C=\text { const }, \quad \partial \varphi=0, \\ \boldsymbol{S} a \varphi \partial n=a \boldsymbol{S} \varphi \partial n, \quad \boldsymbol{S} u g \partial n=u \boldsymbol{S} g \partial n+\boldsymbol{S}(g-\boldsymbol{S} g \partial n) \delta u \end{array}$ | $\boldsymbol{S} \Sigma=\Sigma \boldsymbol{S}, \quad \boldsymbol{S} \partial \varphi=C=\text { const }, \quad \partial \varphi=0,$ $\boldsymbol{S} a \varphi \partial n=a \boldsymbol{S} \varphi \partial n, \quad \boldsymbol{S} u g \partial n=u \boldsymbol{S} g \partial n+\boldsymbol{S}(g-\boldsymbol{S} g \partial n) \delta u$ | $\boldsymbol{S} \Sigma=\Sigma \boldsymbol{S}, \quad \boldsymbol{S} \partial \varphi=C=\text { const }, \quad \partial \varphi=0,$ $\boldsymbol{S} a \varphi \partial n=a \boldsymbol{S} \varphi \partial n, \quad \boldsymbol{S} u g \partial n=u \boldsymbol{S} g \partial n+\boldsymbol{S}(g-\boldsymbol{S} g \partial n) \delta u$ |

**Conf.** MathPix confidence:
 

■ ≥0.9
■ 0.5–0.9
■ <0.5
— no value

**LaTeX source** MathPix’s reading, unchanged;
**Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

**Residual** not shown for this document — no residual measurements are paired with these rows.

**[refined: *basis*]** beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Page | Conf.  | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Scan image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| BH3FR_EQ0135                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 115  | ■0.378 | $\begin{aligned} & D_{(\mu\nu)m}^{(\kappa\lambda)} = \delta(\underline{k}, s); () \sum_{\varepsilon} \left[ \begin{matrix} s \\ k\ m \end{matrix} \right]_{(\alpha\beta)}^{(\gamma\delta)} (1 - \delta) - \\ & - \delta(\underline{k} \underline{l}, sm); () \sum_{\varepsilon} \left[ \begin{matrix} s \\ k\ l \end{matrix} \right]_{(\alpha\beta)}^{(\gamma\delta)} (1 - \delta) + \\ & + Q_{(\mu\nu)t}^{(\kappa\lambda)\underline{i}} \delta(\underline{i} \underline{k}, ts); () \sum_{\varepsilon} Q_{(\alpha\beta)t}^{(\gamma\delta)\underline{s}} \left[ \begin{matrix} t \\ k' m \end{matrix} \right]_{(\alpha\beta)}^{(\gamma\delta)} - \\ & - Q_{(\mu\nu)t}^{(\kappa\lambda)\underline{i}} \delta(\underline{i} \underline{k} \underline{l}, tsm); () \sum_{\varepsilon} Q_{(\alpha\beta)t}^{(\gamma\delta)\underline{s}} \left[ \begin{matrix} t \\ k' l \end{matrix} \right]_{(\alpha\beta)}^{(\gamma\delta)} + \end{aligned}$ | $D_{(\mu\nu)m}^{(\kappa\lambda)} = \delta(\underline{k}, s); () \sum_{\varepsilon} \left[ \begin{matrix} s \\ k\ m \end{matrix} \right]_{(\alpha\beta)}^{(\gamma\delta)} (1 - \delta) -$ $- \delta(\underline{k} \underline{l}, sm); () \sum_{\varepsilon} \left[ \begin{matrix} s \\ k\ l \end{matrix} \right]_{(\alpha\beta)}^{(\gamma\delta)} (1 - \delta) +$ $+ Q_{(\mu\nu)t}^{(\kappa\lambda)\underline{i}} \delta(\underline{i} \underline{k}, ts); () \sum_{\varepsilon} Q_{(\alpha\beta)t}^{(\gamma\delta)\underline{s}} \left[ \begin{matrix} t \\ k' m \end{matrix} \right]_{(\alpha\beta)}^{(\gamma\delta)} -$ $- Q_{(\mu\nu)t}^{(\kappa\lambda)\underline{i}} \delta(\underline{i} \underline{k} \underline{l}, tsm); () \sum_{\varepsilon} Q_{(\alpha\beta)t}^{(\gamma\delta)\underline{s}} \left[ \begin{matrix} t \\ k' l \end{matrix} \right]_{(\alpha\beta)}^{(\gamma\delta)} +$ | $D_{(\mu\nu)m}^{(\kappa\lambda)} = \delta(\underline{k}, s); () \sum_{\varepsilon} \left[ \begin{matrix} s \\ k\ m \end{matrix} \right]_{(\alpha\beta)}^{(\gamma\delta)} (1 - \delta) -$ $- \delta(\underline{k} \underline{l}, sm); () \sum_{\varepsilon} \left[ \begin{matrix} s \\ k\ l \end{matrix} \right]_{(\alpha\beta)}^{(\gamma\delta)} (1 - \delta) +$ $+ Q_{(\mu\nu)t}^{(\kappa\lambda)\underline{i}} \delta(\underline{i} \underline{k}, ts); () \sum_{\varepsilon} Q_{(\alpha\beta)t}^{(\gamma\delta)\underline{s}} \left[ \begin{matrix} t \\ k\ m \end{matrix} \right]_{(\alpha\beta)}^{(\gamma\delta)} -$ $- Q_{(\mu\nu)t}^{(\kappa\lambda)\underline{i}} \delta(\underline{i} \underline{k} \underline{l}, tsm); () \sum_{\varepsilon} Q_{(\alpha\beta)t}^{(\gamma\delta)\underline{s}} \left[ \begin{matrix} t \\ k\ l \end{matrix} \right]_{(\alpha\beta)}^{(\gamma\delta)} +$ |
| BH3FR_EQ0318 (M28a)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 147  | ■0.392 | $\lim_{\substack{2 \\ \gamma \rightarrow 2 \cdot E}} \widehat{\gamma}_2 \widehat{\gamma} \rightarrow 2 \cdot E(\kappa)_\pm^{(s_1)(s_2)} = \widehat{\text{DIV}}_{(x)}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | $\lim_{2\hat{\gamma} \rightarrow 2 \cdot E} (\kappa)_{(\pm)}^{(s_1)(s_2)} = \widehat{\text{DIV}}_{(x)}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | $\lim_{2\hat{\gamma} \rightarrow 2 E} (\kappa)_{(\pm)}^{(s_1)(s_2)} = \widehat{\text{DIV}}_{(x)}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| BH3FR_EQ0165 (75)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 122  | ■0.400 | $N_{+} \left( \left[ \begin{matrix} \alpha \\ \beta \end{matrix} \right], \left[ \begin{matrix} \beta \\ \alpha \end{matrix} \right] \right) (\alpha, \beta) N_{-} \left( \left[ \begin{matrix} \alpha \\ \beta \end{matrix} \right], \left\{ \begin{matrix} \beta \\ \alpha \end{matrix} \right\} \right), \quad \alpha \hat{=} (\mu\nu), \beta \hat{=} (\kappa\lambda)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | $N_{+} \left( \left[ \begin{matrix} \alpha \\ \beta \end{matrix} \right], \left[ \begin{matrix} \beta \\ \alpha \end{matrix} \right] \right) (\alpha, \beta) N_{-} \left( \left[ \begin{matrix} \alpha \\ \beta \end{matrix} \right], \left\{ \begin{matrix} \beta \\ \alpha \end{matrix} \right\} \right), \quad \alpha \hat{=} (\mu\nu), \beta \hat{=} (\kappa\lambda)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | $N_{+} \left( \left[ \begin{matrix} \alpha \\ \beta \end{matrix} \right], \left[ \begin{matrix} \beta \\ \alpha \end{matrix} \right] \right) (\alpha, \beta) N_{-} \left( \left[ \begin{matrix} \alpha \\ \beta \end{matrix} \right], \left[ \begin{matrix} \beta \\ \alpha \end{matrix} \right] \right), \quad \alpha \hat{=} (\mu\nu), \beta \hat{=} (\kappa\lambda)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| BH3FR_EQ0117 (49a)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 113  | ■0.414 | $\begin{aligned} & D_0 \leq D, \quad A > 0, \quad \dot{\tau} < 0, \quad D_{\max} = \Lambda < \infty, \\ & A = 0, \quad \dot{\tau} = 0, \quad A > D \geq D_e > 0, \quad A < 0, \\ & \dot{\tau} > 0 \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | $D_0 \leq D, \quad A > 0, \quad \dot{\tau} < 0, \quad D_{\max} = \Lambda < \infty,$ $A = 0, \quad \dot{\tau} = 0, \quad A > D \geq D_e > 0, \quad A < 0,$ $\dot{\tau} > 0$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | $D_0 \leq D, \quad A > 0, \quad \dot{\tau} < 0, \quad D_{\max} = A < \infty,$ $A = 0, \quad \dot{\tau} = 0, \quad A > D \geq D_e > 0, \quad A < 0,$ $\dot{\tau} > 0$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <div>Conf. MathPix confidence: <span>■</span> ≥0.9 <span>■</span> 0.5–0.9 <span>■</span> &lt;0.5 — no value</div> <div>LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ</div> <div>Residual not shown for this document — no residual measurements are paired with these rows.</div> <div>[refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</div> |      |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

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| BH3FR_EQ0162 (73)                                                                                                                                                                                                                                                                    | 121  | ■ 0.427 | $\begin{aligned} & \{ \}^2 \bar{\kappa}_{(\lambda) \pm} \sim f_{\lambda}^{\pm 1} = f^{\pm 1}(\mu_{\lambda}), \\ & \sim f_{\lambda}^2 + f_{\lambda}^2 = \beta_{\kappa \lambda} = \text{const}, \quad \beta_{\kappa \lambda} \neq 2, \quad \beta_{\lambda \lambda} = 2, \\ & \quad \quad \quad \mu = \mu_{\kappa} + \mu_{\lambda} \end{aligned}$ | ${}^2\bar{\kappa}_{(\lambda)\pm}\sim f_{\lambda}^{\pm 1}=f^{\pm 1}(\mu_{\lambda}),\quad f_{\kappa}f_{\lambda}\sim\left (e^{\lambda_1\mu}-E)^{\underline{a}}\delta_{kl}\right _q$ $\frac{f_{\kappa}^2+f_{\lambda}^2}{f_{\kappa}f_{\lambda}}=\beta_{\kappa\lambda}=\text{const},\quad\beta_{\kappa\lambda}\neq 2,\quad\beta_{\lambda\lambda}=2,$ $\mu=\mu_{\kappa}+\mu_{\lambda}$ | ${}^2\bar{\kappa}_{(\lambda)\pm}\sim f_{\lambda}^{\pm 1}=f^{\pm 1}(\mu_{\lambda}),\quad f_{\kappa}f_{\lambda}\sim\left (e^{\lambda_1\mu}-E)^{\underline{a}}\delta_{kl}\right _q,$ $\frac{f_{\kappa}^2+f_{\lambda}^2}{f_{\kappa}f_{\lambda}}=\beta_{\kappa\lambda}=\text{const},\quad\beta_{\kappa\lambda}\neq 2,\quad\beta_{\lambda\lambda}=2,$ $\mu=\mu_{\kappa}+\mu_{\lambda}$ |
| BH3FR_EQ0057 (19b)                                                                                                                                                                                                                                                                   | 105  | ■ 0.428 | $\operatorname{sp} K; \widehat{\square} = \widehat{\lambda \square}$                                                                                                                                                                                                                                                                           | $\operatorname{sp} K; \widehat{\square} = \widehat{\lambda \square}$                                                                                                                                                                                                                                                                                                            | $\operatorname{sp} K; \widehat{\square} = \widehat{\lambda \square}$                                                                                                                                                                                                                                                                                                             |
| BH3FR_EQ0025 (9a)                                                                                                                                                                                                                                                                    | 102  | ■ 0.436 | $\widehat{\setminus}=\sum_{\mu=1}^3\widehat{\setminus}_{\setminus}_{(\mu)}^{+},\quad\quad\quad\widehat{\setminus}_{\setminus}_{(\mu)}^{-}=\widehat{\setminus}_{\setminus}_{(\mu)}^{+}$                                                                                                                                                         | $(not\ rendered)$                                                                                                                                                                                                                                                                                                                                                               | $\widehat{\setminus}=\sum_{\mu=1}^3\widehat{\setminus}_{(\mu)}^{+},\quad\quad\quad\widehat{\setminus}_{(\mu)}^{-}=\widehat{\setminus}_{(\mu)}^{+}$                                                                                                                                                                                                                               |
| BH3FR_EQ0054 (18a)                                                                                                                                                                                                                                                                   | 105  | ■ 0.447 | $\begin{aligned} & R_{kmp}^i\rightarrow\varsigma_{kmp}^i;n,\quad\quad\quad{}^2\overline{\varsigma}=sp^4\overline{\varsigma},\quad\quad\quad\varsigma=sp^2\overline{\varsigma}, \\ & sp()_{(-)}^{(-,-)};\left({}^2\overline{\varsigma}-\frac{1}{2}{}^2\overline{\gamma};() \varsigma\right)=\overline{0} \end{aligned}$                         | $R_{kmp}^i\rightarrow\varsigma_{kmp}^i;n,\quad\quad\quad{}^2\overline{\varsigma}=sp^4\overline{\varsigma},\quad\quad\quad\varsigma=sp^2\overline{\varsigma}$ $sp()_{(-)}^{(-,-)};\left({}^2\overline{\varsigma}-\frac{1}{2}{}^2\overline{\gamma};() \varsigma\right)=\overline{0}$                                                                                              | $R_{kmp}^i\rightarrow\varsigma_{kmp}^i;n,\quad\quad\quad{}^2\overline{\varsigma}=sp^4\overline{\varsigma},\quad\quad\quad\varsigma=sp^2\overline{\varsigma},$ $sp()_{(-)}^{(-,-)};\left({}^2\overline{\varsigma}-\frac{1}{2}{}^2\overline{\gamma};() \varsigma\right)=\overline{0}$                                                                                              |
| BH3FR_EQ0325 (M31b)                                                                                                                                                                                                                                                                  | 148  | ■ 0.447 | $K_{\{p\}^{\{m\}}}=\left \begin{array}{l}\delta_{m,1}^p\\(),\delta_p^m()\end{array}\right ()_{(-)p}^{(1,6)}$                                                                                                                                                                                                                                   | $K_p^{(m)}=\left \begin{array}{l}\delta_m^p,1\\(),\delta_p^m()\end{array}\right ()_{(-)p}^{(1,6)}$                                                                                                                                                                                                                                                                              | $K_p^{(m)}=\left \begin{array}{l}\delta_m^p,1\\(),\delta_p^m()\end{array}\right ()_{(-)p}^{(1,6)}$                                                                                                                                                                                                                                                                               |
| BH3FR_EQ0035 (12a)                                                                                                                                                                                                                                                                   | 103  | ■ 0.448 | $\gamma m_0^3 Y_1 \varsigma = h^2,\quad\quad\quad 2 Y_1 m_o c \varsigma \not> h$                                                                                                                                                                                                                                                               | $\gamma m_0^3 Y_1 \varsigma = h^2,\quad\quad\quad 2 Y_1 m_o c \varsigma \not> h$                                                                                                                                                                                                                                                                                                | $\gamma m_0^3 Y_1 \varsigma = h^2,\quad\quad\quad 2 Y_1 m_o c \varsigma \gg h$                                                                                                                                                                                                                                                                                                   |
| BH3FR_EQ0085 (32)                                                                                                                                                                                                                                                                    | 109  | ■ 0.466 | $\begin{aligned} & m_L c s_0 = 4 \sqrt[4]{\pi} \sqrt[3]{3 \pi \gamma \hbar s_0} \sqrt{\frac{c \hbar}{3 \mathfrak{A}}}, \quad m_e \eta \sqrt[3]{\eta} = m_L, \\ & \quad \quad \quad m_B \eta^2 \sqrt[3]{\eta} = m_L, \quad m_L < m_e < m_B, \quad s_0 = 1[m] \end{aligned}$                                                                     | $m_L c s_0 = 4 \sqrt[4]{\pi} \sqrt[3]{3 \pi \gamma \hbar s_0} \sqrt{\frac{c \hbar}{3 \mathfrak{A}}}, \quad m_e \eta \sqrt[3]{\eta} = m_L,$ $m_B \eta^2 \sqrt[3]{\eta} = m_L, \quad m_L < m_e < m_B, \quad s_0 = 1[m]$                                                                                                                                                           | $m_L c s_0 = 4 \sqrt[4]{\pi} \sqrt[3]{3 \pi \gamma \hbar s_0} \sqrt{\frac{c \hbar}{3 \mathfrak{A}}}, \quad m_e \eta \sqrt[3]{\eta} = m_L,$ $m_B \eta^2 \sqrt[3]{\eta} = m_L, \quad m_L < m_e < m_B, \quad s_0 = 1[m]$                                                                                                                                                            |
| <b>Conf.</b> MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value                                                                                                                                                                                                                  |      |         |                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>LaTeX source</b> MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ                    |      |         |                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Residual</b> not shown for this document — no residual measurements are paired with these rows.                                                                                                                                                                                   |      |         |                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>[refined: basis]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here. |      |         |                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                  |

| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Scan image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| BH3FR_EQ0224 (101b)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 131  | ■ 0.478 | \begin{aligned} &\& (1) \hat{=}{(\eta)}, \quad (2) \hat{=}{\left(e_0, e^{-}\right)}, \quad (3) \hat{=}{(\mu^{-})}, \\ &\& (4) \hat{=}{(K^{+}, K^0)}, \quad (5) \hat{=}{(\pi^{+}, \pi^0, \pi^{-})}, \quad (6) \hat{=}{(\Lambda)}, \\ &\& (7) \hat{=}{(\Omega^{-})}, \quad (8) \hat{=}{(p, n)}, \quad (9) \hat{=}{(\Xi^0, \Xi^{-})}, \\ &\& (10) \hat{=}{(\Sigma^{+}, \Sigma^0, \Sigma^{-})}, \quad (11) \hat{=}{(o^{+}, o^0, o^{-}, o^{--})}, \\ &\& (12) \hat{=}{(\Delta^{++}, \Delta^{+}, \Delta^0, \Delta^{-})}, \quad k = B + 1 \end{aligned} | $ \begin{aligned} (1) &\hat{=}{(\eta)}, \quad (2) \hat{=}{(e_0, e^{-})}, \quad (3) \hat{=}{(\mu^{-})}, \\ (4) &\hat{=}{(K^{+}, K^0)}, \quad (5) \hat{=}{(\pi^{+}, \pi^0, \pi^{-})}, \quad (6) \hat{=}{(\Lambda)}, \\ (7) &\hat{=}{(\Omega^{-})}, \quad (8) \hat{=}{(p, n)}, \quad (9) \hat{=}{(\Xi^0, \Xi^{-})}, \\ (10) &\hat{=}{(\Sigma^{+}, \Sigma^0, \Sigma^{-})}, \quad (11) \hat{=}{(o^{+}, o^0, o^{-}, o^{--})}, \\ (12) &\hat{=}{(\Delta^{++}, \Delta^{+}, \Delta^0, \Delta^{-})}, \quad k = B + 1 \end{aligned} $ | $ \begin{aligned} (1) &\hat{=}{(\eta)}, \quad (2) \hat{=}{(e_0, e^{-})}, \quad (3) \hat{=}{(\mu^{-})}, \\ (4) &\hat{=}{(K^{+}, K^0)}, \quad (5) \hat{=}{(\pi^{+}, \pi^0, \pi^{-})}, \quad (6) \hat{=}{(\Lambda)}, \\ (7) &\hat{=}{(\Omega^{-})}, \quad (8) \hat{=}{(p, n)}, \quad (9) \hat{=}{(\Xi^0, \Xi^{-})}, \\ (10) &\hat{=}{(\Sigma^{+}, \Sigma^0, \Sigma^{-})}, \quad (11) \hat{=}{(o^{+}, o^0, o^{-}, o^{--})}, \\ (12) &\hat{=}{(\Delta^{++}, \Delta^{+}, \Delta^0, \Delta^{-})}, \quad k = B + 1 \end{aligned} $ |
| BH3FR_EQ0218 (99)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 130  | ■ 0.487 | \begin{aligned} &\& k_{\max }=2, \quad \theta \leq P \leq k+1, \\ &\& Q(P)=k-1, \quad Q\left(P_{\pm}\right)=2 k-1, \quad P_{+}=2 k-1, \quad P_{-}=2-k \end{aligned}                                                                                                                                                                                                                                                                                                                                                                              | $ \begin{aligned} k_{\max } &= 2, \quad 0 \leq P \leq k + 1, \quad Q(P) = k - 1, \\ Q\left(P_{\pm}\right) &= 2 k - 1, \quad P_{+} = 2 k - 1, \quad P_{-} = 2 - k \end{aligned} $                                                                                                                                                                                                                                                                                                                                           | $ \begin{aligned} k_{\max } &= 2, \quad 0 \leq P \leq k + 1, \quad Q(P) = k - 1, \\ Q\left(P_{\pm}\right) &= 2 k - 1, \quad P_{+} = 2 k - 1, \quad P_{-} = 2 - k \end{aligned} $                                                                                                                                                                                                                                                                                                                                           |
| <p><b>Conf.</b> MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ &lt;0.5 — no value</p> <p><b>LaTeX source</b> MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ</p> <p><b>Residual</b> not shown for this document — no residual measurements are paired with these rows.</p> <p><b>[refined: <i>basis</i>]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</p> |      |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Scan image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| BH3FR_EQ0175 (80)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 124  | ■ 0.494 | $\begin{aligned} & \underline{\lambda} = \underline{\alpha} \lambda, \quad \underline{P}_{(\mu\nu)}^{(\kappa\lambda)} = \underline{P}, \quad A(1) = [E - \exp\{-\lambda(iT + i\delta_{xb}Z + \delta_{xc}\mu)\}]^{-\underline{\alpha}} (E - \delta_{xb}e^{-i\lambda Z} - \delta_{xc}e^{-\lambda\mu})^{\underline{\alpha}}, \\ & B(2) = [E - \exp\{-i\lambda(Z + T)\}]^{-\underline{\alpha}} (E - e^{i\lambda Z})^{\underline{\alpha}}, \\ & C(1,2) = [E - \exp\{-\lambda(iT + iZ + \delta_{xd}\mu)\}]^{-\underline{\alpha}} (E - \delta_{xd}e^{-\lambda\mu})^{\underline{\alpha}}, \\ & D(1,3) = [E - \exp\{-\lambda(\mu + iT + i\delta_{xd}Z)\}]^{-\underline{\alpha}} (E - \delta_{xd}e^{-i\lambda Z})^{\underline{\alpha}}, \\ & E(2,3) = [E - \exp\{-\lambda(\mu + iZ + iT)\}]^{-\underline{\alpha}} (E - e^{-i\lambda T})^{\underline{\alpha}}, \\ & F(1,2,3) = [E - \exp\{-\lambda(\mu + iT + iZ)\}]^{-\underline{\alpha}}, \\ & \mu = \sqrt{\tau} \sum_{k=1}^3 ( )_k, \quad Z = \sqrt{\tau} ( )_4, \\ & T = \sqrt{\tau} (( )_5 + ( )_6), \quad x \triangleq (a, b, c, d) \end{aligned}$ | $\underline{\lambda} = \underline{\alpha} \lambda, \quad \underline{P}_{(\mu\nu)}^{(\kappa\lambda)} = \underline{P}, \quad A(1) = [E - \exp\{-\lambda(iT + i\delta_{xb}Z + \delta_{xc}\mu)\}]^{-\underline{\alpha}} (E - \delta_{xb}e^{-i\lambda Z} - \delta_{xc}e^{-\lambda\mu})^{\underline{\alpha}},$ $B(2) = [E - \exp\{-i\lambda(Z + T)\}]^{-\underline{\alpha}} (E - e^{i\lambda Z})^{\underline{\alpha}},$ $C(1,2) = [E - \exp\{-\lambda(iT + iZ + \delta_{xd}\mu)\}]^{-\underline{\alpha}} (E - \delta_{xd}e^{-\lambda\mu})^{\underline{\alpha}},$ $D(1,3) = [E - \exp\{-\lambda(\mu + iT + i\delta_{xd}Z)\}]^{-\underline{\alpha}} (E - \delta_{xd}e^{-i\lambda Z})^{\underline{\alpha}},$ $E(2,3) = [E - \exp\{-\lambda(\mu + iZ + iT)\}]^{-\underline{\alpha}} (E - e^{-i\lambda T})^{\underline{\alpha}},$ $F(1,2,3) = [E - \exp\{-\lambda(\mu + iT + iZ)\}]^{-\underline{\alpha}},$ $\mu = \sqrt{\tau} \sum_{k=1}^3 ( )_k, \quad Z = \sqrt{\tau} ( )_4,$ $T = \sqrt{\tau} (( )_5 + ( )_6), \quad x \triangleq (a, b, c, d)$ | $\underline{\lambda} = \underline{\alpha} \lambda, \quad \underline{P}_{(\mu\nu)}^{(\kappa\lambda)} = \underline{P}, \quad A(1) = [E - \exp\{-\lambda(iT + i\delta_{xb}Z + \delta_{xc}\mu)\}]^{-\underline{\alpha}} (E - \delta_{xb}e^{-i\lambda Z} - \delta_{xc}e^{-\lambda\mu})^{\underline{\alpha}},$ $B(2) = [E - \exp\{-i\lambda(Z + T)\}]^{-\underline{\alpha}} (E - e^{i\lambda Z})^{\underline{\alpha}},$ $C(1,2) = [E - \exp\{-\lambda(iT + iZ + \delta_{xd}\mu)\}]^{-\underline{\alpha}} (E - \delta_{xd}e^{-\lambda\mu})^{\underline{\alpha}},$ $D(1,3) = [E - \exp\{-\lambda(\mu + iT + i\delta_{xd}Z)\}]^{-\underline{\alpha}} (E - \delta_{xd}e^{-i\lambda Z})^{\underline{\alpha}},$ $E(2,3) = [E - \exp\{-\lambda(\mu + iZ + iT)\}]^{-\underline{\alpha}} (E - e^{-i\lambda T})^{\underline{\alpha}},$ $F(1,2,3) = [E - \exp\{-\lambda(\mu + iT + iZ)\}]^{-\underline{\alpha}},$ $\mu = \sqrt{\tau} \sum_{k=1}^3 ( )_k, \quad Z = \sqrt{\tau} ( )_4,$ $T = \sqrt{\tau} (( )_5 + ( )_6), \quad x \triangleq (a, b, c, d)$ |
| BH3FR_EQ0169 (77)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 123  | ■ 0.510 | $\begin{aligned} & \bar{\lambda} \perp \bar{Y}(q), \quad \bar{\lambda}_{(\mu\nu)}^{(\kappa\lambda)} \perp \bar{Y}(z), \quad \bar{Y}(q) = \sum_{i=1}^q \dot{\bar{x}}_i, \\ & z = z(\kappa\lambda\mu\nu) \leq q \leq 6, \quad \bar{Y}(6) = \bar{v} + i\bar{w}, \\ & w^2 = c^2 + \varepsilon^2 + \dot{\eta}^2, \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $\bar{\lambda} \perp \bar{Y}(q), \quad \bar{\lambda}_{(\mu\nu)}^{(\kappa\lambda)} \perp \bar{Y}(z), \quad \bar{Y}(q) = \sum_{i=1}^q \dot{\bar{x}}_i,$ $z = z(\kappa\lambda\mu\nu) \leq q \leq 6, \quad \bar{Y}(6) = \bar{v} + i\bar{w},$ $w^2 = c^2 + \varepsilon^2 + \dot{\eta}^2,$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $\bar{\lambda} \perp \bar{Y}(q), \quad \bar{\lambda}_{(\mu\nu)}^{(\kappa\lambda)} \perp \bar{Y}(z), \quad \bar{Y}(q) = \sum_{i=1}^q \dot{\bar{x}}_i,$ $z = z(\kappa\lambda\mu\nu) \leq q \leq 6, \quad \bar{Y}(6) = \bar{v} + i\bar{w},$ $w^2 = c^2 + \varepsilon^2 + \dot{\eta}^2$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <p><b>Conf.</b> MathPix confidence: ■ <math>\geq 0.9</math> ■ 0.5–0.9 ■ <math>&lt; 0.5</math> — no value</p> <p><b>LaTeX source</b> MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ</p> <p><b>Residual</b> not shown for this document — no residual measurements are paired with these rows.</p> <p><b>[refined: <i>basis</i>]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</p> |      |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |



| Identifier          | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                             | Rendered                                                                                                                                                                                                                                                                                                                                                        | Scan image                                                                                                                                                                                                                                                                                                                                                         |
|---------------------|------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BH3FR_EQ0112 (46b)  | 112  | ■ 0.544 | $\begin{array}{l} s \leq \varrho, \quad z=0, \quad \varrho < s < y, \quad z > 0 \\ s \rightarrow y, \quad z \rightarrow +\infty \end{array}$                                                                                                                                                                                                                                                             | $s \leq \varrho, \quad z=0, \quad \varrho < s < y, \quad z > 0 \\ s \rightarrow y, \quad z \rightarrow +\infty$                                                                                                                                                                                                                                                 | $s \leq \varrho, \quad z=0, \quad \varrho < s < y, \quad z > 0, \\ s \rightarrow y, \quad z \rightarrow +\infty$                                                                                                                                                                                                                                                   |
| BH3FR_EQ0088 (34a)  | 110  | ■ 0.578 | $\begin{array}{l} p \, \underline{\eta} = \pi \hbar, \quad p = \frac{h}{2} \left( \frac{\tau}{3} \cdot \underline{\varepsilon} \right)^{-1/3}, \\ \frac{w}{2 \, p} \geq \underline{\eta}, \quad w = N h \end{array}$                                                                                                                                                                                     | $p \underline{\eta} = \pi \hbar, \quad p = \frac{h}{2} \left( \frac{\tau}{3} \cdot \underline{\varepsilon} \right)^{-1/3}, \\ \frac{w}{2p} \geq \underline{\eta}, \quad w = N h$                                                                                                                                                                                | $p \underline{\eta} = \pi \hbar, \quad p = \frac{h}{2} \left( \frac{\tau}{3} \cdot \underline{\varepsilon} \right)^{-1/3}, \\ \frac{w}{2 \, p} \geq \underline{\eta}, \quad w = N h$                                                                                                                                                                               |
| BH3FR_EQ0096 (39)   | 111  | ■ 0.591 | $\begin{aligned} & h \sim D^{-8/11}, \quad \gamma \sim D^{-13/11}, \quad \varepsilon_0 \sim D^{13/11}, \\ & \mu_0 \sim D^{-3/11} \end{aligned}$                                                                                                                                                                                                                                                          | $h \sim D^{-8/11}, \quad \gamma \sim D^{-13/11}, \quad \varepsilon_0 \sim D^{13/11}, \\ \mu_0 \sim D^{-3/11}$                                                                                                                                                                                                                                                   | $h \sim D^{-8/11}, \quad \gamma \sim D^{-13/11}, \quad \varepsilon_0 \sim D^{13/11}, \\ \mu_0 \sim D^{-3/11}$                                                                                                                                                                                                                                                      |
| BH3FR_EQ0210 (97)   | 128  | ■ 0.594 | $\begin{aligned} & \mu_{\pm} = 4 \mu \alpha_{\pm}, \quad \mu_S = \left( 1 - \frac{\alpha_-}{\alpha_+} \right) \mu_+, \\ & c \mu s_0 = \sqrt[4]{\pi} \sqrt[3]{3 \pi s_0 \gamma \hbar} \cdot \sqrt{\frac{c \hbar}{3 \gamma}}, \\ & \eta^2 (1 + \alpha_+) = \eta (1 + \alpha_-) = \frac{1}{\sqrt[3]{\eta}} \left( 1 - \frac{2}{3} \xi \eta^2 (1 - \sqrt{\eta}) Y_2 \right), \\ & s_0 = 1 [m] \end{aligned}$ | $\mu_{\pm} = 4 \mu \alpha_{\pm}, \quad \mu_S = \left( 1 - \frac{\alpha_-}{\alpha_+} \right) \mu_+ \\ c \mu s_0 = \sqrt[4]{\pi} \sqrt[3]{3 \pi s_0 \gamma \hbar} \cdot \sqrt{\frac{c \hbar}{3 \gamma}} \\ \eta^2 (1 + \alpha_+) = \eta (1 + \alpha_-) = \frac{1}{\sqrt[3]{\eta}} \left( 1 - \frac{2}{3} \xi \eta^2 (1 - \sqrt{\eta}) Y_2 \right) \\ s_0 = 1 [m]$ | $\mu_{\pm} = 4 \mu \alpha_{\pm}, \quad \mu_S = \left( 1 - \frac{\alpha_-}{\alpha_+} \right) \mu_+, \\ c \mu s_0 = \sqrt[4]{\pi} \sqrt[3]{3 \pi s_0 \gamma \hbar} \cdot \sqrt{\frac{c \hbar}{3 \gamma}}, \\ \eta^2 (1 + \alpha_+) = \eta (1 + \alpha_-) = \frac{1}{\sqrt[3]{\eta}} \left( 1 - \frac{2}{3} \xi \eta^2 (1 - \sqrt{\eta}) Y_2 \right), \\ s_0 = 1 [m]$ |
| BH3FR_EQ0069 (23)   | 107  | ■ 0.600 | $\begin{aligned} & H_{(1)} = \pi \underline{\varepsilon} n_1, \quad H_{(2)} = \pi \underline{\varepsilon} n_2, \\ & -\varsigma^2 = x_5^2 + x_6^2 = -\varepsilon^2 - \eta^2, \quad \varsigma \geq \underline{\varepsilon} = \text{const} > 0 \end{aligned}$                                                                                                                                               | $H_{(1)} = \pi \underline{\varepsilon} n_1, \quad H_{(2)} = \pi \underline{\varepsilon} n_2, \\ -\varsigma^2 = x_5^2 + x_6^2 = -\varepsilon^2 - \eta^2, \quad \leq \underline{\varepsilon} = \text{const} > 0$                                                                                                                                                  | $H_{(1)} = \pi \underline{\varepsilon} n_1, \quad H_{(2)} = \pi \underline{\varepsilon} n_2, \\ -\varsigma^2 = x_5^2 + x_6^2 = -\varepsilon^2 - \eta^2, \quad \varsigma \geq \underline{\varepsilon} = \text{const} > 0$                                                                                                                                           |
| BH3FR_EQ0145 (65)   | 117  | ■ 0.607 | $\underline{\beta}_{+}^{(\pm)} = \pm \frac{\pi}{2} (2(\underline{\phantom{a}})_{+} + E), \quad \underline{\beta}_{-}^{(\pm)} = \pm \pi(\underline{\phantom{a}})$                                                                                                                                                                                                                                         | $\underline{\beta}_{+}^{(\pm)} = \pm \frac{\pi}{2} (2(\underline{\phantom{a}})_{+} + E), \quad \underline{\beta}_{-}^{(\pm)} = \pm \pi(\underline{\phantom{a}})$                                                                                                                                                                                                | $\underline{\beta}_{+}^{(\pm)} = \pm \frac{\pi}{2} (2(\underline{\phantom{a}})_{+} + E), \quad \underline{\beta}_{-}^{(\pm)} = \pm \pi(\underline{\phantom{a}})$                                                                                                                                                                                                   |
| BH3FR_EQ0285 (M11b) | 141  | ■ 0.608 | $\bar{Z}(i) = \bar{e}_{\{i\}}(\bar{e}_{\{i\}})_{\{i\}}, \quad \left  \bar{e}_{\{i\}} \right  = 1, \quad (\bar{e}_i \bar{e}_k)_L = \hat{A}(n_i)_1^L$                                                                                                                                                                                                                                                      | $\bar{Z}(i) = \bar{e}_i(\bar{e}_i)_{\{i\}}, \quad \left  \bar{e}_i \right  = 1, \quad (\bar{e}_i \bar{e}_k)_L = \hat{A}(n_i)_1^L$                                                                                                                                                                                                                               | $\bar{Z}(i) = \bar{e}_i(\bar{e}_i)_{\{i\}}, \quad \left  \bar{e}_i \right  = 1, \quad (\bar{e}_i \bar{e}_k)_L = \hat{A}(n_i)_1^L$                                                                                                                                                                                                                                  |
| BH3FR_EQ0002        | 025  | ■ 0.616 | $g_{(\mu \nu)}^{ik} = \gamma_{(\mu \nu)}^{ik}; n \quad \gamma_{(\mu \nu)}^{ik} = \sum_{m=1}^6 \kappa_{(\mu) m}^i \kappa_{(\nu) m}^k$                                                                                                                                                                                                                                                                     | $g_{(\mu \nu)}^{ik} = \gamma_{(\mu \nu)}^{ik}; n \quad \gamma_{(\mu \nu)}^{ik} = \sum_{m=1}^6 \kappa_{(\mu) m}^i \kappa_{(\nu) m}^k$                                                                                                                                                                                                                            | $g_{(\mu \nu)}^{ik} = \gamma_{(\mu \nu)}^{ik}; n \quad \gamma_{(\mu \nu)}^{ik} = \sum_{m=1}^6 \kappa_{(\mu) m}^i \kappa_{(\nu) m}^k$                                                                                                                                                                                                                               |

**Conf.** MathPix confidence:
 

■ ≥0.9
■ 0.5-0.9
■ <0.5
— no value

**LaTeX source** MathPix’s reading, unchanged;
**Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

**Residual** not shown for this document — no residual measurements are paired with these rows.

**[refined: *basis*]** beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.



| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Scan image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| BH3FR_E00305                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 144  | ■ 0.676 | $\begin{array}{ll} \bar{\psi}_{(\nu)}=\boldsymbol{S}^2\bar{\kappa}_{(\nu)};() \partial \bar{x}, & {}^2\bar{\gamma}_{(\nu v)}=\mathrm{sp}\left({}^2\bar{\kappa}_{(\nu)}\times {}^2\bar{\kappa}_{(\nu)}\right), \\ {}^2\bar{\gamma}_{(\nu v)};n={}^2\bar{g}_{(\nu)}\left(x^l\right)_1^N, & {}^2\bar{\gamma}={}^2\bar{\gamma}\left({}^2\bar{\kappa}_{(\nu)}\right)_1^\omega, \quad N=p\omega \end{array}$                                                                                                                                                                                                                                                                                                                                                                                   | $\bar{\psi}_{(\nu)}=\boldsymbol{S}^2\bar{\kappa}_{(\nu)};() \partial \bar{x}, \quad {}^2\bar{\gamma}_{(\nu v)}=\mathrm{sp}\left({}^2\bar{\kappa}_{(\nu)}\times {}^2\bar{\kappa}_{(\nu)}\right), \\ {}^2\bar{\gamma}_{(\nu v)};n={}^2\bar{g}_{(\nu)}\left(x^l\right)_1^N, \quad {}^2\bar{\gamma}={}^2\bar{\gamma}\left({}^2\bar{\kappa}_{(\nu)}\right)_1^\omega, \quad N=p\omega$                                                                                                                                                                                                        | $\bar{\psi}_{(\nu)}=\boldsymbol{S}^2\bar{\kappa}_{(\nu)};() \partial \bar{x}, \quad {}^2\bar{\gamma}_{(\nu v)}=\mathrm{sp}\left({}^2\bar{\kappa}_{(\nu)}\times {}^2\bar{\kappa}_{(\nu)}\right), \\ {}^2\bar{\gamma}_{(\nu v)};n={}^2\bar{g}_{(\nu)}\left(x^l\right)_1^N, \quad {}^2\bar{\gamma}={}^2\bar{\gamma}\left({}^2\bar{\kappa}_{(\nu)}\right)_1^\omega, \quad N=p\omega$                                                                                                                                                                                                        |
| BH3FR_E00166 (76)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 122  | ■ 0.684 | $\begin{aligned} & \& q_{\{1\}}=Q_{\{1\}}, \quad \& q_{\{2,3\}}=Q_{\{2\}}^{\pm} \pm \\ & \& q_{\{4\}}=Q_{\{3\}}, \quad \& q_{\{5,6\}}=Q_{\{4\}}^{\pm} \pm \\ & F_{\{v\}}=\mathrm{binom}\{\alpha\beta\}\{\kappa\lambda\} F\left(q_{\{j\}}\right. \\ & \left.\right)_{\{v\}} \hat{=} N_{\{+\}}(\alpha\beta, \beta, \kappa\lambda) N_{-}, \quad 1 \leq \nu \leq 6, \\ & AF_pBF_rC=AF_vC, \quad p+r=v, \quad A \neq C, \\ & \left(\left(\right)-\frac{1}{2} {}^2\bar{\gamma} sp(\right)\right); sp^4\bar{\varrho}=\sum_{k=1}^{L>1}\left(\left(\right)-\frac{1}{2} {}^2\bar{\gamma}_k sp(\right)\right); sp^4\bar{\varrho}_k, \\ & A=C, \quad \left(\left(\right)-\frac{1}{2} {}^2\bar{\gamma} sp(\right)\right); sp^4\bar{\varrho}=\mathrm{const}\left(\left(\right)_4\right) \end{aligned}$ | $q_1=Q_1, \quad q_{2,3}=Q_2^\pm, \quad q_4=Q_3, \quad q_{5,6}=Q_4^\pm,$ $F_v=\binom{\alpha\beta}{\kappa\lambda}F(q_j)_v\hat{=}N_+(\alpha\beta,\kappa\lambda)N_-, \quad 1\leq\nu\leq 6,$ $AF_pBF_rC=AF_vC, \quad p+r=v, \quad A\neq C,$ $\left(\left(\right)-\frac{1}{2} {}^2\bar{\gamma} sp(\right)\right); sp^4\bar{\varrho}=\sum_{k=1}^{L>1}\left(\left(\right)-\frac{1}{2} {}^2\bar{\gamma}_k sp(\right)\right); sp^4\bar{\varrho}_k,$ $A=C, \quad \left(\left(\right)-\frac{1}{2} {}^2\bar{\gamma} sp(\right)\right); sp^4\bar{\varrho}=\mathrm{const}\left(\left(\right)_4\right)$ | $q_1=Q_1, \quad q_{2,3}=Q_2^\pm, \quad q_4=Q_3, \quad q_{5,6}=Q_4^\pm,$ $F_v=\binom{\alpha\beta}{\kappa\lambda}F(q_j)_v\hat{=}N_+(\alpha\beta,\kappa\lambda)N_-, \quad 1\leq\nu\leq 6,$ $AF_pBF_rC=AF_vC, \quad p+r=v, \quad A\neq C,$ $\left(\left(\right)-\frac{1}{2} {}^2\bar{\gamma} sp(\right)\right); sp^4\bar{\varrho}=\sum_{k=1}^{L>1}\left(\left(\right)-\frac{1}{2} {}^2\bar{\gamma}_k sp(\right)\right); sp^4\bar{\varrho}_k,$ $A=C, \quad \left(\left(\right)-\frac{1}{2} {}^2\bar{\gamma} sp(\right)\right); sp^4\bar{\varrho}=\mathrm{const}\left(\left(\right)_4\right)$ |
| BH3FR_E00306                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 144  | ■ 0.691 | $\{ \}^{\{2\}} \bar{\gamma}_{(\mu \nu)}=\operatorname{sp}\left(\{ \}^{\{2\}} \bar{\kappa}_{(\mu)}\right) \times \left(\{ \}^{\{2\}} \bar{\kappa}_{(\nu)}\right), \quad \hat{\gamma}=\left({}^2\bar{\gamma}_{(\mu \nu)}\right)_\omega$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | ${}^2\bar{\gamma}_{(\mu \nu)}=\mathrm{sp}\left({}^2\bar{\kappa}_{(\mu)}\times {}^2\bar{\kappa}_{(\nu)}\right), \quad \hat{\gamma}=\left({}^2\bar{\gamma}_{(\mu \nu)}\right)_\omega$                                                                                                                                                                                                                                                                                                                                                                                                     | ${}^2\bar{\gamma}_{(\mu \nu)}=\mathrm{sp}\left({}^2\bar{\kappa}_{(\mu)}\times {}^2\bar{\kappa}_{(\nu)}\right), \quad \hat{\gamma}=\left({}^2\bar{\gamma}_{(\mu \nu)}\right)_\omega$                                                                                                                                                                                                                                                                                                                                                                                                     |
| <div>Conf. MathPix confidence: <span>■</span> <math>\geq 0.9</math> <span>■</span> 0.5–0.9 <span>■</span> <math>&lt; 0.5</math> — no value</div> <div>LaTeX source MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ</div> <div>Residual not shown for this document — no residual measurements are paired with these rows.</div> <div>[refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</div> |      |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Scan image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| BH3FR_EQ0159 (71)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 121  | ■ 0.693 | $\begin{aligned} & \backslash\begin{aligned} & K_{11}^{-}\binom{2}{\gamma,\gamma^*} = 6d, 7s, 9s, 2e, 3e \\ & C_{11}^{-} = 1d, 3d, 2s, 5s \\ & K_{12}^{-}\binom{3}{\gamma,\gamma^*} = 6d, 7s, 8s, 2e, 3e \\ & C_{12}^{-} = 2d, 3d, 3s, 6s \\ & K_{13}^{-}\binom{\alpha,\alpha^*}{\beta,\beta^*} = 5d, 4s, 5s, 6s, 3e \\ & C_{13}^{-} = 4d, 1s, 2s, 3s, 2e \\ & K_{14}^{-}\binom{\alpha,\alpha^*}{\gamma,\gamma^*} = 6d, 7s, 8s, 9s, 3e \\ & C_{14}^{-} = 4d, 1s, 2s, 3s, 3e \\ & K_{15}^{-}\binom{\beta,\beta^*}{\gamma,\gamma^*} = 6d, 7s, 8s, 9s, 2e \\ & C_{15}^{-} = 5d, 4s, 5s, 6s, 1e \\ & K_{\lambda}^{\pm} = C_{\lambda}^{\mp}, 1 \leq \lambda \leq 15 \end{aligned} \end{aligned}$ | $\begin{aligned} & K_{11}^{-}\binom{2}{\gamma,\gamma^*} = 6d, 7s, 9s, 2e, 3e \\ & C_{11}^{-} = 1d, 3d, 2s, 5s \\ & K_{12}^{-}\binom{3}{\gamma,\gamma^*} = 6d, 7s, 8s, 2e, 3e \\ & C_{12}^{-} = 2d, 3d, 3s, 6s \\ & K_{13}^{-}\binom{\alpha,\alpha^*}{\beta,\beta^*} = 5d, 4s, 5s, 6s, 3e \\ & C_{13}^{-} = 4d, 1s, 2s, 3s, 2e \\ & K_{14}^{-}\binom{\alpha,\alpha^*}{\gamma,\gamma^*} = 6d, 7s, 8s, 9s, 3e \\ & C_{14}^{-} = 4d, 1s, 2s, 3s, 3e \\ & K_{15}^{-}\binom{\beta,\beta^*}{\gamma,\gamma^*} = 6d, 7s, 8s, 9s, 2e \\ & C_{15}^{-} = 5d, 4s, 5s, 6s, 1e \\ & K_{\lambda}^{\pm} = C_{\lambda}^{\mp}, 1 \leq \lambda \leq 15 \end{aligned}$ | $\begin{aligned} & K_{11}^{-}\binom{2}{\gamma,\gamma^*} = 6d, 7s, 9s, 2e, 3e \\ & C_{11}^{-} = 1d, 3d, 2s, 5s \\ & K_{12}^{-}\binom{3}{\gamma,\gamma^*} = 6d, 7s, 8s, 2e, 3e \\ & C_{12}^{-} = 2d, 3d, 3s, 6s \\ & K_{13}^{-}\binom{\alpha,\alpha^*}{\beta,\beta^*} = 5d, 4s, 5s, 6s, 3e \\ & C_{13}^{-} = 4d, 1s, 2s, 3s, 2e \\ & K_{14}^{-}\binom{\alpha,\alpha^*}{\gamma,\gamma^*} = 6d, 7s, 8s, 9s, 3e \\ & C_{14}^{-} = 4d, 1s, 2s, 3s, 3e \\ & K_{15}^{-}\binom{\beta,\beta^*}{\gamma,\gamma^*} = 6d, 7s, 8s, 9s, 2e \\ & C_{15}^{-} = 5d, 4s, 5s, 6s, 1e \\ & K_{\lambda}^{\pm} = C_{\lambda}^{\mp}, 1 \leq \lambda \leq 15 \end{aligned}$ |
| BH3FR_EQ0183 (84)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 125  | ■ 0.726 | $\begin{array}{l} \backslash\begin{array}{l} \backslash\left(\mu x_{\mu}\right)-\hat{=}\{2\} \\ \backslash\bar{\left\{\kappa\right\}}_{\left(\mu\right)}, \& \left(\pm p\right)-\backslash\left(\mu x_{\mu}\right)-\left(\mp q\right), \\ \backslash\left(\pm p\right)-\left(\mp q\right) \backslash\left(\pm p\right)-\backslash\left(\mu x_{\mu}\right)-\left(\mp q\right) \\ \backslash\left(\pm p\right)-\backslash\left(3 x_3\right)-(+7) \backslash\left(Z((-k))=Z_k\right. \\ \left.0< x_{\mu} \leq Z\right. \end{array} \end{array}$                                                                                                                                                | $\begin{array}{l} \backslash\begin{array}{l} \backslash\left(\mu x_{\mu}\right)-\hat{=}\{2\} \\ \backslash\bar{\left\{\kappa\right\}}_{\left(\mu\right)}, \& \left(\pm p\right)-\backslash\left(\mu x_{\mu}\right)-\left(\mp q\right), \\ \backslash\left(\pm p\right)-\left(\mp q\right) \backslash\left(\pm p\right)-\backslash\left(\mu x_{\mu}\right)-\left(\mp q\right) \\ \backslash\left(\pm p\right)-\backslash\left(3 x_3\right)-(+7) \backslash\left(Z((-k))=Z_k\right. \\ \left.0< x_{\mu} \leq Z\right. \end{array} \end{array}$                                                                                                      | $\begin{array}{l} \backslash\begin{array}{l} \backslash\left(\mu x_{\mu}\right)-\hat{=}\{2\} \\ \backslash\bar{\left\{\kappa\right\}}_{\left(\mu\right)}, \& \left(\pm p\right)-\backslash\left(\mu x_{\mu}\right)-\left(\mp q\right), \\ \backslash\left(\pm p\right)-\left(\mp q\right) \backslash\left(\pm p\right)-\backslash\left(\mu x_{\mu}\right)-\left(\mp q\right) \\ \backslash\left(\pm p\right)-\backslash\left(3 x_3\right)-(+7) \backslash\left(Z((-k))=Z_k\right. \\ \left.0< x_{\mu} \leq Z\right. \end{array} \end{array}$                                                                                                      |
| <div>Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ &lt;0.5 — no value</div> <div>LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ</div> <div>Residual not shown for this document — no residual measurements are paired with these rows.</div> <div>[refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — basis names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</div> |      |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Page | Conf. | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Scan image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| BH3FR_EQ0302                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 144  | 0.742 | $\begin{aligned} & \varphi \left( x^{\underline{k}} \right)_1^N \rightarrow \phi; n, \quad \frac{\partial \varphi}{\partial x^{\underline{k}}} \rightarrow \left( \frac{\partial \left( x^{\underline{k}} \right)^{\phi}}{\partial C^{\underline{k}}} \right); n, \\ & d\varphi \rightarrow \left( \sum_{k=1}^N \partial_{(C^k)} \phi \right); n, \quad \partial C^{\underline{k}} = \sum_{i=1}^N \partial_i C^{\underline{k}}, \quad \partial_i n^{\underline{l}} = \delta_{il}, \\ & \int \varphi d\psi \rightarrow \boldsymbol{S} \phi; n \partial \psi \end{aligned}$ | $\varphi \left( x^{\underline{k}} \right)_1^N \rightarrow \phi; n, \quad \frac{\partial \varphi}{\partial x^{\underline{k}}} \rightarrow \left( \frac{\partial \left( x^{\underline{k}} \right)^{\phi}}{\partial C^{\underline{k}}} \right); n,$ $d\varphi \rightarrow \left( \sum_{k=1}^N \partial_{(C^k)} \phi \right); n, \quad \partial C^{\underline{k}} = \sum_{i=1}^N \partial_i C^{\underline{k}}, \quad \partial_i n^{\underline{l}} = \delta_{il},$ $\int \varphi d\psi \rightarrow \boldsymbol{S} \phi; n \partial \psi$ | $\varphi \left( x^{\underline{k}} \right)_1^N \rightarrow \phi; n, \quad \frac{\partial \varphi}{\partial x^{\underline{k}}} \rightarrow \left( \frac{\partial \left( x^{\underline{k}} \right)^{\phi}}{\partial C^{\underline{k}}} \right); n,$ $d\varphi \rightarrow \left( \sum_{k=1}^N \partial_{(C^k)} \phi \right); n, \quad \partial C^{\underline{k}} = \sum_{i=1}^N \partial_i C^{\underline{k}}, \quad \partial_i n^{\underline{l}} = \delta_{il},$ $\int \varphi d\psi \rightarrow \boldsymbol{S} \phi; n \partial \psi$                                    |
| BH3FR_EQ0290                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 142  | 0.743 | $\begin{array}{l} \text{I m } \mathrm{ROT}_{\underline{L}} = \{ \}^2 \overline{\{0\}}, \text{ \& s p } \mathrm{ROT}_{\underline{L}} = 0, \quad \backslash \backslash = \mathrm{DIV}_{\underline{L}} \\ \mathrm{GRAD}_{\underline{L}} - \mathrm{GRAD}_{\underline{L}} \mathrm{DIV}_{\underline{L}}, \text{ \& } \\ \mathrm{DIV}_{\underline{L}} \mathrm{DIV}_{\underline{L}} \mathrm{ROT}_{\underline{L}} = 0, \quad \backslash \backslash \\ \mathrm{ROT}_{\underline{L}} \mathrm{GRAD}_{\underline{L}} = \{ \}^2 \overline{\{0\}} \text{ \& } \\ \end{array}$            | $\begin{array}{l} \text{I m } \mathrm{ROT}_{\underline{L}} = \{ \}^2 \overline{\{0\}}, \quad \text{sp } \mathrm{ROT}_{\underline{L}} = 0, \\ = \mathrm{DIV}_{\underline{L}} \mathrm{GRAD}_{\underline{L}} - \mathrm{GRAD}_{\underline{L}} \mathrm{DIV}_{\underline{L}}, \quad \mathrm{DIV}_{\underline{L}} \mathrm{DIV}_{\underline{L}} \mathrm{ROT}_{\underline{L}} = 0, \\ \mathrm{ROT}_{\underline{L}} \mathrm{GRAD}_{\underline{L}} = \{ \}^2 \overline{\{0\}} \end{array}$                                                     | $\begin{array}{l} \text{Im } \mathrm{ROT}_{\underline{L}} = \{ \}^2 \overline{\{0\}}, \quad \text{sp } \mathrm{ROT}_{\underline{L}} = 0, \quad \overline{\mathrm{DIV}}_{\underline{L}} \mathrm{ROT}_{\underline{L}} = \\ = \mathrm{DIV}_{\underline{L}} \mathrm{GRAD}_{\underline{L}} - \mathrm{GRAD}_{\underline{L}} \mathrm{DIV}_{\underline{L}}, \quad \mathrm{DIV}_{\underline{L}} \overline{\mathrm{DIV}}_{\underline{L}} \mathrm{ROT}_{\underline{L}} = 0, \\ \mathrm{ROT}_{\underline{L}} \mathrm{GRAD}_{\underline{L}} = \{ \}^2 \overline{\{0\}} \end{array}$ |
| BH3FR_EQ0293                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 142  | 0.746 | $\varphi \left( n_i \right)_1^L = \varphi \left( x_i \right)_1^N, \quad L = \binom{N}{p}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | $\varphi \left( n_i \right)_1^L = \varphi \left( x_i \right)_1^N, \quad L = \binom{N}{p}$                                                                                                                                                                                                                                                                                                                                                                                                                                           | $\varphi \left( n_i \right)_1^L = \varphi \left( x_i \right)_1^N, \quad L = \binom{N}{p}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| BH3FR_EQ0118 (50)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 113  | 0.747 | $\vec{x}_{\{5\}^{\{g\}}} + \vec{x}_{\{6\}^{\{g\}}} = i \vec{M} = \mathrm{const} \left( x_4 \right) < \infty, \quad \dot{\varepsilon}_g^2 + \dot{\eta}_g^2 = 0$                                                                                                                                                                                                                                                                                                                                                                                                            | $\vec{x}_{\{5\}^{\{g\}}} + \vec{x}_{\{6\}^{\{g\}}} = i \vec{M} = \mathrm{const} \left( x_4 \right) < \infty, \quad \dot{\varepsilon}_g^2 + \dot{\eta}_g^2 = 0$                                                                                                                                                                                                                                                                                                                                                                      | $\vec{x}_{\{5\}^{\{g\}}} + \vec{x}_{\{6\}^{\{g\}}} = i \vec{M} = \mathrm{const} \left( x_4 \right) < \infty, \quad \dot{\varepsilon}_g^2 + \dot{\eta}_g^2 = 0$                                                                                                                                                                                                                                                                                                                                                                                                         |
| <p><b>Conf.</b> MathPix confidence: <span style="color: green;">■</span> <math>\geq 0.9</math> <span style="color: orange;">■</span> <math>0.5-0.9</math> <span style="color: red;">■</span> <math>&lt; 0.5</math> — no value</p> <p><b>LaTeX source</b> MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ</p> <p><b>Residual</b> not shown for this document — no residual measurements are paired with these rows.</p> <p><b>[refined: basis]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</p> |      |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |



| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Scan image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| BH3FR_EQ0060 (20a)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 106  | ■ 0.769 | $\begin{aligned} & \Lambda_{kl} = \alpha_l (a(kl) - 1)^{-1} - \sum_{m \neq l} \alpha_m \\ & (a(kl) - q) \cdot \lambda_{kl} = \lambda(kl), \\ & \left( \lambda(kl), a(kl), b_i^{(kl)} \right) = \sum_{m=1}^q \left( \lambda_m(k, l), a_m(kl), b_{mi}^{(kl)} \right), \\ & b_{mi}^{(kl)} = \left( \frac{a_{li}}{a_{lk}} - \frac{a_{mi}}{a_{mk}} \right) \frac{a_{km}}{a_{kl}}, \quad a_m^{(kl)} = \frac{a_{km}}{a_{kl}}, \\ & a_{ml} \lambda_m(m, l) = -\lambda_l(m, m), \quad \lambda_m(\tilde{k}, l) = \lambda_m(k, \tilde{l}) = \\ & = \lambda_m(\tilde{k}, \tilde{l}) = \lambda_{\tilde{m}}(k, l) = 0, \quad \lambda_m(k, m) = \lambda_m(m, m) = 0 \end{aligned}$ | $\Lambda_{kl} = \alpha_l (a(kl) - 1)^{-1} - \sum_{m \neq l} \alpha_m$ $(a(kl) - q) \cdot \lambda_{kl} = \lambda(kl),$ $\left( \lambda(kl), a(kl), b_i^{(kl)} \right) = \sum_{m=1}^q \left( \lambda_m(k, l), a_m(kl), b_{mi}^{(kl)} \right),$ $b_{mi}^{(kl)} = \left( \frac{a_{li}}{a_{lk}} - \frac{a_{mi}}{a_{mk}} \right) \frac{a_{km}}{a_{kl}}, \quad a_m^{(kl)} = \frac{a_{km}}{a_{kl}},$ $a_{ml} \lambda_m(m, l) = -\lambda_l(m, m), \quad \lambda_m(k, m) = \lambda_m(m, m) = 0$ $= \lambda_m(\tilde{k}, \tilde{l}) = \lambda_{\tilde{m}}(k, l) = 0,$ | $A_{kl} = \alpha_l (a(kl) - 1)^{-1} - \sum_{m \neq l} \alpha_m,$ $(a(kl) - q) \cdot \lambda_{kl} = \lambda(kl),$ $(\lambda(kl), a(kl), b_i^{(kl)}) = \sum_{m=1}^q (\lambda_m(k, l), a_m(kl), b_{mi}^{(kl)}),$ $b_{mi}^{(kl)} = \left( \frac{a_{li}}{a_{lk}} - \frac{a_{mi}}{a_{mk}} \right) \frac{a_{km}}{a_{kl}}, \quad a_m^{(kl)} = \frac{a_{km}}{a_{kl}},$ $a_{ml} \lambda_m(m, l) = -\lambda_l(m, m), \quad \lambda_m(\tilde{k}, l) = \lambda_m(k, \tilde{l}) =$ $= \lambda_m(\tilde{k}, \tilde{l}) = \lambda_{\tilde{m}}(k, l) = 0, \quad \lambda_m(k, m) = \lambda_m(m, m) = 0$ |
| BH3FR_EQ0246 (110b)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 135  | ■ 0.772 | $\begin{aligned} & k^P \left( 1 + P + Q + \kappa \eta^{2-q} \right) (1 + R(S_1 + S_2)) b(vx) = \\ & = A_{54} A_{55}^{k-1} \left[ 1 + P A_{56} (1 - \kappa A_{61} A_{62}^{1-k}) (1 + q A_{63} (1 + \right. \\ & \quad \left. + \kappa A_{64})) \right] \left( 1 - k^{-1} (A_{65} (q + k - 1))^{2-k} \binom{P}{2} \left( 1 - \binom{P}{3} \right) \right) \\ & S_1 = \binom{P}{2} (2 - k) (1 - q), \quad S_2 = P \left( 1 - \binom{P}{2} \right) (1 - \\ & \quad - \binom{P}{3}) (k - 1) (1 - \kappa) \end{aligned}$                                                                                                                                                  | $k^P (1 + P + Q + \kappa \eta^{2-q}) (1 + R(S_1 + S_2)) b(vx) =$ $= A_{54} A_{55}^{k-1} [1 + P A_{56} (1 - \kappa A_{61} A_{62}^{1-k}) (1 + q A_{63} (1 +$ $+ \kappa A_{64}))] \left( 1 - k^{-1} (A_{65} (q + k - 1))^{2-k} \binom{P}{2} \left( 1 - \binom{P}{3} \right) \right)$ $S_1 = \binom{P}{2} (2 - k) (1 - q), \quad S_2 = P \left( 1 - \binom{P}{2} \right) (1 -$ $- \binom{P}{3}) (k - 1) (1 - \kappa)$                                                                                                                                          | $k^P (1 + P + Q + \kappa \eta^{2-q}) (1 + R(S_1 + S_2)) b(vx) =$ $= A_{54} A_{55}^{k-1} \left[ 1 + P A_{56} (1 - \kappa A_{61} A_{62}^{1-k}) (1 + q A_{63} (1 + \right.$ $+ \kappa A_{64})) \right] \left( 1 - k^{-1} (A_{65} (q + k - 1))^{2-k} \binom{P}{2} \left( 1 - \binom{P}{3} \right) \right),$ $S_1 = \binom{P}{2} (2 - k) (1 - q), \quad S_2 = P \left( 1 - \binom{P}{2} \right) \left( 1 - \right.$ $\left. - \binom{P}{3} \right) (k - 1) (1 - \kappa)$                                                                                                                   |
| <p><b>Conf.</b> MathPix confidence: ■ <math>\geq 0.9</math> ■ 0.5–0.9 ■ <math>&lt; 0.5</math> — no value</p> <p><b>LaTeX source</b> MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ</p> <p><b>Residual</b> not shown for this document — no residual measurements are paired with these rows.</p> <p><b>[refined: <i>basis</i>]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</p> |      |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

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| BH3FR_E00171 (79)  | 123  | 0.775 | $\begin{aligned} & \left[ \begin{array}{c} i \\ k \ l \end{array} \right]_{(3)} + Q_{(3)m}^i \left[ \begin{array}{c} m \\ k \end{array} \right]_{(3)} = \alpha_{kl}^i e^{\lambda_{kl} \mu} \left[ \frac{1}{2} (E + \cos K) (E + \right. \\ & \left. + (e^{\lambda_{kl} \mu} - \cos K)^2 \sin^{-2} K) \right]^{-a/(2\lambda_{kl})}, \quad K = \lambda_{kl} T, \\ & T = \sum_{k>3}^q \sqrt{\tau}()_k, \quad \mu = \sum_{k=1}^3 \alpha_k()_k, \\ & a = \vec{c}_{(3)} \vec{C}^{-1} = a^* \end{aligned}$                                                                                         | $\left[ \begin{array}{c} i \\ k \ l \end{array} \right]_{(3)} + Q_{(3)m}^i \left[ \begin{array}{c} m \\ k \end{array} \right]_{(3)} = \alpha_{kl}^i e^{\lambda_{kl} \mu} \left[ \frac{1}{2} (E + \cos K) (E + \right. \\ \left. + (e^{\lambda_{kl} \mu} - \cos K)^2 \sin^{-2} K) \right]^{-a/(2\lambda_{kl})}, \quad K = \lambda_{kl} T, \\ T = \sum_{k>3}^q \sqrt{\tau}()_k, \quad \mu = \sum_{k=1}^3 \alpha_k()_k, \\ a = \vec{c}_{(3)} \vec{C}^{-1} = a^*$                                                                                           | $\left[ \begin{array}{c} i \\ k \ l \end{array} \right]_{(3)} + Q_{(3)m}^i \left[ \begin{array}{c} m \\ k \end{array} \right]_{(3)} = \alpha_{kl}^i e^{\lambda_{kl} \mu} \left[ \frac{1}{2} (E + \cos K) (E + \right. \\ \left. + (e^{\lambda_{kl} \mu} - \cos K)^2 \sin^{-2} K) \right]^{-a/(2\lambda_{kl})}, \quad K = \lambda_{kl} T, \\ T = \sum_{k>3}^q \sqrt{\tau}()_k, \quad \mu = \sum_{k=1}^3 \alpha_k()_k, \\ a = \vec{c}_{(3)} \vec{C}^{-1} = a^*$                                                                                           |
| BH3FR_E00150 (68)  | 118  | 0.776 | $\begin{aligned} & Q_{(\mu v)}^{(\kappa \lambda)} = A_{(\mu v)}^{(\kappa \lambda)} \sum_{l=1}^q C_{(\mu v)l}^{(\kappa \lambda)} (E - e^{-\lambda_l \mu})^{-\alpha_l} \cdot \left( \sum_{k=1}^q C_l^{(\mu v)} (E - \right. \\ & \left. - e^{-\lambda_l^{(\mu v)} \mu})^{-1} \right)^{-1} \cdot \left  \delta_{kl} \left( e^{\lambda_l^{(\kappa \lambda)} \mu} - E \right)^{\alpha_{kl}^{(\kappa \lambda)}} \right _q \cdot \left  \delta_{kl} \left( e^{\lambda_l^{(\mu v)} \mu} - \right. \right. \\ & \left. \left. - E \right)^{\alpha_{kl}^{(\mu v)}} \right _q^{-1} - qE \end{aligned}$ | $Q_{(\mu v)}^{(\kappa \lambda)} = A_{(\mu v)}^{(\kappa \lambda)} \sum_{l=1}^q C_{(\mu v)l}^{(\kappa \lambda)} (E - e^{-\lambda_l \mu})^{-\alpha_l} \cdot \left( \sum_{k=1}^q C_l^{(\mu v)} (E - \right. \\ \left. - e^{-\lambda_l^{(\mu v)} \mu})^{-1} \right)^{-1} \cdot \left  \delta_{kl} \left( e^{\lambda_l^{(\kappa \lambda)} \mu} - E \right)^{\alpha_{kl}^{(\kappa \lambda)}} \right _q \cdot \left  \delta_{kl} \left( e^{\lambda_l^{(\mu v)} \mu} - \right. \right. \\ \left. \left. - E \right)^{\alpha_{kl}^{(\mu v)}} \right _q^{-1} - qE$ | $Q_{(\mu v)}^{(\kappa \lambda)} = A_{(\mu v)}^{(\kappa \lambda)} \sum_{l=1}^q C_{(\mu v)l}^{(\kappa \lambda)} (E - e^{-\lambda_l \mu})^{-\alpha_l} \cdot \left( \sum_{k=1}^q C_l^{(\mu v)} (E - \right. \\ \left. - e^{-\lambda_l^{(\mu v)} \mu})^{-1} \right)^{-1} \cdot \left  \delta_{kl} \left( e^{\lambda_l^{(\kappa \lambda)} \mu} - E \right)^{\alpha_{kl}^{(\kappa \lambda)}} \right _q \cdot \left  \delta_{kl} \left( e^{\lambda_l^{(\mu v)} \mu} - \right. \right. \\ \left. \left. - E \right)^{\alpha_{kl}^{(\mu v)}} \right _q^{-1} - qE$ |
| BH3FR_E00148 (66a) | 118  | 0.777 | $q_{(\mu \nu) k l}^{(\kappa \lambda)} = q_{(\mu \nu) k l}^{(\alpha \beta)} \psi_{(\mu \nu) k l}^{(\alpha \beta)} \left( \psi_{(\mu \nu) k l}^{(\kappa \lambda)} \right)^{-1}$                                                                                                                                                                                                                                                                                                                                                                                                               | $q_{(\mu \nu) k l}^{(\kappa \lambda)} = q_{(\mu \nu) k l}^{(\alpha \beta)} \psi_{(\mu \nu) k l}^{(\alpha \beta)} \left( \psi_{(\mu \nu) k l}^{(\kappa \lambda)} \right)^{-1}$                                                                                                                                                                                                                                                                                                                                                                           | $q_{(\mu \nu) k l}^{(\kappa \lambda)} = q_{(\mu \nu) k l}^{(\alpha \beta)} \psi_{(\mu \nu) k l}^{(\alpha \beta)} \left( \psi_{(\mu \nu) k l}^{(\kappa \lambda)} \right)^{-1}$                                                                                                                                                                                                                                                                                                                                                                           |
| BH3FR_E00263 (118) | 137  | 0.791 | $\left( \begin{array}{c} -Q_1, -Q_2 \\ -Q_3, -Q_4 \end{array} \right) \equiv \left( 0 \mid, \quad \hat{S}(v_x) = kP \left( 0 \mid \begin{array}{c} Q \kappa v_x v_x \\ \tilde{C}_0 \varepsilon \end{array} \right) v_x \right.$                                                                                                                                                                                                                                                                                                                                                             | $\left( \begin{array}{c} -Q_1, -Q_2 \\ -Q_3, -Q_4 \end{array} \right) \equiv \left( 0 \mid, \quad \hat{S}(v_x) = kP \left( 0 \mid \begin{array}{c} Q \kappa v_x v_x \\ \tilde{C}_0 \varepsilon \end{array} \right) v_x \right.$                                                                                                                                                                                                                                                                                                                         | $\left( \begin{array}{c} -Q_1, -Q_2 \\ -Q_3, -Q_4 \end{array} \right) \equiv (0 \mid, \quad \hat{S}(v_x) = kP(0 \mid \begin{array}{c} Q \kappa \\ \tilde{C} 0 \end{array}) v_x$                                                                                                                                                                                                                                                                                                                                                                         |
| BH3FR_E00195 (89)  | 126  | 0.799 | $\begin{aligned} & \bar{\sigma}(x) = i(\bar{e}_t s + \bar{e}_r J P_a), \quad \bar{e}_t \perp \bar{e}_r, \quad 2s = P \geq 0, \\ & 2J = Q \geq 0, \quad P_a = \alpha_p (-1)^J, \quad \alpha_p = \pm 1 \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                         | $\bar{\sigma}(x) = i(\bar{e}_t s + \bar{e}_r J P_a), \quad \bar{e}_t \perp \bar{e}_r, \quad 2s = P \geq 0, \\ 2J = Q \geq 0, \quad P_a = \alpha_p (-1)^J, \quad \alpha_p = \pm 1$                                                                                                                                                                                                                                                                                                                                                                       | $\bar{\sigma}(x) = i(\bar{e}_t s + \bar{e}_r J P_a), \quad \bar{e}_t \perp \bar{e}_r, \quad 2s = P \geq 0, \\ 2J = Q \geq 0, \quad P_a = \alpha_p (-1)^J, \quad \alpha_p = \pm 1$                                                                                                                                                                                                                                                                                                                                                                       |

| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Page | Conf. | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Scan image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| BH3FR_EQ0239 (108a)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 133  | 0.804 | $\begin{aligned} & f(N \geq 0) \fallingdotseq 0, \quad \quad \quad \partial_N f > 0, \\ & \partial_N N = \partial N = 1, \\ & 0 \leq N \leq L_N < \infty, \quad w(vx) = 1 + (2 - k)\underline{w}_1 + (k - 1)\underline{w}_2, \\ & w = w^*, \quad \underline{w}_k(n_j = 0) = 0 \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                               | $\begin{aligned} & f(N \geq 0) \doteq 0, \quad \partial_N f > 0, \quad \partial_N N = \partial N = 1, \\ & 0 \leq N \leq L_N < \infty, \quad w(vx) = 1 + (2 - k)\underline{w}_1 + (k - 1)\underline{w}_2, \\ & w = w^*, \quad \underline{w}_k(n_j = 0) = 0 \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                  | $\begin{aligned} & f(N \geq 0) \geq 0, \quad \partial_N f > 0, \quad \partial_N N = \partial N = 1, \\ & 0 \leq N \leq L_N < \infty, \quad w(vx) = 1 + (2 - k)\underline{w}_1 + (k - 1)\underline{w}_2, \\ & w = w^*, \quad \underline{w}_k(n_j = 0) = 0 \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| BH3FR_EQ0129 (56)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 115  | 0.807 | $\begin{aligned} & \gamma_{(\mu v)+}^{\underline{\tilde{k}l}} = \text{const}, \quad \gamma_{(\mu v)+}^{kl} = \gamma_{(\mu v)+}^{kl}(V_r), \quad \gamma_{-\tilde{kl}}^{(\mu v)} = 0, \\ & \gamma_{-kl}^{(\mu v)} = \text{const} \neq 0 \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                       | $\begin{aligned} & \gamma_{(\mu v)+}^{\tilde{kl}} = \text{const}, \quad \gamma_{(\mu v)+}^{kl} = \gamma_{(\mu v)+}^{kl}(V_r), \quad \gamma_{-\tilde{kl}}^{(\mu v)} = 0, \\ & \gamma_{-kl}^{(\mu v)} = \text{const} \neq 0 \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $\begin{aligned} & \gamma_{(\mu v)+}^{\tilde{kl}} = \text{const}, \quad \gamma_{(\mu v)+}^{kl} = \gamma_{(\mu v)+}^{kl}(V_r), \quad \gamma_{-\tilde{kl}}^{(\mu v)} = 0, \\ & \gamma_{-kl}^{(\mu v)} = \text{const} \neq 0 \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| BH3FR_EQ0242                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 133  | 0.807 | $\begin{aligned} & 2 \eta^2 A_{11} = \xi^2 (\pi e \xi)^2 (1 - 4 \pi \alpha^2) Y_4, \quad 3 A_{12} = 2 \pi e \xi^2 \left( \frac{\vartheta}{8} - \right. \\ & \left. - \pi e \eta \frac{\alpha^2}{3} \right) Y_5, \quad A_{13} = 3 (4 + \eta \alpha) \left( 1 - \frac{\eta^2}{5} \left( \frac{1 - \sqrt{\eta}}{1 + \sqrt{\eta}} \right)^2 \right) Y_6, \\ & \alpha A_{14} = 1 + \frac{3 \eta}{4 \xi} \left( 2 \eta \alpha - e^2 \xi \left( \frac{1 - \sqrt{\eta}}{1 + \sqrt{\eta}} \right)^2 \right) Y_7, \\ & 3 A_{15} = e^2 \left( 1 - 2 e \frac{\alpha^2}{\eta} \right) Y_8, \\ & A_{16} = (\pi e)^2 \left( 1 + \frac{\alpha}{5 \eta} \left( 1 + 6 \frac{\alpha}{\pi} \right) \right) Y_9, \end{aligned}$ | $\begin{aligned} & 2 \eta^2 A_{11} = \xi^2 (\pi e \xi)^2 (1 - 4 \pi \alpha^2) Y_4, \quad 3 A_{12} = 2 \pi e \xi^2 \left( \frac{\vartheta}{8} - \right. \\ & \left. - \pi e \eta \frac{\alpha^2}{3} \right) Y_5, \quad A_{13} = 3 (4 + \eta \alpha) \left( 1 - \frac{\eta^2}{5} \left( \frac{1 - \sqrt{\eta}}{1 + \sqrt{\eta}} \right)^2 \right) Y_6, \\ & \alpha A_{14} = 1 + \frac{3 \eta}{4 \xi} \left( 2 \eta \alpha - e^2 \xi \left( \frac{1 - \sqrt{\eta}}{1 + \sqrt{\eta}} \right)^2 \right) Y_7, \\ & 3 A_{15} = e^2 \left( 1 - 2 e \frac{\alpha^2}{\eta} \right) Y_8, \\ & A_{16} = (\pi e)^2 \left( 1 + \frac{\alpha}{5 \eta} \left( 1 + 6 \frac{\alpha}{\pi} \right) \right) Y_9, \end{aligned}$ | $\begin{aligned} & 2 \eta^2 A_{11} = \xi^2 (\pi e \xi)^2 (1 - 4 \pi \alpha^2) Y_4, \quad 3 A_{12} = 2 \pi e \xi^2 \left( \frac{\vartheta}{8} - \right. \\ & \left. - \pi e \eta \frac{\alpha^2}{3} \right) Y_5, \quad A_{13} = 3 (4 + \eta \alpha) \left( 1 - \frac{\eta^2}{5} \left( \frac{1 - \sqrt{\eta}}{1 + \sqrt{\eta}} \right)^2 \right) Y_6, \\ & \alpha A_{14} = 1 + \frac{3 \eta}{4 \xi} \left( 2 \eta \alpha - e^2 \xi \left( \frac{1 - \sqrt{\eta}}{1 + \sqrt{\eta}} \right)^2 \right) Y_7, \\ & 3 A_{15} = e^2 \left( 1 - 2 e \frac{\alpha^2}{\eta} \right) Y_8, \\ & A_{16} = (\pi e)^2 \left( 1 + \frac{\alpha}{5 \eta} \left( 1 + 6 \frac{\alpha}{\pi} \right) \right) Y_9, \end{aligned}$ |
| BH3FR_EQ0133                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 115  | 0.809 | $\begin{aligned} & L_{(\mu \nu)}^{(\kappa \lambda)} = \Lambda_{(\mu \nu)}^{(\kappa \lambda)} - \bar{\lambda}_{(\mu \nu)}^{(\kappa \lambda)} \times \left( 1 + s p^2 \bar{Q}_{(\mu \nu)}^{(\kappa \lambda)} \times () \right), \\ & \Lambda_{(\mu \nu)}^{(\kappa \lambda)} = K_{(\mu \nu)}^{(\kappa \lambda \lambda)} ; \left( 1 + s p^2 \bar{Q}_{(\mu \nu)}^{(\kappa \lambda)} \times () \right) + D_{(\mu \nu)}^{(\kappa \lambda)}, \end{aligned}$                                                                                                                                                                                                                                                        | $\begin{aligned} & L_{(\mu \nu)}^{(\kappa \lambda)} = \Lambda_{(\mu \nu)}^{(\kappa \lambda)} - \bar{\lambda}_{(\mu \nu)}^{(\kappa \lambda)} \times \left( 1 + s p^2 \bar{Q}_{(\mu \nu)}^{(\kappa \lambda)} \times () \right), \\ & \Lambda_{(\mu \nu)}^{(\kappa \lambda)} = K_{(\mu \nu)}^{(\kappa \lambda \lambda)} ; \left( 1 + s p^2 \bar{Q}_{(\mu \nu)}^{(\kappa \lambda)} \times () \right) + D_{(\mu \nu)}^{(\kappa \lambda)}, \end{aligned}$                                                                                                                                                                                                                                                        | $\begin{aligned} & L_{(\mu \nu)}^{(\kappa \lambda)} = A_{(\mu \nu)}^{(\kappa \lambda)} - \bar{\lambda}_{(\mu \nu)}^{(\kappa \lambda)} \times \left( 1 + s p^2 \bar{Q}_{(\mu \nu)}^{(\kappa \lambda)} \times () \right), \\ & A_{(\mu \nu)}^{(\kappa \lambda)} = K_{(\mu \nu)}^{(\kappa \lambda)} ; \left( 1 + s p^2 \bar{Q}_{(\mu \nu)}^{(\kappa \lambda)} \times () \right) + D_{(\mu \nu)}^{(\kappa \lambda)}, \end{aligned}$                                                                                                                                                                                                                                                                            |
| <p><b>Conf.</b> MathPix confidence: <span style="color: green;">■</span> <math>\geq 0.9</math> <span style="color: orange;">■</span> <math>0.5 - 0.9</math> <span style="color: red;">■</span> <math>&lt; 0.5</math> — no value</p> <p><b>LaTeX source</b> MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ</p> <p><b>Residual</b> not shown for this document — no residual measurements are paired with these rows.</p> <p><b>[refined: <i>basis</i>]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</p> |      |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Page | Conf. | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Scan image                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| BH3FR_E00284 (M11a)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 141  | 0.814 | $\begin{aligned} & \& 0; n=0, \quad E; n=1, \quad E; (); n=n, \quad (C_i \times C_k)_\pm \neq 0, \\ & C; n=a = \text{const}(n), \quad C = a \frac{()}{()} \end{aligned}$                                                                                                                                                                                                                                                                                                                                                         | $0; n=0, \quad E; n=1, \quad E; (); n=n, \quad (C_i \times C_k)_\pm \neq 0,$ $C; n=a = \text{const}(n), \quad C = a \frac{()}{()}$                                                                                                                                                                                                                                                                                                                                                   | $0; n=0, \quad E; n=1, \quad E; (); n=n, \quad (C_i \times C_k)_\pm \neq 0,$ $C; n=a = \text{const}(n), \quad C = a \frac{()}{()}$ <b>M1</b>                                                                                                                                                                                                                                                                                                                                         |
| BH3FR_E00289 (M12b)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 141  | 0.816 | $\begin{aligned} & \& \bar{D}; {}^m\bar{C} = {}^{m+1}\bar{W}, \quad {}^{sp}\bar{D}; {}^m\bar{C} = {}^{m-1}\bar{W}, \quad D; {}^m\bar{C} = {}^m\bar{W}, \\ & \bar{\partial} = \sum_{i=1}^L \bar{e}_i \partial_i, \quad -\varphi = \text{GRAD}_L \varphi, \\ & \bar{\partial}; {}^m\bar{C} = \widehat{\text{DIV}}_L {}^m\bar{C}, \quad {}^{sp}\bar{\partial}; {}^m\bar{C} = \overline{\text{DIV}}_L {}^m\bar{C}, \\ & \bar{\partial}; {}^m\bar{C} - (\bar{\partial}; {}^m\bar{C})^\times = \text{ROT}_L {}^m\bar{C} \end{aligned}$ | $\bar{D}; {}^m\bar{C} = {}^{m+1}\bar{W}, \quad {}^{sp}\bar{D}; {}^m\bar{C} = {}^{m-1}\bar{W}, \quad D; {}^m\bar{C} = {}^m\bar{W},$ $\bar{\partial} = \sum_{i=1}^L \bar{e}_i \partial_i, \quad -\varphi = \text{GRAD}_L \varphi,$ $\bar{\partial}; {}^m\bar{C} = \widehat{\text{DIV}}_L {}^m\bar{C}, \quad {}^{sp}\bar{\partial}; {}^m\bar{C} = \overline{\text{DIV}}_L {}^m\bar{C},$ $\bar{\partial}; {}^m\bar{C} - (\bar{\partial}; {}^m\bar{C})^\times = \text{ROT}_L {}^m\bar{C}$ | $\bar{D}; {}^m\bar{C} = {}^{m+1}\bar{W}, \quad {}^{sp}\bar{D}; {}^m\bar{C} = {}^{m-1}\bar{W}, \quad D; {}^m\bar{C} = {}^m\bar{W},$ $\bar{\partial} = \sum_{i=1}^L \bar{e}_i \partial_i, \quad -\varphi = \text{GRAD}_L \varphi,$ $\bar{\partial}; {}^m\bar{C} = \widehat{\text{DIV}}_L {}^m\bar{C}, \quad {}^{sp}\bar{\partial}; {}^m\bar{C} = \overline{\text{DIV}}_L {}^m\bar{C},$ $\bar{\partial}; {}^m\bar{C} - (\bar{\partial}; {}^m\bar{C})^\times = \text{ROT}_L {}^m\bar{C}$ |
| BH3FR_E00277 (M6a)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 140  | 0.827 | $\boldsymbol{S} \frac{\varphi}{\psi} \partial n = \frac{u}{v}, \quad \frac{\varphi}{\psi} = \frac{v \partial u - u \partial v}{v(n) v(n-1)}$                                                                                                                                                                                                                                                                                                                                                                                     | $\boldsymbol{S} \frac{\varphi}{\psi} \partial n = \frac{u}{v}, \quad \frac{\varphi}{\psi} = \frac{v \partial u - u \partial v}{v(n) v(n-1)}$                                                                                                                                                                                                                                                                                                                                         | $\boldsymbol{S} \frac{\varphi}{\psi} \partial n = \frac{u}{v}, \quad \frac{\varphi}{\psi} = \frac{v \partial u - u \partial v}{v(n) v(n-1)}$                                                                                                                                                                                                                                                                                                                                         |
| BH3FR_E00197 (90)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 127  | 0.843 | $\mathrm{s}_v = s+1-v, \quad 1 \leq v \leq I = P+1$                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $s_v = s+1-v, \quad 1 \leq v \leq I = P+1$                                                                                                                                                                                                                                                                                                                                                                                                                                           | $s_v = s+1-v, \quad 1 \leq v \leq I = P+1$                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| BH3FR_E00274                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 139  | 0.844 | $\stackrel{n_2}{\underset{n_1}{\boldsymbol{S}}} + \stackrel{n_1}{\underset{n_2}{\boldsymbol{S}}} = \stackrel{n_1}{\underset{n_1}{\boldsymbol{S}}} + \stackrel{n_2}{\underset{n_2}{\boldsymbol{S}}}$                                                                                                                                                                                                                                                                                                                              | $\stackrel{n_2}{\underset{n_1}{\boldsymbol{S}}} + \stackrel{n_1}{\underset{n_2}{\boldsymbol{S}}} = \stackrel{n_1}{\underset{n_1}{\boldsymbol{S}}} + \stackrel{n_2}{\underset{n_2}{\boldsymbol{S}}}$                                                                                                                                                                                                                                                                                  | $\stackrel{n_2}{\underset{n_1}{\boldsymbol{S}}} + \stackrel{n_1}{\underset{n_2}{\boldsymbol{S}}} = \stackrel{n_1}{\underset{n_1}{\boldsymbol{S}}} + \stackrel{n_2}{\underset{n_2}{\boldsymbol{S}}}$                                                                                                                                                                                                                                                                                  |
| BH3FR_E00061 (20b)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 106  | 0.851 | $\begin{aligned} & \& \varepsilon_+^{(\pm)} = \pm \frac{\pi}{2} (2()_+ + E), \quad \varepsilon_-^{(\pm)} = \pm \pi()_-, \quad \varepsilon_\pm^{(\pm)}; n = \beta_\pm^{(\pm)}, \\ & n_\pm \geq 0, \quad \beta_+^{(\pm)} + \beta_-^{(\mp)} = \pm \frac{\pi}{2}, \end{aligned}$                                                                                                                                                                                                                                                     | $\varepsilon_+^{(\pm)} = \pm \frac{\pi}{2} (2()_+ + E), \quad \varepsilon_-^{(\pm)} = \pm \pi()_-,$ $n_\pm \geq 0, \quad \beta_+^{(\pm)} + \beta_-^{(\mp)} = \pm \frac{\pi}{2},$                                                                                                                                                                                                                                                                                                     | $\varepsilon_+^{(\pm)} = \pm \frac{\pi}{2} (2()_+ + E), \quad \varepsilon_-^{(\pm)} = \pm \pi()_-, \quad \varepsilon_\pm^{(\pm)}; n = \beta_\pm^{(\pm)},$ $n_\pm \geq 0, \quad \beta_+^{(\pm)} + \beta_-^{(\mp)} = \pm \frac{\pi}{2},$ <b>(2)</b>                                                                                                                                                                                                                                    |
| BH3FR_E00099 (40a)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 111  | 0.851 | $\begin{aligned} & \& a s_0 \eta \sqrt[3]{2\eta} = 4 \sqrt[4]{\pi} \sqrt[3]{3s_0}, \quad s_0 = 1[m], \\ & 4\pi^2 b = 3 \sqrt{\vartheta/\pi}, \quad \vartheta = 5\eta + 2 \sqrt{\eta} + 1 \end{aligned}$                                                                                                                                                                                                                                                                                                                          | $a s_0 \eta \sqrt[3]{2\eta} = 4 \sqrt[4]{\pi} \sqrt[3]{3s_0}, \quad s_0 = 1[ \text{ m}],$ $4\pi^2 b = 3 \sqrt{\vartheta/\pi}, \quad \vartheta = 5\eta + 2 \sqrt{\eta} + 1$                                                                                                                                                                                                                                                                                                           | $a s_0 \eta \sqrt[3]{2\eta} = 4 \sqrt[4]{\pi} \sqrt[3]{3s_0}, \quad s_0 = 1[m],$ $4\pi^2 b = 3 \sqrt{\vartheta/\pi}, \quad \vartheta = 5\eta + 2 \sqrt{\eta} + 1$                                                                                                                                                                                                                                                                                                                    |
| <b>Conf.</b> MathPix confidence: <span style="color: green;">■</span> ≥0.9 <span style="color: orange;">■</span> 0.5–0.9 <span style="color: red;">■</span> <0.5 <span style="color: gray;">—</span> no value<br><b>LaTeX source</b> MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ<br><b>Residual</b> not shown for this document — no residual measurements are paired with these rows.<br><b>[refined: basis]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here. |      |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                | Rendered                                                                                                                                                                                                                                                                                                                                                                                 | Scan image                                                                                                                                                                                                                                                                                                                                                                                         |
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| BH3FR_E00124 (53a)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 114  | ■ 0.858 | $\begin{aligned} & \{ \}^{\{2\}} \bar{\kappa}_{\{1\}} = \{ \}^{\{2\}} \bar{\kappa}_{\{1\}} \left( C_5, C_6 \right), \quad \{ \}^{\{2\}} \bar{\kappa}_{\{2\}} = \{ \}^{\{2\}} \bar{\kappa}_{\{2\}} \left( C_4 \right), \\ & \{ \}^{\{2\}} \bar{\kappa}_{\{3\}} = \{ \}^{\{2\}} \bar{\kappa}_{\{3\}} \left( C_1, C_2, C_3 \right) \end{aligned}$                                              | $\begin{aligned} {}^2\bar{\kappa}_{(1)} &= {}^2\bar{\kappa}_{(1)}(C_5, C_6), \quad {}^2\bar{\kappa}_{(2)} = {}^2\bar{\kappa}_{(2)}(C_4), \\ {}^2\bar{\kappa}_{(3)} &= {}^2\bar{\kappa}_{(3)}(C_1, C_2, C_3) \end{aligned}$                                                                                                                                                               | $\begin{aligned} {}^2\bar{\kappa}_{(1)} &= {}^2\bar{\kappa}_{(1)}(C_5, C_6), \quad {}^2\bar{\kappa}_{(2)} = {}^2\bar{\kappa}_{(2)}(C_4), \\ {}^2\bar{\kappa}_{(3)} &= {}^2\bar{\kappa}_{(3)}(C_1, C_2, C_3) \end{aligned}$                                                                                                                                                                         |
| BH3FR_E00250 (111a)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 136  | ■ 0.863 | $\begin{aligned} & F_1 = P(P+Q)(-1)^{P+Q} (2-k+\eta^3(k-1)), \\ & F_2 = 1 + 4 \frac{\xi}{k} \binom{P}{2} (\xi/6)^q Y_{41}, \\ & F_q \sqrt{\eta_{qk}} = (3-\alpha + \frac{\pi}{2}(k-1)3^{2-q/2}) (2\sqrt{\eta_{11}\eta_{qk}} + q\eta^2(k-1)Y_{42}), \\ & F_\kappa = 1 + \frac{2k\kappa}{3\eta^2} \xi(1+\pi\xi^2(P-Q)(\pi-5q/4))Y_{43} \end{aligned}$                                         | $\begin{aligned} F_1 &= P(P+Q)(-1)^{P+Q} (2-k+\eta^3(k-1)), \\ F_2 &= 1 + 4 \frac{\xi}{k} \binom{P}{2} (\xi/6)^q Y_{41}, \\ F_q \sqrt{\eta_{qk}} &= \left( 3 - \alpha + \frac{\pi}{2}(k-1)3^{2-q/2} \right) (2\sqrt{\eta_{11}\eta_{qk}} + q\eta^2(k-1)Y_{42}), \\ F_\kappa &= 1 + \frac{2k\kappa}{3\eta^2} \xi(1+\pi\xi^2(P-Q)(\pi-5q/4))Y_{43} \end{aligned}$                           | $\begin{aligned} F_1 &= P(P+Q)(-1)^{P+Q} (2-k+\eta^3(k-1)), \\ F_2 &= 1 + 4 \frac{\xi}{k} \binom{P}{2} (\xi/6)^q Y_{41}, \\ F_q \sqrt{\eta_{qk}} &= (3-\alpha + \frac{\pi}{2}(k-1)3^{2-q/2}) (2\sqrt{\eta_{11}\eta_{qk}} + q\eta^2(k-1)Y_{42}), \\ F_\kappa &= 1 + \frac{2k\kappa}{3\eta^2} \xi(1+\pi\xi^2(P-Q)(\pi-5q/4))Y_{43} \end{aligned} \quad (11)$                                         |
| BH3FR_E00113 (47)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 113  | ■ 0.867 | $\begin{aligned} & \eta^7 - \eta = \pm a, \quad 2\eta^2 = f_{(0)} \sqrt[6]{6/\pi}, \quad a\sqrt{\pi} = \sqrt[6]{\pi/6}, \\ & f_{(0)} (eD_0^2(E\sqrt{\pi})^{-1} - 1)^{1/2} = \sqrt[4]{eD_0^2(E\sqrt{\pi})^{-1}} \end{aligned}$                                                                                                                                                               | $\begin{aligned} \eta^7 - \eta &= \pm a, \quad 2\eta^2 = f_{(0)} \sqrt[6]{6/\pi}, \quad a\sqrt{\pi} = \sqrt[6]{\pi/6} \\ f_{(0)} (eD_0^2(E\sqrt{\pi})^{-1} - 1)^{1/2} &= \sqrt[4]{eD_0^2(E\sqrt{\pi})^{-1}} \end{aligned}$                                                                                                                                                               | $\begin{aligned} \eta^7 - \eta &= \pm a, \quad 2\eta^2 = f_{(0)} \sqrt[6]{6/\pi}, \quad a\sqrt{\pi} = \sqrt[6]{\pi/6}, \\ f_{(0)} (eD_0^2(E\sqrt{\pi})^{-1} - 1)^{1/2} &= \sqrt[4]{eD_0^2(E\sqrt{\pi})^{-1}} \end{aligned}$                                                                                                                                                                        |
| BH3FR_E00204 (94)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 127  | ■ 0.876 | $\begin{aligned} & \bar{\lambda}_X - [X, Y] - \bar{\lambda}_Y, \quad J_X > 0, \quad J_Y > 0, \\ & \cos(\bar{s}_X, \bar{s}_Y) \approx \pm 1 \end{aligned}$                                                                                                                                                                                                                                   | $\begin{aligned} \bar{\lambda}_X - [X, Y] - \bar{\lambda}_Y, \quad J_X > 0, \quad J_Y > 0 \\ \cos(\bar{s}_X, \bar{s}_Y) &\approx \pm 1 \end{aligned}$                                                                                                                                                                                                                                    | $\begin{aligned} \bar{\lambda}_X - [X, Y] - \bar{\lambda}_Y, \quad J_X > 0, \quad J_Y > 0, \\ \cos(\bar{s}_X, \bar{s}_Y) &\approx \pm 1 \end{aligned}$                                                                                                                                                                                                                                             |
| BH3FR_E00174 (79 c)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 123  | ■ 0.876 | $r=r(v), \quad \delta r \rightarrow \beta = \text{const} > 0$                                                                                                                                                                                                                                                                                                                               | $r=r(v), \quad \delta r \rightarrow \beta = \text{const} > 0$                                                                                                                                                                                                                                                                                                                            | $r=r(v), \quad \delta r \rightarrow \beta = \text{const} > 0$                                                                                                                                                                                                                                                                                                                                      |
| BH3FR_E00215 (98c)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 129  | ■ 0.876 | $\begin{aligned} & 2\alpha_1 = 1 + \sqrt{\eta_{qk}}, \quad \alpha_2 \eta_{qk} = 1, \quad k\alpha_3 = e^{k-1} - kqF, \\ & \alpha_4 = 1, \quad F = H + G, \quad 3H = \alpha(1 + \sqrt{\eta_{qk}}) \left( \frac{\xi}{\eta_{qk}^2} \right)^{2k+1} \eta_{qk}^3, \\ & e\eta_{qk}G = \eta_{11} (2\xi\eta_{qk})^k \left( 1 - \frac{\sqrt{\eta_{qk}}}{1 + \sqrt{\eta_{qk}}} \right)^2 \end{aligned}$ | $\begin{aligned} 2\alpha_1 &= 1 + \sqrt{\eta_{qk}}, \quad \alpha_2 \eta_{qk} = 1, \quad k\alpha_3 = e^{k-1} - kqF, \\ \alpha_4 &= 1, \quad F = H + G, \quad 3H = \alpha(1 + \sqrt{\eta_{qk}}) \left( \frac{\xi}{\eta_{qk}^2} \right)^{2k+1} \eta_{qk}^3, \\ e\eta_{qk}G &= \eta_{11} (2\xi\eta_{qk})^k \left( 1 - \frac{\sqrt{\eta_{qk}}}{1 + \sqrt{\eta_{qk}}} \right)^2 \end{aligned}$ | $\begin{aligned} 2\alpha_1 &= 1 + \sqrt{\eta_{qk}}, \quad \alpha_2 \eta_{qk} = 1, \quad k\alpha_3 = e^{k-1} - kqF, \\ \alpha_4 &= 1, \quad F = H + G, \quad 3H = \alpha(1 + \sqrt{\eta_{qk}}) \left( \frac{\xi}{\eta_{qk}^2} \right)^{2k+1} \eta_{qk}^3, \\ e\eta_{qk}G &= \eta_{11} (2\xi\eta_{qk})^k \left( 1 - \frac{\sqrt{\eta_{qk}}}{1 + \sqrt{\eta_{qk}}} \right)^2 \end{aligned} \quad (9)$ |
| <p><b>Conf.</b> MathPix confidence: ■ <math>\geq 0.9</math> ■ 0.5–0.9 ■ <math>&lt; 0.5</math> — no value</p> <p><b>LaTeX source</b> MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ</p> <p><b>Residual</b> not shown for this document — no residual measurements are paired with these rows.</p> <p><b>[refined: <i>basis</i>]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</p> |      |         |                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                    |

| Identifier           | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                           | Rendered                                                                                                                                                                                                                                                         | Scan image                                                                                                                                                                                                                                                        |
|----------------------|------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BH3FR_EQ00123 (53)   | 114  | ■ 0.879 | $\begin{aligned} &1\leqq(\mu,v)\leqq 3,\quad\{ }^{\wedge 2}\bar{\kappa}_{\{\mu\}}\neq\{ }^{\wedge 2}\bar{\kappa}_{\{\mu\}}\}^{\wedge}\times,\quad\{ }^{\wedge 2}\bar{\gamma}_{\{\mu v\}}=s\,p\left(\{ }^{\wedge 2}\bar{\kappa}_{\{\mu\}}\times\{ }^{\wedge 2}\bar{\kappa}_{\{v\}}\right)\}\\ &\hat{\gamma}=\left(\{ }^2\bar{\gamma}_{\{\mu v\}}\right)_3\end{aligned}$ | $1\leq(\mu,v)\leq 3,\quad\{ }^2\bar{\kappa}_{(\mu)}\neq\{ }^2\bar{\kappa}_{(\mu)}^{\times},\quad\{ }^2\bar{\gamma}_{(\mu v)}=sp\left(\{ }^2\bar{\kappa}_{(\mu)}\times\{ }^2\bar{\kappa}_{(v)}\right)$ $\hat{\gamma}=\left(\{ }^2\bar{\gamma}_{(\mu v)}\right)_3$ | $1\leq(\mu,v)\leq 3,\quad\{ }^2\bar{\kappa}_{(\mu)}\neq\{ }^2\bar{\kappa}_{(\mu)}^{\times},\quad\{ }^2\bar{\gamma}_{(\mu v)}=sp\left(\{ }^2\bar{\kappa}_{(\mu)}\times\{ }^2\bar{\kappa}_{(v)}\right),$ $\hat{\gamma}=\left(\{ }^2\bar{\gamma}_{(\mu v)}\right)_3$ |
| BH3FR_EQ00000 (28a)  | 109  | ■ 0.885 | $\begin{aligned} &e_r=\varepsilon_{\pm}\sqrt{\eta},\quad e_d=\varepsilon_{\pm}(1-\sqrt{\eta}),\quad 2e_w=\varepsilon_{\pm}(1+\sqrt{\eta}),\\ &\eta\sqrt[4]{4+\pi^4}=\pi\end{aligned}$                                                                                                                                                                                  | $e_r=\varepsilon_{\pm}\sqrt{\eta},\quad e_d=\varepsilon_{\pm}(1-\sqrt{\eta}),\quad 2e_w=\varepsilon_{\pm}(1+\sqrt{\eta}),$ $\eta\sqrt[4]{4+\pi^4}=\pi$                                                                                                           | $e_r=\varepsilon_{\pm}\sqrt{\eta},\quad e_d=\varepsilon_{\pm}(1-\sqrt{\eta}),\quad 2e_w=\varepsilon_{\pm}(1+\sqrt{\eta}),$ $\eta\sqrt[4]{4+\pi^4}=\pi$                                                                                                            |
| BH3FR_EQ00262 (117)  | 137  | ■ 0.886 | $\begin{aligned} &h\,\omega_R>\left(M(N)-M_x\right)c^2,\quad\omega_R=\omega_x=1/\theta_x,\\ &T>\theta_x>0\end{aligned}$                                                                                                                                                                                                                                                | $h\omega_R>(M(N)-M_x)c^2,\quad\omega_R=\omega_x=1/\theta_x,$ $T>\theta_x>0$                                                                                                                                                                                      | $h\omega_R>(M(N)-M_x)c^2,\quad\omega_R=\omega_x=1/\theta_x,$ $T>\theta_x>0$                                                                                                                                                                                       |
| BH3FR_EQ00153 (69)   | 118  | ■ 0.886 | $\begin{aligned} &\kappa_{(1)}^{\wedge\{\prime\}}Q_{-}=Q_{+}=C\left(e^{\lambda\mu}-E\right)^a\left(e^{\Lambda\mu}-E\right)^b,\\ &\hat{S}^2\bar{\kappa}_{(1)}\hat{S}^{\times}=\{ }^2\bar{\kappa}_{(1)}^{\prime},\quad\kappa_{(1)ik}^{\prime}\sim\delta_{ik}\end{aligned}$                                                                                               | $\kappa_{(1)}^{\prime}Q_{-}=Q_{+}=C\left(e^{\lambda\mu}-E\right)^a\left(e^{\Lambda\mu}-E\right)^b,$ $\hat{S}^2\bar{\kappa}_{(1)}\hat{S}^{\times}=\{ }^2\bar{\kappa}_{(1)}^{\prime},\quad\kappa_{(1)ik}^{\prime}\sim\delta_{ik}$                                  | $\kappa_{(1)}^{\prime}Q_{-}=Q_{+}=C\left(e^{\lambda\mu}-E\right)^a\left(e^{\Lambda\mu}-E\right)^b,$ $\hat{S}^2\bar{\kappa}_{(1)}\hat{S}^{\times}=\{ }^2\bar{\kappa}_{(1)}^{\prime},\quad\kappa_{(1)ik}^{\prime}\sim\delta_{ik}$                                   |
| BH3FR_EQ00256 (113a) | 136  | ■ 0.887 | $\begin{aligned} &\alpha_2(L_2+Q_2)\left(2(L_2+Q_2)^2+3(L_2+Q_2)+1\right)=6\alpha_1(L_1+Q_1)^3,\\ &\alpha_3(L_3+Q_3)(1+L_3+Q_3)=2\alpha_2(L_2+Q_2)^2,\\ &L_4+Q_4=TRC(\alpha_3(L_3+Q_3))\end{aligned}$                                                                                                                                                                  | $\alpha_2(L_2+Q_2)\left(2(L_2+Q_2)^2+3(L_2+Q_2)+1\right)=6\alpha_1(L_1+Q_1)^3,$ $\alpha_3(L_3+Q_3)(1+L_3+Q_3)=2\alpha_2(L_2+Q_2)^2,$ $L_4+Q_4=TRC(\alpha_3(L_3+Q_3))$                                                                                            | $\alpha_2(L_2+Q_2)(2(L_2+Q_2)^2+3(L_2+Q_2)+1)=6\alpha_1(L_1+Q_1)^3,$ $\alpha_3(L_3+Q_3)(1+L_3+Q_3)=2\alpha_2(L_2+Q_2)^2,$ $L_4+Q_4=TRC(\alpha_3(L_3+Q_3))$                                                                                                        |
| BH3FR_EQ00073 (26a)  | 108  | ■ 0.887 | $\begin{aligned} &\eta=\lambda/2,\quad r=\varepsilon=\frac{\Lambda}{2},\quad\lambda=\frac{h}{mc},\\ &\Lambda=\frac{h}{\mu\omega}=2\varsigma,\quad\varsigma=\frac{h^2}{\gamma m^3}\end{aligned}$                                                                                                                                                                        | $\eta=\lambda/2,\quad r=\varepsilon=\frac{\Lambda}{2},\quad\lambda=\frac{h}{mc},$ $\Lambda=\frac{h}{\mu\omega}=2\varsigma,\quad\varsigma=\frac{h^2}{\gamma m^3}$                                                                                                 | $\eta=\lambda/2,\quad r=\varepsilon=\frac{\Lambda}{2},\quad\lambda=\frac{h}{mc},$ $\Lambda=\frac{h}{\mu\omega}=2\varsigma,\quad\varsigma=\frac{h^2}{\gamma m^3}$                                                                                                  |
| BH3FR_EQ00050 (16c)  | 104  | ■ 0.890 | $R_{6(0)}\equiv R_{6(0)}\left(x^{\underline{k}}\right)_1^6,\quad x^{\underline{k}}=C^{\underline{k}};n,$                                                                                                                                                                                                                                                               | $R_{6(0)}\equiv R_{6(0)}\left(x^{\underline{k}}\right)_1^6,\quad x^{\underline{k}}=C^{\underline{k}};n,$                                                                                                                                                         | $R_{6(0)}\equiv R_{6(0)}\left(x^{\underline{k}}\right)_1^6,\quad x^{\underline{k}}=C^{\underline{k}};n,$                                                                                                                                                          |

**Conf.** MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

**LaTeX source** MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

**Residual** not shown for this document — no residual measurements are paired with these rows.

**[refined: *basis*]** beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Page | Conf. | LaTeX source                                                                                                                                                                                                                                                                                | Rendered                                                                                                                                                                                                                                              | Scan image                                                                                                                                                                                                                                 |
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| BH3FR_E00013 (4)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 101  | 0.892 | $\begin{aligned} & \left(x_1, x_2, x_3\right)=\left(x_1^*, x_2^*, x_3^*\right) \equiv R_3, \\ & \left(x_1^{\wedge *}, x_2^{\wedge *}, x_3^{\wedge *}\right) \equiv R_3, \\ & x_4=\mathrm{ict}, \quad x_5=i \varepsilon, \quad x_6=i \eta \end{aligned}$                                     | $\left(x_1, x_2, x_3\right)=\left(x_1^*, x_2^*, x_3^*\right) \equiv R_3, \quad x_4=\mathrm{ict},$ $x_5=i \varepsilon, \quad x_6=i \eta$                                                                                                               | $\left(x_1, x_2, x_3\right)=\left(x_1^*, x_2^*, x_3^*\right) \equiv R_3, \quad x_4=\mathrm{ict},$ $x_5=i \varepsilon, \quad x_6=i \eta$                                                                                                    |
| BH3FR_E00032 (11b)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 102  | 0.894 | $r\,\varphi(r)=\gamma\,\mathrm{Lm}(r)(1-r/\varsigma)^2$                                                                                                                                                                                                                                     | $r\varphi(r)=\gamma\,\mathrm{Lm}(r)(1-r/\varsigma)^2$                                                                                                                                                                                                 | $r\varphi(r)=\gamma Lm(r)(1-r/\varsigma)^2$                                                                                                                                                                                                |
| BH3FR_E00199 (91)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 127  | 0.894 | $\begin{array}{l} Q=2\,n, \quad \sigma_r=\pm iJ_T, \quad J_T=n, \quad 0\leq n\leq 6, \\ Q=2\,n+1, \quad \sigma_r=\pm J_S, \quad 2J_S=2\,n+1, \\ n\geq 0 \end{array}$                                                                                                                        | $Q=2n, \quad \sigma_r=\pm iJ_T, \quad J_T=n, \\ Q=2n+1, \quad \sigma_r=\pm J_S, \quad 2J_S=2n+1 \\ n\geq 0$                                                                                                                                           | $Q=2n, \quad \sigma_r=\pm iJ_T, \quad J_T=n, \quad 0\leq n\leq 6,$ $Q=2n+1, \quad \sigma_r=\pm J_S, \quad 2J_S=2n+1, \\ n\geq 0$                                                                                                           |
| BH3FR_E00122 (52)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 114  | 0.899 | $\left(\frac{A}{D}\right)^{3/11}\sqrt{E}=2^9e\sqrt[4]{\pi}\left(9\pi^2\sqrt[3]{2}\right)^{-1}\left(\frac{\tau}{E}\right)^{11/6}D$                                                                                                                                                           | $\left(\frac{A}{D}\right)^{3/11}\sqrt{E}=2^9e\sqrt[4]{\pi}\left(9\pi^2\sqrt[3]{2}\right)^{-1}\left(\frac{\tau}{E}\right)^{11/6}D$                                                                                                                     | $\left(\frac{A}{D}\right)^{3/11}\sqrt{E}=2^9e\sqrt[4]{\pi}\left(9\pi^2\sqrt[3]{2}\right)^{-1}\left(\frac{\tau}{E}\right)^{11/6}D$                                                                                                          |
| BH3FR_E00226 (102)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 131  | 0.899 | $\hat{I}\equiv kP\left[\begin{array}{l} Q,\kappa \\ C,q_x \end{array}\right]_{\varepsilon}$                                                                                                                                                                                                 | $\hat{I}\equiv kP\left[\begin{array}{l} Q,\kappa \\ C,q_x \end{array}\right]_{\varepsilon}$                                                                                                                                                           | $\hat{I}\equiv kP\left[\begin{array}{l} Q,\kappa \\ C,q_x \end{array}\right]_{\varepsilon}$                                                                                                                                                |
| BH3FR_E00128 (55a)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 115  | 0.901 | $\{ \}^2\bar{\gamma}_{(p\,q)-}=\{ \}^2\bar{P}_{(pq)}, \quad \tau\alpha_m\gamma_{+\tilde{k}m}^{(pq)}=\partial_m\phi_{\tilde{k}}^{(pq)}$                                                                                                                                                      | $^2\bar{\gamma}_{(pq)-}=\,^2\bar{P}_{(pq)}, \quad \tau\alpha_m\gamma_{+\tilde{k}m}^{(pq)}=\partial_m\phi_{\tilde{k}}^{(pq)}$                                                                                                                          | $^2\bar{\gamma}_{(pq)-}=\,^2\bar{P}_{(pq)}, \quad \tau\alpha_m\gamma_{+\tilde{k}m}^{(pq)}=\delta_m\phi_{\tilde{k}}^{(pq)}$                                                                                                                 |
| BH3FR_E00154 (69a)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 118  | 0.908 | $\begin{aligned} & Q_{+}(y)=\lim_{\tau\rightarrow o}Q_{+};n=C\left(e^{\pm i\lambda y}-1\right)^a, \\ & \beta_{(a)}^{+}=\pm2\pi n_{(a)}, \quad y^2=\varepsilon^2+\eta^2, \quad \beta_{(a)}^{-}(Q_{-})\equiv\beta_{(a)}^{+}, \\ & \kappa_{(1)}^{'}=\kappa_{\mathrm{ext}}^{(1)} \end{aligned}$ | $Q_{+}(y)=\lim_{\tau\rightarrow o}Q_{+};n=C\left(e^{\pm i\lambda y}-1\right)^a,$ $\beta_{(a)}^{+}=\pm2\pi n_{(a)}, \quad y^2=\varepsilon^2+\eta^2, \quad \beta_{(a)}^{-}(Q_{-})\equiv\beta_{(a)}^{+},$ $\kappa_{(1)}^{'}=\kappa_{\mathrm{ext}}^{(1)}$ | $Q_{+}(y)=\lim_{\tau\rightarrow o}Q_{+};n=C(e^{\pm i\lambda y}-1)^a,$ $\beta_{(a)}^{+}=\pm2\pi n_{(a)}, \quad y^2=\varepsilon^2+\eta^2, \quad \beta_{(a)}^{-}(Q_{-})\equiv\beta_{(a)}^{+},$ $\kappa_{(1)}^{'}=\kappa_{\mathrm{ext}}^{(1)}$ |
| BH3FR_E00203 (93)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 127  | 0.914 | $\begin{aligned} & \varepsilon\left(R_4^{\pm}\right)=\cos\left(\vec{S}(\vec{\lambda}),\vec{x}_4\right)=\pm1, \quad C_{\pm}\left(R_4^{\pm}\right)=-C_{\mp}, \\ & \varepsilon C_{\pm}=C>0 \end{aligned}$                                                                                      | $\varepsilon\left(R_4^{\pm}\right)=\cos\left(\vec{S}(\vec{\lambda}),\vec{x}_4\right)=\pm1, \quad C_{\pm}\left(R_4^{\pm}\right)=-C_{\mp}$ $\varepsilon C_{\pm}=C>0$                                                                                    | $\varepsilon\left(R_4^{\pm}\right)=\cos\left(\vec{S}(\vec{\lambda}),\vec{x}_4\right)=\pm1, \quad C_{\pm}\left(R_4^{\pm}\right)=-C_{\mp},$ $\varepsilon C_{\pm}=C>0$                                                                        |
| BH3FR_E00168 (76b)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 122  | 0.919 | $q_j\hat{=}(j), \quad F_v\hat{=}(j)\omega$                                                                                                                                                                                                                                                  | $q_j\hat{=}(j), \quad F_v\hat{=}(j)\omega$                                                                                                                                                                                                            | $q_j\hat{=}(j), \quad F_v\hat{=}(j)\omega$                                                                                                                                                                                                 |
| BH3FR_E00236 (107a)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 132  | 0.922 | $\begin{aligned} & j>1, \quad \beta_j=\check{\delta}_{j-1}G_{j-1}-G_j\geq1, \quad \beta_j=0, \\ & G_j=0, \quad n_{j-1}\rightarrow1+n_{j-1}, \quad j=1, \quad \partial_1G_1>0, \\ & -Q_j\leq n_j\leq L_j<\infty \end{aligned}$                                                               | $j>1, \quad \beta_j=\check{\delta}_{j-1}G_{j-1}-G_j\geq1, \quad \beta_j=0,$ $G_j=0, \quad n_{j-1}\rightarrow1+n_{j-1}, \quad j=1, \quad \partial_1G_1>0,$ $-Q_j\leq n_j\leq L_j<\infty$                                                               | $j>1, \quad \beta_j=\check{\delta}_{j-1}G_{j-1}-G_j\geq1, \quad \beta_j=0,$ $G_j=0, \quad n_{j-1}\rightarrow1+n_{j-1}, \quad j=1, \quad \check{\delta}_1G_1>0,$ $-Q_j\leq n_j\leq L_j<\infty$                                              |
| <p><b>Conf.</b> MathPix confidence: 0.9 0.5-0.9 &lt;0.5 — no value</p> <p><b>LaTeX source</b> MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ</p> <p><b>Residual</b> not shown for this document — no residual measurements are paired with these rows.</p> <p><b>[refined: <i>basis</i>]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</p> |      |       |                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                            |

| Identifier          | Page | Conf.  | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                              | Rendered                                                                                                                                                                                                                                                                                                                                                                                    | Scan image                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------|------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BH3FR_EQ0038 (14)   | 103  | ■0.924 | $R_{\pm}=\varsigma(1+\alpha)\left(1\pm\sqrt{1-(1+\alpha)^{-2}}\right),\quad \alpha e\gamma M_0=2\omega c\varsigma$                                                                                                                                                                                                                                                                                        | $R_{\pm}=\varsigma(1+\alpha)\left(1\pm\sqrt{1-(1+\alpha)^{-2}}\right),\quad \alpha e\gamma M_0=2\omega c\varsigma$                                                                                                                                                                                                                                                                          | $R_{\pm}=\varsigma(1+\alpha)(1\pm\sqrt{1-(1+\alpha)^{-2}}),\quad \alpha e\gamma M_0=2\omega c\varsigma,$                                                                                                                                                                                                                                                                                    |
| BH3FR_EQ0010 (2a)   | 100  | ■0.926 | $\eta_{ik}=\begin{gathered}\Delta N_{ik}\\\Delta\Omega\end{gathered},\quad \Delta\Omega=w\Delta x^1\Delta x^2\Delta x^3\Delta x^4$                                                                                                                                                                                                                                                                        | $\eta_{ik}=\frac{\Delta N_{ik}}{\Delta\Omega},\quad \Delta\Omega=w\Delta x^1\Delta x^2\Delta x^3\Delta x^4$                                                                                                                                                                                                                                                                                 | $\eta_{ik}=\frac{\Delta N_{ik}}{\Delta\Omega},\quad \Delta\Omega=w\Delta x^1\Delta x^2\Delta x^3\Delta x^4$                                                                                                                                                                                                                                                                                 |
| BH3FR_EQ0093 (37 a) | 110  | ■0.928 | $\left(\frac{eD\sqrt{\tau}}{\pi E}-1\right)f^2=\sqrt{\frac{eD\sqrt{\tau}}{\pi E}},\quad E=1\left[m^2\right]$                                                                                                                                                                                                                                                                                              | $\left(\frac{eD\sqrt{\tau}}{\pi E}-1\right)f^2=\sqrt{\frac{eD\sqrt{\tau}}{\pi E}},\quad E=1\left[m^2\right]$                                                                                                                                                                                                                                                                                | $\left(\frac{eD\sqrt{\tau}}{\pi E}-1\right)f^2=\sqrt{\frac{eD\sqrt{\tau}}{\pi E}},\quad E=1\left[m^2\right]$                                                                                                                                                                                                                                                                                |
| BH3FR_EQ0223 (101)  | 130  | ■0.928 | $(v)(kPQ\kappa)\varepsilon C(\varepsilon q_x)_0^P,\quad 1\leq v\leq 12$                                                                                                                                                                                                                                                                                                                                   | $(v)(kPQ\kappa)\varepsilon C(\varepsilon q_x)_0^P,\quad 1\leq v\leq 12$                                                                                                                                                                                                                                                                                                                     | $(v)(kPQ\kappa)\varepsilon C(\varepsilon q_x)_0^P,\quad 1\leq v\leq 12$                                                                                                                                                                                                                                                                                                                     |
| BH3FR_EQ0152 (68b)  | 118  | ■0.932 | $Q_{(\mu\nu)\mathrm{ext}}^{\left(\kappa\lambda\right)}\equiv\binom{\kappa\lambda}{\mu\nu}$                                                                                                                                                                                                                                                                                                                | $Q_{(\mu\nu)\mathrm{ext}}^{\left(\kappa\lambda\right)}\equiv\binom{\kappa\lambda}{\mu\nu}$                                                                                                                                                                                                                                                                                                  | $Q_{(\mu\nu)\mathrm{ext}}^{\left(\kappa\lambda\right)}\equiv\binom{\kappa\lambda}{\mu\nu}$                                                                                                                                                                                                                                                                                                  |
| BH3FR_EQ0046        | 104  | ■0.934 | $\begin{aligned} & \{ \}^{\{2\}} \bar{\tau} &= \left( \operatorname{ROT}_{\{6\}} \bar{\phi} \right) ; 1, \\ & \{ \}^{\{2\}} \bar{\tau} &= \tau \begin{vmatrix} 0 & 1 & 1 & i & i & i \\ -1 & 0 & 1 & i & i & i \\ -1 & -1 & 0 & i & i & i \\ -i & -i & -i & 0 & -1 & -1 \\ -i & -i & -i & 1 & 0 & -1 \\ -i & -i & -i & 1 & 1 & 0 \end{vmatrix} \end{aligned}$                                             | $\begin{aligned} & {}^2\bar{\tau} = \left( \operatorname{ROT}_6 \bar{\phi} \right) ; 1, \\ & {}^2\bar{\tau} = \tau \begin{vmatrix} 0 & 1 & 1 & i & i & i \\ -1 & 0 & 1 & i & i & i \\ -1 & -1 & 0 & i & i & i \\ -i & -i & -i & 0 & -1 & -1 \\ -i & -i & -i & 1 & 0 & -1 \\ -i & -i & -i & 1 & 1 & 0 \end{vmatrix} \end{aligned}$                                                           | $\begin{aligned} & {}^2\bar{\tau} = \left( \operatorname{ROT}_6 \bar{\phi} \right) ; 1, \\ & {}^2\bar{\tau} = \tau \begin{vmatrix} 0 & 1 & 1 & i & i & i \\ -1 & 0 & 1 & i & i & i \\ -1 & -1 & 0 & i & i & i \\ -i & -i & -i & 0 & -1 & -1 \\ -i & -i & -i & 1 & 0 & -1 \\ -i & -i & -i & 1 & 1 & 0 \end{vmatrix} \end{aligned}$                                                           |
| BH3FR_EQ0127 (55)   | 114  | ■0.939 | $\begin{aligned} & \{ \}^{\{2\}} \bar{\gamma}_{(p\,q)+} &= \text{const} \left( Z_{\tilde{k}} \right), \\ & \partial_{\tilde{k}} \gamma_{-il}^{(pq)} &= \frac{\alpha_{\tilde{k}}}{\alpha_i} \partial_i \gamma_{\tilde{k}l}^{(pq)} - \\ & - \frac{\alpha_{\tilde{k}}}{\alpha_l} \partial_l \gamma_{\tilde{k}i}^{(pq)}, \quad {}^2\bar{\gamma}_{(pq)-} &\equiv f \left( Z_{\tilde{k}} \right) \end{aligned}$ | $\begin{aligned} & {}^2\bar{\gamma}_{(pq)+} = \text{const} \left( Z_{\tilde{k}} \right), \quad \partial_{\tilde{k}} \gamma_{-il}^{(pq)} = \frac{\alpha_{\tilde{k}}}{\alpha_i} \partial_i \gamma_{\tilde{k}l}^{(pq)} - \\ & - \frac{\alpha_{\tilde{k}}}{\alpha_l} \partial_l \gamma_{\tilde{k}i}^{(pq)}, \quad {}^2\bar{\gamma}_{(pq)-} \equiv f \left( Z_{\tilde{k}} \right) \end{aligned}$ | $\begin{aligned} & {}^2\bar{\gamma}_{(pq)+} = \text{const} \left( Z_{\tilde{k}} \right), \quad \partial_{\tilde{k}} \gamma_{-il}^{(pq)} = \frac{\alpha_{\tilde{k}}}{\alpha_i} \partial_i \gamma_{\tilde{k}l}^{(pq)} - \\ & - \frac{\alpha_{\tilde{k}}}{\alpha_l} \partial_l \gamma_{\tilde{k}i}^{(pq)}, \quad {}^2\bar{\gamma}_{(pq)-} \equiv f \left( Z_{\tilde{k}} \right) \end{aligned}$ |
| BH3FR_EQ0116 (49)   | 113  | ■0.942 | $\begin{aligned} & T_1^+ \left( D_{fmp}^{xy+} \right) \frac{D \left( R_3^+ \right)}{\cos \alpha_4 > 0} \rightarrow T_2^+ \left( D_{fmp}^{yx-} \right) \equiv \\ & \equiv T_1^- \frac{D \left( R_3^- \right)}{\cos \alpha_4 < 0} \rightarrow T_2^- \equiv T_1^+ \end{aligned}$                                                                                                                             | $\begin{aligned} & T_1^+ \left( D_{fmp}^{xy+} \right) \frac{D \left( R_3^+ \right)}{\cos \alpha_4 > 0} \rightarrow T_2^+ \left( D_{fmp}^{yx-} \right) \equiv \\ & \equiv T_1^- \frac{D \left( R_3^- \right)}{\cos \alpha_4 < 0} \rightarrow T_2^- \equiv T_1^+ \end{aligned}$                                                                                                               | $\begin{aligned} & T_1^+ \left( D_{fmp}^{xy+} \right) \frac{D \left( R_3^+ \right)}{\cos \alpha_4 > 0} \rightarrow T_2^+ \left( D_{fmp}^{yx-} \right) \equiv \\ & \equiv T_1^- \frac{D \left( R_3^- \right)}{\cos \alpha_4 < 0} \rightarrow T_2^- \equiv T_1^+ \end{aligned}$                                                                                                               |

| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Page | Conf.  | LaTeX source                                                                                                                                                                                                                                                              | Rendered                                                                                                                                                                                                                                              | Scan image                                                                                                                                                                                                                                            |
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| BH3FR_E00020 (7)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 101  | ■0.943 | $\left.\left.\left.\left.\left.\begin{array}{c} i \\ k \end{array} \right.\right.\right.\right.\right)_m = \widehat{\left\{ \right\}}, \quad C = \sum_{p=1}^6 C_p, \quad \vec{\lambda} = \sum_{p=1}^6 \vec{e}_p \lambda_p$                                                | $\left(\left(\begin{array}{c} i \\ km \end{array}\right)\right)_6 = \widehat{\left\{ \right\}}, \quad C = \sum_{p=1}^6 C_p, \quad \vec{\lambda} = \sum_{p=1}^6 \vec{e}_p \lambda_p$                                                                   | $\left(\left(\begin{array}{c} i \\ k \end{array} \right)_m\right)_6 = \widehat{\left\{ \right\}}, \quad C = \sum_{p=1}^6 C_p, \quad \vec{\lambda} = \sum_{p=1}^6 \vec{e}_p \lambda_p$                                                                 |
| BH3FR_E00176 (80a)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 124  | ■0.943 | $\begin{aligned} \hat{\gamma}_{(x)} &= \left(\gamma_{(\mu\nu)}\right)_3, \quad n\left(\gamma_{(\mu\nu)} \neq \bar{E}\right) \leq 9, \\ Z &= n(n-1) \end{aligned}$                                                                                                         | $\hat{\gamma}_{(x)} = \left(\gamma_{(\mu\nu)}\right)_3, \quad n\left(\gamma_{(\mu\nu)} \neq \bar{E}\right) \leq 9, \\ Z = n(n-1)$                                                                                                                     | $\hat{\gamma}_{(x)} = \left(\gamma_{(\mu\nu)}\right)_3, \quad n\left(\gamma_{(\mu\nu)} \neq \bar{E}\right) \leq 9, \\ Z = n(n-1)$                                                                                                                     |
| BH3FR_E00220 (100)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 130  | ■0.944 | $\begin{aligned} Ck(1+\kappa) &= 2(P\varepsilon_P + Q\varepsilon_Q)(k-1+\kappa), \quad \varepsilon_{P,Q} = \varepsilon \cos \alpha_{P,Q}, \\ \alpha_P &= \pi Q\left(\kappa + \binom{P}{2}\right), \quad \alpha_Q = \pi Q\left(Q(k-1) + \binom{P}{2}\right) \end{aligned}$ | $Ck(1+\kappa) = 2(P\varepsilon_P + Q\varepsilon_Q)(k-1+\kappa), \quad \varepsilon_{P,Q} = \varepsilon \cos \alpha_{P,Q}, \\ \alpha_P = \pi Q\left(\kappa + \binom{P}{2}\right), \quad \alpha_Q = \pi Q\left(Q(k-1) + \binom{P}{2}\right)$             | $Ck(1+\kappa) = 2(P\varepsilon_P + Q\varepsilon_Q)(k-1+\kappa), \quad \varepsilon_{P,Q} = \varepsilon \cos \alpha_{P,Q}, \\ \alpha_P = \pi Q\left(\kappa + \binom{P}{2}\right), \quad \alpha_Q = \pi Q\left(Q(k-1) + \binom{P}{2}\right)$             |
| BH3FR_E00228 (103a)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 131  | ■0.945 | $\begin{aligned} G(1) &= 2, \quad b_1 \equiv Z(n), \quad b_2 \equiv Z(m) + Z(p) + Z(\sigma), \\ G(2) &= 3, \quad b_1 \equiv Z(n), \quad b_2 \equiv Z(m), \\ b_3 &\equiv Z(p) + Z(\sigma) \end{aligned}$                                                                   | $\begin{aligned} G(1) &= 2, \quad b_1 \equiv Z(n), \quad b_2 \equiv Z(m) + Z(p) + Z(\sigma), \\ G(2) &= 3, \quad b_1 \equiv Z(n), \quad b_2 \equiv Z(m), \\ b_3 &\equiv Z(p) + Z(\sigma) \end{aligned}$                                               | $\begin{aligned} G(1) &= 2, \quad b_1 \equiv Z(n), \quad b_2 \equiv Z(m) + Z(p) + Z(\sigma), \\ G(2) &= 3, \quad b_1 \equiv Z(n), \quad b_2 \equiv Z(m), \\ b_3 &\equiv Z(p) + Z(\sigma) \end{aligned}$                                               |
| BH3FR_E00214 (98b)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 129  | ■0.947 | $\begin{aligned} N_{(j)}(t) &= n_j(t) + Q_j, \quad (j \leq 4) \triangleq (n, m, p, \sigma), \\ Q_n &= 3 \cdot 2^{s-2}, \quad Q_m = 2^s - 1, \quad Q_p = 2^s + 2(-1)^k, \\ Q_\sigma &= 2^{s-1} - 1, \quad s = k^2 + 1 = \text{const}(t) \end{aligned}$                     | $\begin{aligned} N_{(j)}(t) &= n_j(t) + Q_j, \quad (j \leq 4) \triangleq (n, m, p, \sigma), \\ Q_n &= 3 \cdot 2^{s-2}, \quad Q_m = 2^s - 1, \quad Q_p = 2^s + 2(-1)^k, \\ Q_\sigma &= 2^{s-1} - 1, \quad s = k^2 + 1 = \text{const}(t) \end{aligned}$ | $\begin{aligned} N_{(j)}(t) &= n_j(t) + Q_j, \quad (j \leq 4) \triangleq (n, m, p, \sigma), \\ Q_n &= 3 \cdot 2^{s-2}, \quad Q_m = 2^s - 1, \quad Q_p = 2^s + 2(-1)^k, \\ Q_\sigma &= 2^{s-1} - 1, \quad s = k^2 + 1 = \text{const}(t) \end{aligned}$ |
| BH3FR_E00001                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 021  | ■0.948 | $S \, U(5) \, \text{mit} \, S \, U(3) \, \times S \, U(2) \, \times U(1) \subset S \, U(5)$                                                                                                                                                                               | $SU(5) \text{ mit } SU(3) \times SU(2) \times U(1) \subset SU(5).$                                                                                                                                                                                    | $SU(5) \text{ mit } SU(3) \times SU(2) \times U(1) \subset SU(5).$                                                                                                                                                                                    |
| BH3FR_E00005 (*b)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 100  | ■0.950 | $\operatorname{divgrad} \vec{p} + \alpha \beta \ddot{\vec{p}} = \vec{0}, \quad \omega^2 \alpha \beta = 1, \quad 0 < \omega < \infty$                                                                                                                                      | $\operatorname{divgrad} \vec{p} + \alpha \beta \ddot{\vec{p}} = \vec{0}, \quad \omega^2 \alpha \beta = 1, \quad 0 < \omega < \infty$                                                                                                                  | $\operatorname{divgrad} \vec{p} + \alpha \beta \ddot{\vec{p}} = \vec{0}, \quad \omega^2 \alpha \beta = 1, \quad 0 < \omega < \infty$                                                                                                                  |
| <b>Conf.</b> MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value<br><b>LaTeX source</b> MathPix's reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ<br><b>Residual</b> not shown for this document — no residual measurements are paired with these rows.<br><b>[refined: basis]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here. |      |        |                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                       |

| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Page | Conf.  | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Scan image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| BH3FR_EQ0157                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 119  | ■0.953 | $\begin{aligned} & \& K_1^{\--}(\alpha)=4\,d,\,1\,s,\,2\,s,\,3\,s,\,1\,e,\,2\,e \\ & \& K_2^{\--}(\beta)=5\,d,\,4\,s,\,5\,s,\,6\,s,\,1\,e,\,3\,e \\ & \& K_3^{\--}(\gamma)=6\,d,\,7\,s,\,8\,s,\,9\,s,\,2\,e,\,3\,e \\ & \& K_1^{\++}(\alpha)=1\,d,\,2\,d,\,3\,d,\,5\,d,\,6\,d,\,4\,s,\,5\,s,\,7\,s,\,8\,s,\,9\,s,\,3\,e \\ & \& K_2^{\++}(\beta)=1\,d,\,2\,d,\,3\,d,\,4\,d,\,6\,d,\,1\,s,\,2\,s,\,3\,s,\,7\,s,\,8\,s,\,9\,s,\,2\,e \\ & \& K_3^{\++}(\gamma)=1\,d,\,2\,d,\,3\,d,\,4\,d,\,5\,d,\,1\,s,\,2\,s,\,3\,s,\,4\,s,\,5\,s,\,6\,s,\,1\,e \\ & \& K_4^{\++}(1,2)=4\,d,\,5\,d,\,6\,d,\,3\,s,\,6\,s,\,9\,s,\,1\,e,\,2\,e,\,3\,e \\ & \& K_5^{\++}(1,3)=4\,d,\,5\,d,\,6\,d,\,2\,s,\,5\,s,\,8\,s,\,1\,e,\,2\,e,\,3\,e \\ & \& K_6^{\++}(2,3)=4\,d,\,5\,d,\,6\,d,\,1\,s,\,4\,s,\,7\,s,\,1\,e,\,2\,e,\,3\,e \\ & \& K_7^{\++}(1,\alpha)=3\,d,\,5\,d,\,6\,d,\,5\,s,\,6\,s,\,8\,s,\,9\,s,\,3\,e \\ & \& K_8^{\++}(2,\alpha)=2\,d,\,5\,d,\,6\,d,\,4\,s,\,6\,s,\,7\,s,\,9\,s,\,3\,e \\ & \& K_9^{\++}(3,\alpha)=1\,d,\,5\,d,\,6\,d,\,4\,s,\,5\,s,\,7\,s,\,8\,s,\,3\,e \\ & \& K_{10}^{\++}(1,\beta)=3\,d,\,4\,d,\,6\,d,\,2\,s,\,3\,s,\,8\,s,\,9\,s,\,2\,e \\ & \& K_{11}^{\++}(2,\beta)=2\,d,\,4\,d,\,6\,d,\,1\,s,\,3\,s,\,7\,s,\,9\,s,\,2\,e \\ & \& K_{12}^{\++}(3,\beta)=1\,d,\,4\,d,\,6\,d,\,1\,s,\,2\,s,\,7\,s,\,8\,s,\,2\,e \\ & \& K_{13}^{\++}(1,\gamma)=3\,d,\,4\,d,\,5\,d,\,2\,s,\,3\,s,\,5\,s,\,6\,s,\,1\,e \end{aligned}$ | $\begin{aligned} & K_1^{\--}(\alpha)=4d,1s,2s,3s,1e,2e \\ & K_2^{\--}(\beta)=5d,4s,5s,6s,1e,3e \\ & K_3^{\--}(\gamma)=6d,7s,8s,9s,2e,3e \\ & K_1^{\++}(\alpha)=1d,2d,3d,5d,6d,4s,5s,6s,7s,8s,9s,3e \\ & K_2^{\++}(\beta)=1d,2d,3d,4d,6d,1s,2s,3s,7s,8s,9s,2e \\ & K_3^{\++}(\gamma)=1d,2d,3d,4d,5d,1s,2s,3s,4s,5s,6s,1e \\ & K_4^{\++}(1,2)=4d,5d,6d,3s,6s,9s,1e,2e,3e \\ & K_5^{\++}(1,3)=4d,5d,6d,2s,5s,8s,1e,2e,3e \\ & K_6^{\++}(2,3)=4d,5d,6d,1s,4s,7s,1e,2e,3e \\ & K_7^{\++}(1,\alpha)=3d,5d,6d,5s,6s,8s,9s,3e \\ & K_8^{\++}(2,\alpha)=2d,5d,6d,4s,6s,7s,9s,3e \\ & K_9^{\++}(3,\alpha)=1d,5d,6d,4s,5s,7s,8s,3e \\ & K_{10}^{\++}(1,\beta)=3d,4d,6d,2s,3s,8s,9s,2e \\ & K_{11}^{\++}(2,\beta)=2d,4d,6d,1s,3s,7s,9s,2e \\ & K_{12}^{\++}(3,\beta)=1d,4d,6d,1s,2s,7s,8s,2e \\ & K_{13}^{\++}(1,\gamma)=3d,4d,5d,2s,3s,5s,6s,1e \end{aligned}$ | $\begin{aligned} & K_1^{\--}(\alpha)=4d,1s,2s,3s,1e,2e \\ & K_2^{\--}(\beta)=5d,4s,5s,6s,1e,3e \\ & K_3^{\--}(\gamma)=6d,7s,8s,9s,2e,3e \\ & K_1^{\++}(\alpha)=1d,2d,3d,5d,6d,4s,5s,6s,7s,8s,9s,3e \\ & K_2^{\++}(\beta)=1d,2d,3d,4d,6d,1s,2s,3s,7s,8s,9s,2e \\ & K_3^{\++}(\gamma)=1d,2d,3d,4d,5d,1s,2s,3s,4s,5s,6s,1e \\ & K_4^{\++}(1,2)=4d,5d,6d,3s,6s,9s,1e,2e,3e \\ & K_5^{\++}(1,3)=4d,5d,6d,2s,5s,8s,1e,2e,3e \\ & K_6^{\++}(2,3)=4d,5d,6d,1s,4s,7s,1e,2e,3e \\ & K_7^{\++}(1,\alpha)=3d,5d,6d,5s,6s,8s,9s,3e \\ & K_8^{\++}(2,\alpha)=2d,5d,6d,4s,6s,7s,9s,3e \\ & K_9^{\++}(3,\alpha)=1d,5d,6d,4s,5s,7s,8s,3e \\ & K_{10}^{\++}(1,\beta)=3d,4d,6d,2s,3s,8s,9s,2e \\ & K_{11}^{\++}(2,\beta)=2d,4d,6d,1s,3s,7s,9s,2e \\ & K_{12}^{\++}(3,\beta)=1d,4d,6d,1s,2s,7s,8s,2e \\ & K_{13}^{\++}(1,\gamma)=3d,4d,5d,2s,3s,5s,6s,1e \end{aligned}$ |
| BH3FR_EQ0180 (83)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 124  | ■0.953 | $\begin{aligned} & \& E_N^{\pm}(\pm)=2^{N-2}\cdot(N!)^2, \\ & \& Z_N=2^{N-1}\cdot N!, \\ & \& S_N=N! \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $\begin{aligned} & E_N^{(\pm)}=2^{N-2}\cdot(N!)^2, \quad Z_N=2^{N-1}\cdot N! \\ & S_N=N! \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | $\begin{aligned} & E_N^{(\pm)}=2^{N-2}\cdot(N!)^2, \quad Z_N=2^{N-1}\cdot N!, \\ & S_N=N! \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| BH3FR_EQ0208 (96a)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 128  | ■0.962 | $\begin{aligned} & \underline{\varrho_n}=\underline{n}\,\underline{\varepsilon}, \quad \bar{\lambda}_0=\bar{\lambda}_0(k), \quad  \bar{\lambda} \underline{\varepsilon}=1, \quad \underline{n}\geq 1 \\ & \underline{\varrho_n}=\underline{n}\,\underline{\varepsilon}, \quad \bar{\lambda}_0=\bar{\lambda}_0(k), \quad  \bar{\lambda} \underline{\varepsilon}=1, \quad \underline{n}\geq 1 \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | $\underline{\varrho_n}=\underline{n}\,\underline{\varepsilon}, \quad \bar{\lambda}_0=\bar{\lambda}_0(k), \quad  \bar{\lambda} \underline{\varepsilon}=1, \quad \underline{n}\geq 1$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | $\underline{\varrho_n}=\underline{n}\,\underline{\varepsilon}, \quad \bar{\lambda}_0=\bar{\lambda}_0(k), \quad  \bar{\lambda} \underline{\varepsilon}=1, \quad \underline{n}\geq 1$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <p><b>Conf.</b> MathPix confidence: ■≥0.9 ■0.5-0.9 ■&lt;0.5 — no value</p> <p><b>LaTeX source</b> MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ</p> <p><b>Residual</b> not shown for this document — no residual measurements are paired with these rows.</p> <p><b>[refined: <i>basis</i>]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</p> |      |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

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| BH3FR_EQ0212 (98)   | 129  | ■0.964 | $\begin{aligned} & \text{e}_{\varrho}=\varepsilon_{\pm}\sqrt{\eta_{qk}},\quad\quad\quad 2\text{e}_{\omega}=\varepsilon_{\pm}(1+\sqrt{\eta_{qk}}),\\ & \text{e}_{\delta}=\varepsilon_{\pm}(1-\sqrt{\eta_{qk}}),\quad\quad\quad \text{e}_C=\varepsilon_{\pm}\sqrt{\vartheta_{qk}/8},\\ & \vartheta_{qk}=5\eta_{qk}+2\sqrt{\eta_{qk}}+1,\quad\quad\quad \eta_{qk}\sqrt[4]{\pi^4+q^4(4+k)}=\pi \end{aligned}$ |                                                                                                                                                                |                                                                                                                                                                |
| BH3FR_EQ0151 (68a)  | 118  | ■0.965 | $\begin{aligned} & \underline{\partial} Q_{(\mu\nu)}^{(\kappa\lambda)}=0,\quad \underline{\partial}\psi_{ll}=0,\quad \underline{\partial}^2\left({}^2\tilde{\gamma}_{(\kappa\lambda)}\right)={}^2\overline{0},\\ & \underline{\partial}^2\left({}^2\tilde{\gamma}_{(\mu\nu)}\right)={}^2\overline{0} \end{aligned}$                                                                                       |                                                                                                                                                                |                                                                                                                                                                |
| BH3FR_EQ0324 (M31a) | 148  | ■0.966 | $\{ \}^{\{4\}}\bar{\{\zeta\}}=K\;;[]$                                                                                                                                                                                                                                                                                                                                                                     | ${}^4\overline{\zeta}=K;[]$                                                                                                                                    | ${}^4\overline{\zeta}=K;[]$                                                                                                                                    |
| BH3FR_EQ0063 (21a)  | 106  | ■0.967 | $\left[\begin{array}{cc} \tilde{r} & \\ k & l \end{array}\right]=B_{\bar{k}l}^{\tilde{r}}\cdot\exp\lambda_{kl}\left(\mu-\mathbf{S}\psi_{kl}\partial\mu\right)$                                                                                                                                                                                                                                            | $\left[\begin{array}{cc} \tilde{r} & \\ k & l \end{array}\right]=B_{\bar{k}l}^{\tilde{r}}\cdot\exp\lambda_{kl}\left(\mu-\mathbf{S}\psi_{kl}\partial\mu\right)$ | $\left[\begin{array}{cc} \tilde{r} & \\ k & l \end{array}\right]=B_{\bar{k}l}^{\tilde{r}}\cdot\exp\lambda_{kl}\left(\mu-\mathbf{S}\psi_{kl}\partial\mu\right)$ |
| BH3FR_EQ0201 (92)   | 127  | ■0.967 | $\begin{aligned} & K\left(R_4^{\pm}\right)=K^{\pm},\quad\cos\left(\vec{Y}^{+},\vec{Y}^{-}\right)=-1,\quad\bar{\lambda}^{\pm}\perp\vec{Y}^{\pm},\\ & \bar{\lambda}^{+}\ \bar{\lambda}^{-},\quad a^{+}(\omega)=a^{-}(\omega),\quad b^{+}=b^{-},\\ & m^{\pm}>0 \end{aligned}$                                                                                                                                |                                                                                                                                                                |                                                                                                                                                                |
| BH3FR_EQ0045 (16)   | 104  | ■0.968 | $\begin{aligned} & \bar{x}_k=\bar{X}_k;n,\quad\bar{X}_k=\bar{e}_k\alpha_k(\,)_k,\quad(\bar{e}_i\bar{e}_k)_6=\hat{E},\\ & \sum_{k=1}^6v_{kk}=1,\quad\alpha_k=\sqrt{\tau},\quad k\leqslant3,\quad\alpha_k=i\sqrt{\tau},\\ & k>3,\quad\Delta=\tau^3 \end{aligned}$                                                                                                                                           |                                                                                                                                                                |                                                                                                                                                                |
| BH3FR_EQ0034 (12)   | 103  | ■0.968 | $Y_{1\varsigma}\left(1-h(2Y_1m_0c\varsigma)^{-1}\right)^2=\frac{h^2}{\gamma m_0^3}.\quad 2Y_1m_0c\varsigma>h$                                                                                                                                                                                                                                                                                             |                                                                                                                                                                |                                                                                                                                                                |



| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Page | Conf.  | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Rendered                                                                                                                                                                                                                                                                                                                    | Scan image                                                                                                                                                                                                                                                                                                      |
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| BH3FR_E00281 (M9a)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 140  | ■0.977 | $\begin{array}{ll} \varphi=\varphi\left(\psi_1,\ldots\psi_\lambda\right), & \psi_j=\psi_j\left(n_1\ldots n_L\right), \\ \psi_{\left\{\lambda\right\}}=\psi_{\left\{j\right\}}\left(n_{\left\{1\right\}}\ldots n_{\left\{L\right\}}\right), & \partial\varphi=\sum_{j=1}^{\lambda}\partial\psi_j\varphi\partial\psi_j, \\ \partial\varphi=\sum_{j=1}^{\lambda}\partial\psi_j\varphi\partial\psi_j, & \partial\psi_j\varphi=\left(\varphi-\varphi\left(\ldots,\psi_j-\partial\psi_j,\ldots\right)\right)\left(\partial\psi_j\right)^{-1} \end{array}$ | $\varphi=\varphi\left(\psi_1,\ldots\psi_\lambda\right), \quad \psi_j=\psi_j\left(n_1\ldots n_L\right),$ $\partial\varphi=\sum_{j=1}^{\lambda}\partial\psi_j\varphi\partial\psi_j, \quad \partial\psi_j\varphi=\left(\varphi-\varphi\left(\ldots,\psi_j-\partial\psi_j,\ldots\right)\right)\left(\partial\psi_j\right)^{-1}$ | $\varphi=\varphi\left(\psi_1,\ldots\psi_\lambda\right), \quad \psi_j=\psi_j\left(n_1\ldots n_L\right),$ $\delta\varphi=\sum_{j=1}^{\lambda}\delta\psi_j\varphi\delta\psi_j, \quad \delta\psi_j\varphi=\left(\varphi-\varphi\left(\ldots,\psi_j-\delta\psi_j,\ldots\right)\right)\left(\delta\psi_j\right)^{-1}$ |
| BH3FR_E00267 (M1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 138  | ■0.980 | $\begin{aligned} & y=f(x), \quad \lim_{\Delta F_v \rightarrow \tau} \sum_{v=1}^n \tilde{y}_v \Delta x_v = n \tau, \\ & \Delta F_v = \tilde{y}_v \Delta x_v = \int_{x_{v-1}}^{x_v} y dx, \quad \sum_{v=1}^n \int_{v-1}^v = \int_0^n, \\ & \int_{x_0}^{x_n} f(x) dx = n \tau, \quad x_n = x(n), \quad y_n = f(n) \end{aligned}$                                                                                                                                                                                                                       | $y=f(x), \quad \lim_{\Delta F_v \rightarrow \tau} \sum_{v=1}^n \tilde{y}_v \Delta x_v = n \tau,$ $\Delta F_v = \tilde{y}_v \Delta x_v = \int_{x_{v-1}}^{x_v} y dx, \quad \sum_{v=1}^n \int_{v-1}^v = \int_0^n,$ $\int_{x_0}^{x_n} f(x) dx = n \tau, \quad x_n = x(n), \quad y_n = f(n)$                                     | $y=f(x), \quad \lim_{\Delta F_v \rightarrow \tau} \sum_{v=1}^n \tilde{y}_v \Delta x_v = n \tau,$ $\Delta F_v = \tilde{y}_v \Delta x_v = \int_{x_{v-1}}^{x_v} y dx, \quad \sum_{v=1}^n \int_{v-1}^v = \int_0^n,$ $\int_{x_0}^{x_n} f(x) dx = n \tau, \quad x_n = x(n), \quad y_n = f(n)$                         |
| BH3FR_E00064 (21b)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 106  | ■0.981 | $\gamma_{kl}^{(w)} \equiv \psi_{kl}^{(\min)} = 0, \quad \psi_{kl}^{(\max)} = E$                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $\gamma_{kl}^{(w)} \equiv \psi_{kl}^{(\min)} = 0, \quad \psi_{kl}^{(\max)} = E$                                                                                                                                                                                                                                             | $\gamma_{kl}^{(w)} \equiv \psi_{kl}^{(\min)} = 0, \quad \psi_{kl}^{(\max)} = E$                                                                                                                                                                                                                                 |
| BH3FR_E00235 (107)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 132  | ■0.982 | $\partial_j G_j > G_{j+1}, \quad \partial_j G_j \geq \partial_{j+1} G_{j+1}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | $\partial_j G_j > G_{j+1}, \quad \partial_j G_j \geq \partial_{j+1} G_{j+1}$                                                                                                                                                                                                                                                | $\delta_j G_j > G_{j+1}, \quad \delta_j G_j \geq \delta_{j+1} G_{j+1}$                                                                                                                                                                                                                                          |
| BH3FR_E00076 (27a)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 108  | ■0.984 | $m(n)=\mu\,\frac{\sqrt[4]{2n}}{\sqrt{2n-1}}, \quad \mu=\sqrt{ch/\gamma}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $m(n)=\mu\,\frac{\sqrt[4]{2n}}{\sqrt{2n-1}}, \quad \mu=\sqrt{ch/\gamma}$                                                                                                                                                                                                                                                    | $m(n)=\mu\,\frac{\sqrt[4]{2n}}{\sqrt{2n-1}}, \quad \mu=\sqrt{ch/\gamma}$                                                                                                                                                                                                                                        |
| BH3FR_E00308 (M23a)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 145  | ■0.984 | $\{ \}^{\{2\}} \bar{K} = \{ \}^{\{2\}} \bar{\kappa} ; n$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $^2\bar{K} = ^2\bar{\kappa}; n$                                                                                                                                                                                                                                                                                             | $^2\bar{K} = ^2\bar{\kappa}; n$                                                                                                                                                                                                                                                                                 |
| BH3FR_E00182 (83b)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 125  | ■0.987 | $Q_{\{x\}^{\{k\}}=n_{\{x\}^{\{k\}}}\cdot(\pm k)_x\sim\bar{\lambda}, \quad n_x^{(k)}\geq 1$                                                                                                                                                                                                                                                                                                                                                                                                                                                          | $Q_x^{(k)}=n_x^{(k)}\cdot(\pm k)_x\sim\bar{\lambda}, \quad n_x^{(k)}\geq 1$                                                                                                                                                                                                                                                 | $Q_x^{(k)}=n_x^{(k)}\cdot(\pm k)_x\sim\bar{\lambda}, \quad n_x^{(k)}\geq 1$                                                                                                                                                                                                                                     |
| BH3FR_E00190 (86b)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 126  | ■0.987 | $\begin{array}{ll} (\pm p)-(\mu)-(\pm q), & (\pm p)-(\mu V_\mu^x)-(\pm k)-(\mu V_\mu^y)-(\pm q), \\ \left(\mu V_{\{\mu\}^{\{x\}}\right)-(\pm k)-\left(\mu V_{\{\mu\}^{\{y\}}\right)-(\pm q), & \left(\mu V_{\mu'}^x V_{\mu'}^y\right), \\ \left(V_{\{\mu\}^{\{x\}}\right), & I(k)>\frac{1}{2}[I(p)+I(q)] \end{array}$                                                                                                                                                                                                                               | $\begin{array}{ll} (\pm p)-(\mu)-(\pm q), & (\pm p)-(\mu V_\mu^x)-(\pm k)-(\mu V_\mu^y)-(\pm q), \\ \left(\mu V_{\mu'}^x V_{\mu'}^y\right), & I(k)>\frac{1}{2}[I(p)+I(q)] \end{array}$                                                                                                                                      | $\begin{array}{ll} (\pm p)-(\mu)-(\pm q), & (\pm p)-(\mu V_\mu^x)-(\pm k)-(\mu V_\mu^y)-(\pm q), \\ \left(\mu V_{\mu'}^x V_{\mu'}^y\right), & I(k)>\frac{1}{2}[I(p)+I(q)] \end{array}$                                                                                                                          |
| <p><b>Conf.</b> MathPix confidence: ■≥0.9 ■0.5–0.9 ■&lt;0.5 — no value</p> <p><b>LaTeX source</b> MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ</p> <p><b>Residual</b> not shown for this document — no residual measurements are paired with these rows.</p> <p><b>[refined: <i>basis</i>]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</p> |      |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                 |

| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Page | Conf.  | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Scan image                                                                                                                                                                                                                                                                         |
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| BH3FR_EQ0247 (110c)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 135  | ■0.987 | $\begin{aligned} & (2\ \beta\!-\!\alpha)\ A_{41}\!=\!\beta\!\left(\!\xi\!\left(2\!+\!(\xi\alpha)^2\right)\!-\!2\beta\right)Y_{24},\ \& \\ & \& 2\ A_{42}\!=\!\pi\ \xi^2\eta(\beta\!-\!3\alpha)Y_{25},\ \& 2\ A_{43}\!=\!\xi Y_{26}, \\ & 2\ A_{43}\!=\!\xi\ Y_{26},\ \& \& A_{44}\!=\!2\!\left(\frac{\eta}{\xi}\right)^2Y_{27},\ \& 6\xi A_{45}\!=\!(3\beta\!-\!\alpha)Y_{28}, \\ & \& \& A_{46}\!=\!\left(\frac{\pi e}{\xi\eta}\!-\!e\eta^2\alpha/2\right)Y_{29},\ \& A_{51}\!=\!(2\alpha\!+\!1)^2Y_{30}, \\ & \eta^2A_{52}\!=\!6\alpha Y_{31},\ \& A_{53}\!=\!\left(\frac{\xi}{\eta}\right)^3Y_{32},\ \& A_{54}\sqrt{2}\!=\!\alpha(\beta\!-\! \\ & \!-\!\alpha)\sqrt{3}Y_{33},\ \& A_{55}\!=\!\xi^3Y_{34},\ \& A_{56}\!=\!\left(\frac{\xi}{\eta}\right)^4Y_{35}, \\ & 12\beta A_{61}\!=\!\pi\xi(2\beta\!-\!\alpha)Y_{36},\ \& 12A_{62}\!=\!\pi^2(\beta\!-\!2\alpha)Y_{37}, \\ & 9A_{63}\!=\!\sqrt{\eta}Y_{38},\ \& 3\eta A_{64}\!=\!\pi Y_{39},\ \& 3\xi A_{65}\!=\!\pi Y_{40} \end{aligned}$ | $\begin{aligned} & (2\beta-\alpha)A_{41}=\beta\left(\xi\left(2+(\xi\alpha)^2\right)-2\beta\right)Y_{24}, \\ & 2A_{42}=\pi\xi^2\eta(\beta-3\alpha)Y_{25},\quad 2A_{43}=\xi Y_{26}, \\ & A_{44}=2\left(\frac{\eta}{\xi}\right)^2Y_{27},\quad 6\xi A_{45}=(3\beta-\alpha)Y_{28}, \\ & A_{46}=\left(\frac{\pi e}{\xi\eta}-e\eta^2\alpha/2\right)Y_{29},\quad A_{51}=(2\alpha+1)^2Y_{30}, \\ & \eta^2A_{52}=6\alpha Y_{31},\quad A_{53}=\left(\frac{\xi}{\eta}\right)^3Y_{32},\quad A_{54}\sqrt{2}=\alpha(\beta- \\ & -\alpha)\sqrt{3}Y_{33},\quad A_{55}=\xi^3Y_{34},\quad A_{56}=\left(\frac{\xi}{\eta}\right)^4Y_{35}, \\ & 12\beta A_{61}=\pi\xi(2\beta-\alpha)Y_{36},\quad 12A_{62}=\pi^2(\beta-2\alpha)Y_{37}, \\ & 9A_{63}=\sqrt{\eta}Y_{38},\quad 3\eta A_{64}=\pi Y_{39},\quad 3\xi A_{65}=\pi Y_{40} \end{aligned}$ |                                                                                                                                                                                                                                                                                    |
| BH3FR_EQ0283 (M11)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 140  | ■0.989 | $\begin{aligned} & \& Z(i)=(\ )_i,\quad \varphi(n_i)_1^L=\phi;n,\quad \phi=\phi(C_k,Z(i))_{i,k=1}^{L,K} \\ & C_k;n_i=f_k(n_i) \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $\begin{aligned} & Z(i)=(\ )_i,\quad \varphi(n_i)_1^L=\phi;n,\quad \phi=\phi(C_k,Z(i))_{i,k=1}^{L,K} \\ & C_k;n_i=f_k(n_i) \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | $\begin{aligned} & Z(i)=(\ )_i,\quad \varphi(n_i)_1^L=\phi;n,\quad \phi=\phi(C_k,Z(i))_{i,k=1}^{L,K} \\ & C_k;n_i=f_k(n_i) \end{aligned}$                                                                                                                                          |
| BH3FR_EQ0211 (97a)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 128  | ■0.989 | $\{ \ }^{\{2\}}\bar{P}_{\{(\kappa\lambda)\}}^{\{(\mu\nu)\}};N=\{ \ }^{\{2\}}\bar{A}_{\{(\kappa\lambda)\}}^{\{(\mu\nu)\}}+i^{\{2\}}\bar{F}_{\{(\kappa\lambda)\}}^{\{(\mu\nu)\}};N,\quad \{ \ }^{\{2\}}\bar{A}_{\{(\kappa\lambda)\}}^{\{(\mu\nu)\}}=\operatorname{const}\left(R_6\right)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | ${}^2\bar{P}_{(\kappa\lambda)}^{(\mu\nu)};N={}^2\bar{A}_{(\kappa\lambda)}^{(\mu\nu)}+i^2\bar{F}_{(\kappa\lambda)}^{(\mu\nu)};N,\quad {}^2\bar{A}_{(\kappa\lambda)}^{(\mu\nu)}=\operatorname{const}(R_6)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | ${}^2\bar{P}_{(\kappa\lambda)}^{(\mu\nu)};N={}^2\bar{A}_{(\kappa\lambda)}^{(\mu\nu)}+i^2\bar{F}_{(\kappa\lambda)}^{(\mu\nu)};N,\quad {}^2\bar{A}_{(\kappa\lambda)}^{(\mu\nu)}=\operatorname{const}(R_6)$                                                                           |
| BH3FR_EQ0280 (M9)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 140  | ■0.990 | $\begin{array}{l} \varphi=\varphi(n_1\dots n_L)=\varphi(n_i)_1^L,\quad 1\leqslant\kappa_i\leqslant n_i\leqslant N_i<\infty, \\ \partial_i\varphi=\varphi-\varphi(\dots,n_i-1,\dots),\quad \partial=\sum_{i=1}^L\partial_i, \\ (\partial_i\times\partial_k)_-=0 \end{array}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $\begin{aligned} & \varphi=\varphi(n_1\dots n_L)=\varphi(n_i)_1^L,\quad 1\leqslant\kappa_i\leqslant n_i\leqslant N_i<\infty, \\ & \partial_i\varphi=\varphi-\varphi(\dots,n_i-1,\dots),\quad \partial=\sum_{i=1}^L\partial_i, \\ & (\partial_i\times\partial_k)_-=0 \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | $\begin{aligned} & \varphi=\varphi(n_1\dots n_L)=\varphi(n_i)_1^L,\quad 1\leqslant\kappa_i\leqslant n_i\leqslant N_i<\infty, \\ & \partial_i\varphi=\varphi-\varphi(\dots,n_i-1,\dots),\quad \partial=\sum_{i=1}^L\partial_i, \\ & (\partial_i\times\partial_k)_-=0 \end{aligned}$ |
| <div>Conf. MathPix confidence: <span>■</span>≥0.9 <span>■</span>0.5-0.9 <span>■</span>&lt;0.5 — no value</div> <div>LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ</div> <div>Residual not shown for this document — no residual measurements are paired with these rows.</div> <div>[refined: <b>basis</b>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <b>basis</b> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</div> |      |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                    |

| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Page | Conf.  | LaTeX source                                                                                                                                                                                                                                                                                                                              | Rendered | Scan image                                                                                                                                                                                                                                                                                                                                |
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| BH3FR_EQ0067 (22a)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 107  | ■0.991 | $\begin{aligned} & \tau \rightarrow 0 \psi_{kl}(x_s)_1^q = (1 + C_{kl} e^{\pm i \lambda_{kl} y})^{-1}, \quad y^2 = \varsigma^2 - r^2, \\ & -\varsigma^2 = \sum_{s=4}^q x_s^2, \quad r^2 = \sum_{i=1}^3 x_i^2 \end{aligned}$                                                                                                               |          |                                                                                                                                                                                                                                                                                                                                           |
| BH3FR_EQ0023 (8b)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 102  | ■0.992 | $\vec{\lambda} \parallel \widehat{sp\{\}} \rightarrow \operatorname{grad}_6 \ln w$                                                                                                                                                                                                                                                        |          | $\vec{\lambda} \parallel sp\{\} \rightarrow \operatorname{grad}_6 \ln w$                                                                                                                                                                                                                                                                  |
| BH3FR_EQ00167 (76a)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 122  | ■0.993 | $F_{\{v\}} \equiv \binom{\alpha \beta}{\kappa \lambda} F(q_j)_v \omega$                                                                                                                                                                                                                                                                   |          | $F_v \equiv \binom{\alpha \beta}{\kappa \lambda} F(q_j)_v \omega$                                                                                                                                                                                                                                                                         |
| BH3FR_EQ0004 (*a)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 100  | ■0.993 | $\operatorname{rot}\{\vec{G}\} \sim \beta \dot{\vec{\mu}} + (\sigma - \sigma_{(0)}) \vec{f}, \quad \operatorname{div} \vec{\mu} \sim \sigma - \sigma_{(0)}, \quad \beta \neq 0$                                                                                                                                                           |          | $\operatorname{rot} \vec{G} \sim \beta \dot{\vec{\mu}} + (\sigma - \sigma_{(0)}) \vec{f}, \quad \operatorname{div} \vec{\mu} \sim \sigma - \sigma_{(0)}, \quad \beta \neq 0$                                                                                                                                                              |
| BH3FR_EQ0039 (14a)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 103  | ■0.994 | $R_{\{+ \}=2 \alpha \varsigma}, \quad 2 \alpha R_{\{- \}}=\varsigma, \quad R_{+} R_{-}=\varsigma^2$                                                                                                                                                                                                                                       |          | $R_{+}=2 \alpha \varsigma, \quad 2 \alpha R_{-}=\varsigma, \quad R_{+} R_{-}=\varsigma^2$                                                                                                                                                                                                                                                 |
| BH3FR_EQ00200 (91a)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 127  | ■0.995 | $\begin{aligned} & a_g \hat{=} a(R_4), \quad P(a_g) = P(b) = 0, \quad J_T(a_g) = 2, \\ & J_T(b) = 1 \end{aligned}$                                                                                                                                                                                                                        |          | $\begin{aligned} & a_g \hat{=} a(R_4), \quad P(a_g) = P(b) = 0, \quad J_T(a_g) = 2, \\ & J_T(b) = 1 \end{aligned}$                                                                                                                                                                                                                        |
| BH3FR_EQ00090 (35a)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 110  | ■0.995 | $\begin{aligned} & \vartheta = \frac{3b}{4c} \tau^{5/6}, \quad 2 \omega c^2 \tau = \gamma h, \quad \gamma \mu^2 = ch, \\ & \pi^4 \kappa^2 R_{-} = 9 \hbar \end{aligned}$                                                                                                                                                                  |          | $\begin{aligned} & \vartheta = \frac{3b}{4c} \tau^{5/6}, \quad 2 \omega c^2 \tau = \gamma h, \quad \gamma \mu^2 = ch, \\ & \pi^4 \kappa^2 R_{-} = 9 \hbar \end{aligned}$                                                                                                                                                                  |
| BH3FR_EQ00188 (86)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 126  | ■0.996 | $\begin{aligned} & \bar{S}[(\pm p)_x - (\mu) - (\pm q)_x] = \frac{p(\mu)}{x} \pm, \\ & \cos[(\mu)_+, (\mu)_-] = -1 \end{aligned}$                                                                                                                                                                                                         |          | $\begin{aligned} & \bar{S}[(\pm p)_x - (\mu) - (\pm q)_x] = \frac{p(\mu)}{x} \pm, \\ & \cos[(\mu)_+, (\mu)_-] = -1 \end{aligned}$                                                                                                                                                                                                         |
| BH3FR_EQ00144 (64)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 117  | ■0.996 | $\underline{\psi}_{kl} = \psi_{kl}^{\alpha_{kl}}, \quad \underline{\lambda}_{kl} = \alpha_{kl} \lambda_{kl}$                                                                                                                                                                                                                              |          | $\underline{\psi}_{kl} = \psi_{kl}^{\alpha_{kl}}, \quad \underline{\lambda}_{kl} = \alpha_{kl} \lambda_{kl}$                                                                                                                                                                                                                              |
| BH3FR_EQ00245 (110a)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 134  | ■0.996 | $\begin{aligned} & ka(vx) = A_{41}(1 + a_n a_q), \quad a_n = PA_{42}[1 - \kappa A_{43}(1 + \\ & + A_{44}(-\alpha)^{2-k} A_{45}^{k-1})(1 - \kappa Q A_{46}(2 - k)) - A_{51}(k - 1)(1 - \kappa)], \\ & a_q = 1 - q A_{52}(1 - 2 \kappa A_{53}^k) \left(1 + \frac{q \kappa}{6}(3 - \varepsilon q_x)(k - 1)(1 - \kappa)\right) \end{aligned}$ |          | $\begin{aligned} & ka(vx) = A_{41}(1 + a_n a_q), \quad a_n = PA_{42}[1 - \kappa A_{43}(1 + \\ & + A_{44}(-\alpha)^{2-k} A_{45}^{k-1})(1 - \kappa Q A_{46}(2 - k)) - A_{51}(k - 1)(1 - \kappa)], \\ & a_q = 1 - q A_{52}(1 - 2 \kappa A_{53}^k) \left(1 + \frac{q \kappa}{6}(3 - \varepsilon q_x)(k - 1)(1 - \kappa)\right) \end{aligned}$ |
| <p><b>Conf.</b> MathPix confidence: ■≥0.9 ■0.5–0.9 ■&lt;0.5 — no value</p> <p><b>LaTeX source</b> MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ</p> <p><b>Residual</b> not shown for this document — no residual measurements are paired with these rows.</p> <p><b>[refined: <i>basis</i>]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</p> |      |        |                                                                                                                                                                                                                                                                                                                                           |          |                                                                                                                                                                                                                                                                                                                                           |

| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Page | Conf.  | LaTeX source                                                                                                                                                                                                                                                                                                                                                                     | Rendered                                                                                                                             | Scan image                                                                                                                                                                    |
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| BH3FR_EQ0091 (36)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 110  | ■0.996 | $\begin{aligned} & h=b\sqrt{3/2}\frac{\mu}{\vartheta}\tau^{4/3},\quad\quad\gamma=\frac{3}{4}\sqrt{3/2}(b\tau/\vartheta)^2\cdot\frac{\sqrt[6]{\tau}}{\mu}\\ & \varepsilon_o=(\pi\kappa)^2\left(\frac{2}{3}\pi\right)^3\sqrt{2/3}\left(\frac{\vartheta}{b\tau}\right)^2(\mu\sqrt[6]{\tau})^{-1},\\ & \mu_o=\frac{6}{\pi^5}\sqrt{3/2}\frac{\mu\sqrt{\tau}}{\kappa^2} \end{aligned}$ |                                                                                                                                      |                                                                                                                                                                               |
| BH3FR_EQ0062 (21)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 106  | ■0.996 | $\sum_{k=1}^q\gamma_{ik}=A_i\exp\left(\sum_{k,l=1}^qc_{kl}^{(i)}\boldsymbol{S}\boldsymbol{\psi}_{kl}\partial\mu\right),\quad c_{kl}^{(i)}b_i^{(kl)}=2\lambda(kl).$                                                                                                                                                                                                               |                                                                                                                                      | $\sum_{k=1}^q\gamma_{ik}=A_i\exp\left(\sum_{k,l=1}^qc_{kl}^{(i)}\boldsymbol{S}\boldsymbol{\psi}_{kl}\boldsymbol{\delta}\mu\right),\quad c_{kl}^{(i)}b_i^{(kl)}=2\lambda(kl).$ |
| BH3FR_EQ0164 (74a)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 122  | ■0.997 | $s\,p()_{(-)}^{\{ (6,6) \}};\,\left( ()-\frac{1}{2}{}^2\overline{\gamma} sp() \right); sp^4\overline{\varrho}=\overline{0}$                                                                                                                                                                                                                                                      | $sp()_{(-)}^{(6,6)};\left( ()-\frac{1}{2}{}^2\overline{\gamma} sp() \right); sp^4\overline{\varrho}=\overline{0}$                    | $sp()_{(-)}^{(6,6)};\left( ()-\frac{1}{2}{}^2\overline{\gamma} sp() \right); sp^4\overline{\varrho}=\overline{0}$                                                             |
| BH3FR_EQ0086 (33)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 110  | ■0.997 | $m_{\{c\}}\!\!\left(1\!-\!\eta_{\!q}\right)\!>\!q\,m_{\!e}$                                                                                                                                                                                                                                                                                                                      | $m_c(1-\eta_q)>qm_e$                                                                                                                 | $m_c(1-\eta_q)>qm_e$                                                                                                                                                          |
| BH3FR_EQ0030 (11)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 102  | ■0.998 | $\begin{aligned} & r\,q\,e^{-q}=A(1-r/\varsigma)^2,\quad 16c^2A=3\gamma M_o,\quad M=Lm,\\ & 1\leqslant L<\infty \end{aligned}$                                                                                                                                                                                                                                                   | $r q e^{-q}=A(1-r/\varsigma)^2,\quad 16c^2A=3\gamma M_o,\quad M=Lm$ $1\leqslant L<\infty$                                            | $r q e^{-q}=A(1-r/\varsigma)^2,\quad 16c^2A=3\gamma M_o,\quad M=Lm,$ $1\leqslant L<\infty$                                                                                    |
| BH3FR_EQ0119 (50a)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 113  | ■0.998 | $T_{\{0\}}\!\!\left(D_{\{f\,m\,p\}}\right)\!=\!0\,\leqq\,t\,\leqq\,\theta\!\!\left(D_{\{p\,m\,f\}}\right)\!=\!2\,T_A\!\!<\!\infty$                                                                                                                                                                                                                                               | $T_0\left(D_{fmp}\right)=0\leq t\leq \theta\left(D_{pmf}\right)=2T_A<\infty$                                                         | $T_0\left(D_{fmp}\right)=0\leq t\leq \theta\left(D_{pmf}\right)=2T_A<\infty$                                                                                                  |
| BH3FR_EQ0181 (83a)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 125  | ■0.998 | $\begin{aligned} & N_x^{(\pm)}=2^y\cdot\prod_{k=1}^n(N_k!)^2,\quad y=\sum_{k=1}^nN_k-2n\\ & n=n(x)\leq 6 \end{aligned}$                                                                                                                                                                                                                                                          | $N_x^{(\pm)}=2^y\cdot\prod_{k=1}^n(N_k!)^2,\quad y=\sum_{k=1}^nN_k-2n$ $n=n(x)\leq 6$                                                | $N_x^{(\pm)}=2^y\cdot\prod_{k=1}^n(N_k!)^2,\quad y=\sum_{k=1}^nN_k-2n,$ $n=n(x)\leq 6$                                                                                        |
| BH3FR_EQ0268 (M2)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 138  | ■0.998 | $\partial\varphi=\varphi(n)-\varphi(n-1),\quad 1\leq n\leq N$                                                                                                                                                                                                                                                                                                                    | $\partial\varphi=\varphi(n)-\varphi(n-1),\quad 1\leq n\leq N$                                                                        | $\delta\varphi=\varphi(n)-\varphi(n-1),\quad 1\leq n\leq N$                                                                                                                   |
| BH3FR_EQ0037 (13)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 103  | ■0.998 | $\begin{array}{lcr} \varphi(r<\varrho)>O, & G(r<\varrho)<O, & \varphi(\varrho)=O, \\ G(\varrho)=O, & \varphi(r>\varrho)>O, & G(r>\varrho)>O \end{array}$                                                                                                                                                                                                                         | $\varphi(r<\varrho)>O,\quad G(r<\varrho)<O,\quad \varphi(\varrho)=O,$ $G(\varrho)=O,\quad \varphi(r>\varrho)>O,\quad G(r>\varrho)>O$ | $\varphi(r<\varrho)>O,\quad G(r<\varrho)<O,\quad \varphi(\varrho)=O,$ $G(\varrho)=O,\quad \varphi(r>\varrho)>O,\quad G(r>\varrho)>O$                                          |
| BH3FR_EQ0055 (19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 105  | ■0.998 | $L\,;\,\widehat{\big[\big]}=\overline{{}^40}$                                                                                                                                                                                                                                                                                                                                    | $L;\widehat{\big[\big]}=\overline{{}^40}$                                                                                            | $L;\widehat{\big[\big]}=\overline{{}^40}$                                                                                                                                     |
| BH3FR_EQ0221 (100a)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 130  | ■0.998 | $\begin{aligned} & 2\,q_x=(P-2\,x)(1-\kappa\,Q(2-k))+\\ & +\varepsilon(k-1-(1+\kappa)\,Q(2-k))+C,\quad 0\leqq x\leqq P,\\ &  q_x =q \end{aligned}$                                                                                                                                                                                                                               | $2q_x=(P-2x)(1-\kappa Q(2-k))+$ $+\varepsilon(k-1-(1+\kappa)Q(2-k))+C,\quad 0\leq x\leq P,$ $ q_x =q$                                | $2q_x=(P-2x)(1-\kappa Q(2-k))+$ $+\varepsilon(k-1-(1+\kappa)Q(2-k))+C,\quad 0\leq x\leq P,$ $ q_x =q$                                                                         |
| <p><b>Conf.</b> MathPix confidence: ■<math>\geq 0.9</math> ■0.5–0.9 ■<math>&lt; 0.5</math> — no value</p> <p><b>LaTeX source</b> MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ</p> <p><b>Residual</b> not shown for this document — no residual measurements are paired with these rows.</p> <p><b>[refined: <i>basis</i>]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</p> |      |        |                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                      |                                                                                                                                                                               |

| Identifier          | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Rendered                                                                                                                                                                                                                                                                                                                        | Scan image                                                                                                                                                                                                                                                                                                        |
|---------------------|------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BH3FR_EQ0237 (107b) | 132  | ■ 0.998 | $\backslash\mathrm{beta}_{\{4\}}=\backslash\mathrm{alpha}_{\{3\}}\ N_{\{(3)\}}\text{-}N_{\{(4)\}}>0,\ \backslash\mathrm{quad}\ \backslash\mathrm{mu}_{\{+\}}\ c^{\{2\}}\ \backslash\mathrm{approx}\ 9,28717[\backslash\mathrm{mathrm{KeV}}]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | $\beta_4 = \alpha_3 N_{(3)} - N_{(4)} > 0, \quad \mu_+ c^2 \approx 9,28717[\mathrm{KeV}]$                                                                                                                                                                                                                                       | $\beta_4 = \alpha_3 N_{(3)} - N_{(4)} > 0, \quad \mu_+ c^2 \approx 9,28717[KeV]$                                                                                                                                                                                                                                  |
| BH3FR_EQ0288 (M12a) | 141  | ■ 0.998 | $\backslash\mathrm{begin}\{\mathrm{aligned}\}\ \&\ s\ p^{\{2\}}\ \backslash\mathrm{bar}\{C\}=\backslash\mathrm{sum}_{\{i=1\}}^{\{L\}}\ C_{\{i\}}^{\{2\}},\ \backslash\mathrm{quad}\ s\ p_{\{i=k\}}^{\{m\}}\ \backslash\mathrm{bar}\{C\}_{\{+(i,\ k)\}}=\{ \ }^{\{m-2\}}\ \backslash\mathrm{bar}\{C\},\ \backslash\backslash\ \&\ s\ p_{\{i=k\}}^{\{m\}}\ \backslash\mathrm{bar}\{C\}_{\{-(i,\ k)\}}=\{ \ }^{\{m-2\}}\ \backslash\mathrm{overline}\{0\}\ \backslash\mathrm{end}\{\mathrm{aligned}\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | $sp^2\bar{C} = \sum_{i=1}^L C_i^2, \quad sp_{i=k}{}^m\bar{C}_{+(i,k)} = {}^{m-2}\bar{C},$<br>$sp_{i=k}{}^m\bar{C}_{-(i,k)} = {}^{m-2}\bar{0}$                                                                                                                                                                                   | $sp^2\overline{C} = \sum_{i=1}^L C_i^2, \quad sp_{i=k}{}^m\overline{C}_{+(i,k)} = {}^{m-2}\overline{C},$<br>$sp_{i=k}{}^m\overline{C}_{-(i,k)} = {}^{m-2}\overline{0}$                                                                                                                                            |
| BH3FR_EQ0026 (10)   | 102  | ■ 0.999 | $3\ \omega\mathrm{e}\mathrm{a}=4\ c,\ \backslash\mathrm{quad}\ \omega\mathrm{e}\mathrm{a}^{\{2\}}\ \backslash\mathrm{alpha} \backslash\mathrm{beta} =1$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | $3\omega = 4c, \quad \omega^2\alpha \beta  = 1$                                                                                                                                                                                                                                                                                 | $3\omega = 4c, \quad \omega^2\alpha \beta  = 1$                                                                                                                                                                                                                                                                   |
| BH3FR_EQ0217 (98e)  | 129  | ■ 0.999 | $\backslash\mathrm{begin}\{\mathrm{aligned}\}\ \&\ \mathrm{eta}_{\{j\}}\ N_{\{(j)\}}=\mathrm{G}_{\{j\}},\ \backslash\mathrm{quad}\ 4\ \mathrm{G}_{\{1\}}=\mathrm{N}_{\{(1)\}}^{\{2\}}\backslash\mathrm{left}(1+\mathrm{N}_{\{(1)\}}\backslash\mathrm{right})^{\{2\}},\ \backslash\backslash\ \&\ 6\ \mathrm{G}_{\{2\}}=\mathrm{N}_{\{(2)\}}\backslash\mathrm{left}(2\ \mathrm{N}_{\{(2)\}}^{\{2\}}+3\ \mathrm{N}_{\{(2)\}}+1\backslash\mathrm{right}),\ \backslash\mathrm{quad}\ 2\ \mathrm{G}_{\{3\}}=\mathrm{N}_{\{(3)\}}\backslash\mathrm{left}(1+\mathrm{N}_{\{(3)\}}\backslash\mathrm{right})\ \backslash\backslash\ \&\ \mathrm{G}_{\{4\}}=\mathrm{N}_{\{(4)\}},\ \backslash\mathrm{quad}\ \mathrm{eta}_{\{1\}}\ \backslash\mathrm{geqq}\ \mathrm{eta}_{\{2\}}\ \backslash\mathrm{geqq}\ \mathrm{eta}_{\{3\}}\ \backslash\mathrm{geqq}\ \mathrm{eta}_{\{4\}}=1\ \backslash\mathrm{end}\{\mathrm{aligned}\}$                                                                                                                             | $\eta_j N_{(j)} = G_j, \quad 4G_1 = N_{(1)}^2 (1 + N_{(1)})^2,$<br>$6G_2 = N_{(2)} \left( 2N_{(2)}^2 + 3N_{(2)} + 1 \right), \quad 2G_3 = N_{(3)} (1 + N_{(3)})$<br>$G_4 = N_{(4)}, \quad \eta_1 \geq \eta_2 \geq \eta_3 \geq \eta_4 = 1$                                                                                       | $\eta_j N_{(j)} = G_j, \quad 4G_1 = N_{(1)}^2 (1 + N_{(1)})^2,$<br>$6G_2 = N_{(2)} (2N_{(2)}^2 + 3N_{(2)} + 1), \quad 2G_3 = N_{(3)} (1 + N_{(3)})$<br>$G_4 = N_{(4)}, \quad \eta_1 \geq \eta_2 \geq \eta_3 \geq \eta_4 = 1$                                                                                      |
| BH3FR_EQ0252 (112)  | 136  | ■ 0.999 | $M(N)=\backslash\mathrm{mu}\ \backslash\mathrm{alpha}_{\{+\}}(K+F+H+\backslash\mathrm{phi})$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | $M(N) = \mu\alpha_+(K + F + H + \phi)$                                                                                                                                                                                                                                                                                          | $M(N) = \mu\alpha_+(K + F + H + \phi)$                                                                                                                                                                                                                                                                            |
| BH3FR_EQ0207 (M12)  | 141  | ■ 0.999 | $\backslash\mathrm{begin}\{\mathrm{aligned}\}\ \&\ \{ \ }^{\{2\}}\ \backslash\mathrm{bar}\{C\}=\backslash\mathrm{left}[C_{\{i\}}\ ;\ C_{\{k\}}\ \backslash\mathrm{right}]_{\{L\}},\ \backslash\mathrm{quad}\{ \ }^{\{2\}}\ \backslash\mathrm{bar}\{C\}=\{ \ }^{\{2\}}\ \backslash\mathrm{bar}\{C\}_{\{+\}}+\{ \ }^{\{2\}}\ \backslash\mathrm{bar}\{C\}_{\{-}\},\ \backslash\backslash\ \&\ \{ \ }^{\{2\}}\ \backslash\mathrm{bar}\{C\}_{\{ \ \mathrm{pm}\}}=\backslash\mathrm{frac}\{1\}{2}\ \backslash\mathrm{left}[\backslash\mathrm{left}(C_{\{i\}}\ \backslash\mathrm{times}\ C_{\{k\}}\backslash\mathrm{right})_{\{ \ \mathrm{pm}\}}\backslash\mathrm{right}]_{\{L\}},\ \backslash\backslash\ \&\ \{ \ }^{\{2\}}\ \backslash\mathrm{bar}\{C\}_{\{+\}}=\{ \ }^{\{2\}}\ \backslash\mathrm{bar}\{C\}_{\{+\}}^{\{ \ \mathrm{times}\}},\ \backslash\mathrm{quad}\{ \ }^{\{2\}}\ \backslash\mathrm{bar}\{C\}_{\{-}\}=-\{ \ }^{\{2\}}\ \backslash\mathrm{bar}\{C\}_{\{-}\}^{\{ \ \mathrm{times}\}}\ \backslash\mathrm{end}\{\mathrm{aligned}\}$ | ${}^2\bar{C} = [C_i; C_k]_L, \quad {}^2\bar{C} = {}^2\bar{C}_+ + {}^2\bar{C}_-,$<br>${}^2\bar{C}_{\pm} = \frac{1}{2} [(C_i \times C_k)_{\pm}]_L,$<br>${}^2\bar{C}_+ = {}^2\bar{C}_+^{\times}, \quad {}^2\bar{C}_- = -{}^2\bar{C}_-^{\times}$                                                                                    | ${}^2\overline{C} = [C_i; C_k]_L, \quad {}^2\overline{C} = {}^2\overline{C}_+ + {}^2\overline{C}_-,$<br>${}^2\overline{C}_{\pm} = \frac{1}{2} [(C_i \times C_k)_{\pm}]_L,$<br>${}^2\overline{C}_+ = {}^2\overline{C}_+^{\times}, \quad {}^2\overline{C}_- = -{}^2\overline{C}_-^{\times}$                         |
| BH3FR_EQ0047        | 104  | ■ 0.999 | $s\ p\backslash\mathrm{left}(\{ \ }^{\{2\}}\ \backslash\mathrm{bar}\{\backslash\mathrm{tau}\}\ \backslash\mathrm{times}\backslash\mathrm{left}(s\ p^{\{2\}}\ \backslash\mathrm{bar}\{\backslash\mathrm{tau}\}\ \backslash\mathrm{times}\{ \ }^{\{2\}}\ \backslash\mathrm{bar}\{\backslash\mathrm{tau}\}\backslash\mathrm{right})\backslash\mathrm{right})=\backslash\mathrm{Delta}\backslash\mathrm{left} \backslash\mathrm{begin}\{\mathrm{array}\}\{\backslash\mathrm{llllll}\}\ 0\ \&\ 3\ \&\ 9\ \&\ 9\ i\ \&\ 3\ i\ \&\ i\ \backslash\backslash\ -3\ \&\ 0\ \&\ 3\ \&\ 3\ i\ \&\ 3\ i\ \backslash\backslash\ -9\ \&\ -3\ \&\ 0\ \&\ i\ \&\ 3\ i\ \&\ 9\ i\ \backslash\backslash\ -9\ i\ \&\ -3\ i\ \&\ -i\ \&\ 0\ \&\ -3\ \&\ -9\ \backslash\backslash\ -3\ i\ \&\ -i\ \&\ -3\ i\ \&\ 3\ \&\ 0\ \&\ -3\ \backslash\backslash\ -i\ \&\ -3\ i\ \&\ -9\ i\ \&\ 9\ \&\ 3\ \&\ 0\ \backslash\mathrm{end}\{\mathrm{array}\}\backslash\mathrm{right} ,$                                                                                          | $sp\left({}^2\bar{\tau} \times \left(sp^2\bar{\tau} \times {}^2\bar{\tau}\right)\right) = \Delta \left  \begin{array}{cccccc} 0 & 3 & 9 & 9i & 3i & i \\ -3 & 0 & 3 & 3i & 3i & \\ -9 & -3 & 0 & i & 3i & 9i \\ -9i & -3i & -i & 0 & -3 & -9 \\ -3i & -i & -3i & 3 & 0 & -3 \\ -i & -3i & -9i & 9 & 3 & 0 \end{array} \right ,$ | $sp({}^2\bar{\tau} \times (sp^2\bar{\tau} \times {}^2\bar{\tau})) = \mathcal{A} \left  \begin{array}{cccccc} 0 & 3 & 9 & 9i & 3i & i \\ -3 & 0 & 3 & 3i & 3i & 3i \\ -9 & -3 & 0 & i & 3i & 9i \\ -9i & -3i & -i & 0 & -3 & -9 \\ -3i & -i & -3i & 3 & 0 & -3 \\ -i & -3i & -9i & 9 & 3 & 0 \end{array} \right ,$ |
| BH3FR_EQ0238 (108)  | 132  | ■ 0.999 | $\backslash\mathrm{begin}\{\mathrm{aligned}\}\ \&\ \backslash\mathrm{alpha}_{\{1\}}\ N_{\{(1)\}}^{\{3\}}+\backslash\mathrm{alpha}_{\{2\}}\ N_{\{(2)\}}^{\{2\}}+\backslash\mathrm{alpha}_{\{3\}}\ N_{\{(3)\}}+\backslash\mathrm{exp}\ \backslash\mathrm{left}(-\backslash\mathrm{frac}\{2\ k-1\}{3}\ Q_{\{4\}}\ \backslash\backslash\ N_{\{(4)\}}\backslash\mathrm{right})=\ \backslash\backslash\ \&=W(v\ x)(1+f(N)),\ \backslash\mathrm{quad}\ W(v\ x)=g(k,\ q)\ w(v\ x),\ \backslash\backslash\ \&\ g(k,\ q)=\backslash\mathrm{alpha}_{\{1\}}\ Q_{\{1\}}^{\{3\}}+\backslash\mathrm{alpha}_{\{2\}}\ Q_{\{2\}}^{\{2\}}+\backslash\mathrm{alpha}_{\{3\}}\ Q_{\{3\}}+\backslash\mathrm{exp}\ \backslash\mathrm{left}(-\backslash\mathrm{frac}\{1\}{3}\{2\ k-1\}\ \backslash\mathrm{right})\ \backslash\mathrm{end}\{\mathrm{aligned}\}$                                                                                                                                                                                                         | $\alpha_1 N_{(1)}^3 + \alpha_2 N_{(2)}^2 + \alpha_3 N_{(3)} + \exp\left(-\frac{2k-1}{3Q_4} N_{(4)}\right) =$<br>$= W(vx)(1 + f(N)), \quad W(vx) = g(k, q)w(vx),$<br>$g(k, q) = \alpha_1 Q_1^3 + \alpha_2 Q_2^2 + \alpha_3 Q_3 + \exp\left(-\frac{1}{3}(2k-1)\right)$                                                            | $\alpha_1 N_{(1)}^3 + \alpha_2 N_{(2)}^2 + \alpha_3 N_{(3)} + \exp\left(-\frac{2k-1}{3Q_4} N_{(4)}\right) =$<br>$= W(vx)(1 + f(N)), \quad W(vx) = g(k, q)w(vx),$<br>$g(k, q) = \alpha_1 Q_1^3 + \alpha_2 Q_2^2 + \alpha_3 Q_3 + \exp\left(-\frac{1}{3}(2k-1)\right)$                                              |

| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Page | Conf.  | LaTeX source                                                                                                                                                                                                                                                                                                                                                             | Rendered                                                                                                                                                                                                                                                                                                                      | Scan image                                                                                                                                                                                                                                                                                                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BH3FR_EQ0014 (4a)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 101  | ■1.000 | $1\leq k\leq 6,\quad \vec{x}_k=\vec{e}_k\,x_k,\quad \vec{e}_i\vec{e}_k=\delta_{ik}$                                                                                                                                                                                                                                                                                      | $1\leq k\leq 6,\quad \vec{x}_k=\vec{e}_kx_k,\quad \vec{e}_i\vec{e}_k=\delta_{ik}$                                                                                                                                                                                                                                             | $1\leq k\leq 6,\quad \vec{x}_k=\vec{e}_kx_k,\quad \vec{e}_i\vec{e}_k=\delta_{ik}$                                                                                                                                                                                                                                             |
| BH3FR_EQ0021 (8)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 101  | ■1.000 | $C\,\widehat{\{\}}=\vec{\lambda}\times\widehat{\{\}}$                                                                                                                                                                                                                                                                                                                    | $C\widehat{\{}}=\vec{\lambda}\times\widehat{\{}}$                                                                                                                                                                                                                                                                             | $C\widehat{\{}}=\vec{\lambda}\times\widehat{\{}}$                                                                                                                                                                                                                                                                             |
| BH3FR_EQ0028 (10b)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 102  | ■1.000 | $8\,\pi\,\gamma\,\alpha^{\prime}=1,\quad c^2 \beta^{\prime} =8\pi\gamma,\quad \omega^{\prime}=c$                                                                                                                                                                                                                                                                         | $8\pi\gamma\alpha^{\prime}=1,\quad c^2 \beta^{\prime} =8\pi\gamma,\quad \omega^{\prime}=c$                                                                                                                                                                                                                                    | $8\pi\gamma\alpha^{\prime}=1,\quad c^2 \beta^{\prime} =8\pi\gamma,\quad \omega^{\prime}=c$                                                                                                                                                                                                                                    |
| BH3FR_EQ0049 (16b)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 104  | ■1.000 | $\Delta_{\{\pm\}}\hat{=} \pm^2\bar{\tau},\quad \Delta_{\pm}\hat{=} R_{6(\pm)},\quad R_{6(+)}+R_{6(-)}\equiv R_{6(0)}$                                                                                                                                                                                                                                                    | $\Delta_{\pm}\hat{=} \pm^2\bar{\tau},\quad \Delta_{\pm}\hat{=} R_{6(\pm)},\quad R_{6(+)}+R_{6(-)}\equiv R_{6(0)}$                                                                                                                                                                                                             | $\mathcal{A}_{\pm}\hat{=} \pm^2\bar{\tau},\quad \mathcal{A}_{\pm}\hat{=} R_{6(\pm)},\quad R_{6(+)}+R_{6(-)}\equiv R_{6(0)}$                                                                                                                                                                                                   |
| BH3FR_EQ0107 (44a)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 112  | ■1.000 | $\begin{aligned} &\&4\,\lambda_{\scriptscriptstyle{r}}^{\scriptscriptstyle{3}}\,\approx\,\tau\,D,\quad \frac{\sqrt{h}}{m_{\scriptscriptstyle{r}}^3}=\text{const},\\ &c\,\lambda_{\scriptscriptstyle{r}}^{\scriptscriptstyle{3}}=\text{const},\quad \left(\frac{\lambda_{\scriptscriptstyle{r}}}{m_{\scriptscriptstyle{r}}}\right)^3c\sqrt{h}=\text{const} \end{aligned}$ | $4\lambda_{\scriptscriptstyle{r}}^{\scriptscriptstyle{3}}\approx\tau D,\quad c\lambda_{\scriptscriptstyle{r}}^{\scriptscriptstyle{3}}=\text{const},\quad \frac{\sqrt{h}}{m_{\scriptscriptstyle{r}}^3}=\text{const},\\ \left(\frac{\lambda_{\scriptscriptstyle{r}}}{m_{\scriptscriptstyle{r}}}\right)^3c\sqrt{h}=\text{const}$ | $4\lambda_{\scriptscriptstyle{r}}^{\scriptscriptstyle{3}}\approx\tau D,\quad c\lambda_{\scriptscriptstyle{r}}^{\scriptscriptstyle{3}}=\text{const},\quad \frac{\sqrt{h}}{m_{\scriptscriptstyle{r}}^3}=\text{const},\\ \left(\frac{\lambda_{\scriptscriptstyle{r}}}{m_{\scriptscriptstyle{r}}}\right)^3c\sqrt{h}=\text{const}$ |
| BH3FR_EQ0194 (88a)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 126  | ■1.000 | $\bar{s}(x)=\pm\hbar\bar{\sigma}(x),\quad 2\bar{\sigma}(x)=\bar{s}_0m_x;v,\quad \bar{\sigma}\neq\bar{\sigma}^*$                                                                                                                                                                                                                                                          | $\bar{s}(x)=\pm\hbar\bar{\sigma}(x),\quad 2\bar{\sigma}(x)=\bar{s}_0m_x;v,\quad \bar{\sigma}\neq\bar{\sigma}^*$                                                                                                                                                                                                               | $\bar{s}(x)=\pm\hbar\bar{\sigma}(x),\quad 2\bar{\sigma}(x)=\bar{s}_0m_x;v,\quad \bar{\sigma}\neq\bar{\sigma}^*$                                                                                                                                                                                                               |
| BH3FR_EQ0251 (111b)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 136  | ■1.000 | $\begin{aligned} &\&\beta\pi A_v=(1-\alpha_-/\alpha_+)\Big((\pi/3)^2+(\eta/\eta_{11})^2\frac{\alpha}{3\xi}\Big),\\ &\xi^2B_v=\alpha(1-\alpha_-/\alpha_+)^2Y_{44} \end{aligned}$                                                                                                                                                                                          | $\beta\pi A_v=(1-\alpha_-/\alpha_+)\Big((\pi/3)^2+(\eta/\eta_{11})^2\frac{\alpha}{3\xi}\Big)\\ \xi^2B_v=\alpha(1-\alpha_-/\alpha_+)^2Y_{44}$                                                                                                                                                                                  | $\beta\pi A_v=(1-\alpha_-/\alpha_+)\Big((\pi/3)^2+(\eta/\eta_{11})^2\frac{\alpha}{3\xi}\Big),\\ \xi^2B_v=\alpha(1-\alpha_-/\alpha_+)^2Y_{44}$                                                                                                                                                                                 |
| BH3FR_EQ0296                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 143  | ■1.000 | $\begin{aligned} &\&\gamma;n=\left ^2\bar{\gamma}\right _N;n=w^2,\quad w=W;n,\quad V=\kappa\tau^MSW;n\partial n,\\ &\kappa=\prod_{k=1}^N\kappa^{\underline{k}} \end{aligned}$                                                                                                                                                                                            | $\gamma;n=\left ^2\bar{\gamma}\right _N;n=w^2,\quad w=W;n,\quad V=\kappa\tau^MSW;n\partial n,\\ \kappa=\prod_{k=1}^N\kappa^{\underline{k}}$                                                                                                                                                                                   | $\gamma;n=\left ^2\bar{\gamma}\right _N;n=w^2,\quad w=W;n,\quad V=\kappa\tau^MSW;n\partial n,\\ \kappa=\prod_{k=1}^N\kappa^{\underline{k}}$                                                                                                                                                                                   |
| BH3FR_EQ0003 (*)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 100  | ■1.000 | $\operatorname{rot}\vec{\mu}\sim\alpha\dot{\vec{G}}+\sigma\vec{v},\quad\operatorname{div}\vec{G}\sim\sigma(r),\quad\alpha>0$                                                                                                                                                                                                                                             | $\operatorname{rot}\vec{\mu}\sim\alpha\dot{\vec{G}}+\sigma\vec{v},\quad\operatorname{div}\vec{G}\sim\sigma(r),\quad\alpha>0$                                                                                                                                                                                                  | $\operatorname{rot}\vec{\mu}\sim\alpha\dot{\vec{G}}+\sigma\vec{v},\quad\operatorname{div}\vec{G}\sim\sigma(r),\quad\alpha>0$                                                                                                                                                                                                  |
| BH3FR_EQ0006 (1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 100  | ■1.000 | $R_{ik}-\frac{1}{2}g_{ik}R\sim T_{ik}$                                                                                                                                                                                                                                                                                                                                   | $R_{ik}-\frac{1}{2}g_{ik}R\sim T_{ik}$                                                                                                                                                                                                                                                                                        | $R_{ik}-\frac{1}{2}g_{ik}R\sim T_{ik}$                                                                                                                                                                                                                                                                                        |
| BH3FR_EQ0007 (1a)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 100  | ■1.000 | $g_{ik}=\sum_{\beta=1}^3g_{ik}^{(\beta)}\neq g_{ki}^*$                                                                                                                                                                                                                                                                                                                   | $g_{ik}=\sum_{\beta=1}^3g_{ik}^{(\beta)}\neq g_{ki}^*$                                                                                                                                                                                                                                                                        | $g_{ik}=\sum_{\beta=1}^3g_{ik}^{(\beta)}\neq g_{ki}^*$                                                                                                                                                                                                                                                                        |
| BH3FR_EQ0008 (1b)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 100  | ■1.000 | $\ddot{x}^i+\Gamma_{km}^i\dot{x}^k\dot{x}^m=0$                                                                                                                                                                                                                                                                                                                           | $\ddot{x}^i+\Gamma_{km}^i\dot{x}^k\dot{x}^m=0$                                                                                                                                                                                                                                                                                | $\ddot{x}^i+\Gamma_{km}^i\dot{x}^k\dot{x}^m=0$                                                                                                                                                                                                                                                                                |
| BH3FR_EQ0009 (2)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 100  | ■1.000 | $R_{ik}\sim w\eta_{ik},\quad w=\sqrt{-g},\quad g= g_{ik} _4$                                                                                                                                                                                                                                                                                                             | $R_{ik}\sim w\eta_{ik},\quad w=\sqrt{-g},\quad g= g_{ik} _4$                                                                                                                                                                                                                                                                  | $R_{ik}\sim w\eta_{ik},\quad w=\sqrt{-g},\quad g= g_{ik} _4$                                                                                                                                                                                                                                                                  |
| <p><b>Conf.</b> MathPix confidence: ■<math>\geq 0.9</math> ■<math>0.5\text{--}0.9</math> ■<math>&lt; 0.5</math> — no value</p> <p><b>LaTeX source</b> MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ</p> <p><b>Residual</b> not shown for this document — no residual measurements are paired with these rows.</p> <p><b>[refined: <i>basis</i>]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</p> |      |        |                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                               |

| Identifier          | Page | Conf.  | LaTeX source                                                                                                                                                                                                                                                                                    | Rendered                                                                                                              | Scan image                                                                                                 |
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| BH3FR_EQ0011 (3)    | 100  | ■1.000 | $C_{\{p\}}\varphi_{k\,m}^{\{p\}}=\lambda_{\{p\}}(k,\,m)\,\varphi_{k\,m}^{\{p\}}$                                                                                                                                                                                                                | $C_{(p)}\varphi_{km}^{(p)}=\lambda_{(p)}(k,m)\varphi_{km}^{(p)}$                                                      | $C_{(p)}\varphi_{km}^{(p)}=\lambda_{(p)}(k,m)\varphi_{km}^{(p)}$                                           |
| BH3FR_EQ0012 (3a)   | 100  | ■1.000 | $\varphi_{k(m)}^{\{k(m)\}^{\{m\}}}=\emptyset$                                                                                                                                                                                                                                                   | $\varphi_{k(m)}^{(m)}=0$                                                                                              | $\varphi_{k(m)}^{(m)}=0$                                                                                   |
| BH3FR_EQ0015 (5)    | 101  | ■1.000 | $\sum_{k=1}^6\cos^2\alpha_k=1$                                                                                                                                                                                                                                                                  | $\sum_{k=1}^6\cos^2\alpha_k=1$                                                                                        | $\sum_{k=1}^6\cos^2\alpha_k=1$                                                                             |
| BH3FR_EQ0016 (5a)   | 101  | ■1.000 | $j\not\equiv 4,\quad 2\,\alpha_j=\pi,\quad \cos\alpha_4=\pm 1$                                                                                                                                                                                                                                  | $j\neq 4,\quad 2\alpha_j=\pi,\quad \cos\alpha_4=\pm 1$                                                                | $j\neq 4,\quad 2\alpha_j=\pi,\quad \cos\alpha_4=\pm 1$                                                     |
| BH3FR_EQ0017 (5b)   | 101  | ■1.000 | $\varphi(v)-\varphi(u)=\pi\,/\,2$                                                                                                                                                                                                                                                               | $\varphi(v)-\varphi(u)=\pi/2$                                                                                         | $\varphi(v)-\varphi(u)=\pi/2$                                                                              |
| BH3FR_EQ0019 (6)    | 101  | ■1.000 | $g_{\{i\,k\}}\backslash\!\!\!\left(x_{\{1\}}\,\ldots\,x_{\{6\}}\right)=g_{\{k\,i\}}^{\{*\}}$                                                                                                                                                                                                    | $g_{ik}(x_1\ldots x_6)=g_{ki}^*$                                                                                      | $g_{ik}(x_1\ldots x_6)=g_{ki}^*$                                                                           |
| BH3FR_EQ0024 (9)    | 102  | ■1.000 | $g_{\{i\,k\}}\backslash\!\!\!\left(g_{\{i\,k\}}^{\{1\}}\,\ldots\,g_{\{i\,k\}}^{\{3\}}\right)=g_{\{k\,i\}}^{\{*\}},\quad \text{\rm \!}\backslash\!\!\!\text{\rm \!}\quad g_{\{i\,k\}}^{\{(\mu)\}}\backslash\!\!\!\left(x_{\{1\}}\,\ldots\,x_{\{6\}}\right)\not\equiv g_{\{k\,i\}}^{\{(\mu)\}^*}$ | $g_{ik}\left(g_{ik}^{(1)}\ldots g_{ik}^{(3)}\right)=g_{ki}^*,\quad g_{ik}^{(\mu)}(x_1\ldots x_6)\neq g_{ki}^{(\mu)*}$ | $g_{ik}(g_{ik}^{(1)}\ldots g_{ik}^{(3)})=g_{ki}^*,\quad g_{ik}^{(\mu)}(x_1\ldots x_6)\neq g_{ki}^{(\mu)*}$ |
| BH3FR_EQ0027 (10a)  | 102  | ■1.000 | $64\,\pi\,\gamma\alpha=3,\quad c^2 \beta =12\,\pi\,\gamma$                                                                                                                                                                                                                                      | $64\pi\gamma\alpha=3,\quad c^2 \beta =12\pi\gamma$                                                                    | $64\pi\gamma\alpha=3,\quad c^2 \beta =12\pi\gamma$                                                         |
| BH3FR_EQ0031 (11a)  | 102  | ■1.000 | $q=1-\sqrt{1-\varepsilon\varphi},\quad 8\,c^2\varepsilon=3$                                                                                                                                                                                                                                     | $q=1-\sqrt{1-\varepsilon\varphi},\quad 8c^2\varepsilon=3$                                                             | $q=1-\sqrt{1-\varepsilon\varphi},\quad 8c^2\varepsilon=3$                                                  |
| BH3FR_EQ0033 (11c)  | 103  | ■1.000 | $r^{\{2\}}\,G=-\gamma M_{\{0\}}\backslash\!\!\!\left(1-(r\,/\,\varsigma)^2\right)\,e^{\{q\}(1+q)^{-1}}$                                                                                                                                                                                         | $r^2G=-\gamma M_0\left(1-(r/\varsigma)^2\right)e^q(1+q)^{-1}$                                                         | $r^2G=-\gamma M_0(1-(r/\varsigma)^2)e^q(1+q)^{-1}$                                                         |
| BH3FR_EQ0036 (12b)  | 103  | ■1.000 | $A_{\{t\}}^3\varsigma\,\varapprox 46[M\,\text{p}\,\text{c}]$                                                                                                                                                                                                                                    | $A_t^3\varsigma\approx 46[Mpc]$                                                                                       | $A_t^3\varsigma\approx 46[Mpc]$                                                                            |
| BH3FR_EQ0040 (14b)  | 103  | ■1.000 | $4\,\omega c\,R_{-}=e\,\gamma M_{\{0\}}$                                                                                                                                                                                                                                                        | $4\omega cR_{-}=e\gamma M_0$                                                                                          | $4\omega cR_{-}=e\gamma M_0$                                                                               |
| BH3FR_EQ0042 (15)   | 103  | ■1.000 | $\tau\,\omega c^2=\pi\,\gamma\hbar$                                                                                                                                                                                                                                                             | $\tau\omega c^2=\pi\gamma\hbar$                                                                                       | $\tau\omega c^2=\pi\gamma\hbar$                                                                            |
| BH3FR_EQ0043 (15a') | 103  | ■1.000 | $F\backslash\!\!\!\left(x_{\{1\}}\,\ldots\,x_{\{6\}}\right)=N\,\tau,\quad \text{\rm \!}\backslash\!\!\!\text{\rm \!}\quad N>0$                                                                                                                                                                  | $F(x_1\ldots x_6)=N\tau,\quad N>0$                                                                                    | $F(x_1\ldots x_6)=N\tau,\quad N>0$                                                                         |
| BH3FR_EQ0044 (15b)  | 104  | ■1.000 | $m\,/\,p=M\,\geqq 1,\quad (M)\,M\,0\,D(1)=0$                                                                                                                                                                                                                                                    | $m/p=M\geq 1,\quad (M)MOD(1)=O$                                                                                       | $m/p=M\geq 1,\quad (M)MOD(1)=O$                                                                            |
| BH3FR_EQ0056 (19a)  | 105  | ■1.000 | $L=K-\bar{\lambda}\backslash\!\!\!\left(\lambda\right)\,\text{\rm \!}\times\,()$                                                                                                                                                                                                                | $L=K-\bar{\lambda}\times()$                                                                                           | $L=K-\bar{\lambda}\times()$                                                                                |
| BH3FR_EQ0066 (22)   | 107  | ■1.000 | $\psi_{\{k\,l\}}\,\varapprox\backslash\!\!\!\left(E+C_{\{k\,l\}}\,\cdot\,e^{\{-\lambda_{kl}\mu\}}\backslash\!\!\!\left(\mu\right)\right)^{-1},\quad \text{\rm \!}\backslash\!\!\!\text{\rm \!}\quad \Lambda_{kl}\approx 0$                                                                      | $\psi_{kl}\approx\left(E+C_{kl}\cdot e^{-\lambda_{kl}\mu}\right)^{-1},\quad \Lambda_{kl}\approx 0$                    | $\psi_{kl}\approx\left(E+C_{kl}\cdot e^{-\lambda_{kl}\mu}\right)^{-1},\quad A_{kl}\approx 0$               |
| BH3FR_EQ0068 (22b)  | 107  | ■1.000 | $r^{\{2\}}\,\gg\,\varsigma^2,\quad \psi=\left(1+C\,e^{\pm\lambda r}\backslash\!\!\!\left(\lambda\right)\right)^{-1}$                                                                                                                                                                            | $r^2\gg\varsigma^2,\quad \psi=\left(1+Ce^{\pm\lambda r}\right)^{-1}$                                                  | $r^2\gg\varsigma^2,\quad \psi=\left(1+Ce^{\pm\lambda r}\right)^{-1}$                                       |
| BH3FR_EQ0070 (24)   | 107  | ■1.000 | $a\,\text{\rm \!}\equiv G\backslash\!\!\!\left(x_{\{5\}},\,x_{\{6\}}\right)\,\text{\rm \!}\rightleftharpoons P\,\backslash\!\!\!\left(x_{\{4\}},\,x_{\{5\}},\,x_{\{6\}}\right)\,\text{\rm \!}\equiv b$                                                                                          | $a\equiv G(x_5,x_6)\rightleftharpoons P(x_4,x_5,x_6)\equiv b$                                                         | $a\equiv G(x_5,x_6)\rightleftharpoons P(x_4,x_5,x_6)\equiv b$                                              |
| BH3FR_EQ0071 (25)   | 108  | ■1.000 | $r=\frac{2\,n_{\{r\}}+1}{2\,n_{\{\varsigma\}}+1}\cdot\varsigma$                                                                                                                                                                                                                                 | $r=\frac{2n_r+1}{2n_\varsigma+1}\cdot\varsigma$                                                                       | $r=\frac{2n_r+1}{2n_\varsigma+1}\cdot\varsigma$                                                            |
| BH3FR_EQ0072 (26)   | 108  | ■1.000 | $m=2\,\sqrt{\frac{c\,h}{\gamma}}\,\frac{\sqrt[4]{2n}}{\sqrt{2n-1}}$                                                                                                                                                                                                                             | $m=2\sqrt{\frac{ch}{\gamma}}\frac{\sqrt[4]{2n}}{\sqrt{2n-1}}$                                                         | $m=2\sqrt{\frac{ch}{\gamma}}\frac{\sqrt[4]{2n}}{\sqrt{2n-1}}$                                              |
| BH3FR_EQ0074 (26b)  | 108  | ■1.000 | $n=n(t),\quad \frac{d\,n}{d\,t}\,\geqq 0$                                                                                                                                                                                                                                                       | $n=n(t),\quad \frac{dn}{dt}\geq 0$                                                                                    | $n=n(t),\quad \frac{dn}{dt}\geq 0$                                                                         |
| BH3FR_EQ0075 (27)   | 108  | ■1.000 | $m(n,\,q)=m(n)\,\eta_{\{q\}},\quad \text{\rm \!}\backslash\!\!\!\text{\rm \!}\quad \eta_{\{q\}}\,\sqrt[4]{4\,q^4+\pi^4}=\pi$                                                                                                                                                                    | $m(n,q)=m(n)\eta_q,\quad \eta_q\sqrt[4]{4q^4+\pi^4}=\pi$                                                              | $m(n,q)=m(n)\eta_q,\quad \eta_q\sqrt[4]{4q^4+\pi^4}=\pi,$                                                  |

**Conf.** MathPix confidence: ■≥0.9   ■0.5–0.9   ■<0.5   — no value

**LaTeX source** MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

**Residual** not shown for this document — no residual measurements are paired with these rows.

**[refined: *basis*]** beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Page | Conf.  | LaTeX source                                                                                                                                 | Rendered | Scan image |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|--------|----------------------------------------------------------------------------------------------------------------------------------------------|----------|------------|
| BH3FR_EQ0077 (27b)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 108  | ■1.000 | $\mathrm{a}\,\pi^2\,\mathrm{\varepsilon}_{\pm}=\pm 3\sqrt{2\hbar/R_-},\quad R_-=\sqrt{\mu_0/\varepsilon_0},\quad (a=\sqrt{2})$               |          |            |
| BH3FR_EQ0078 (27 $\mathrm{mathrm{c}}$ )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 108  | ■1.000 | $\mathrm{a}\,\pi^2\,\mathrm{C}_{\pm}=\pm\sqrt{2\hbar/R_-},\quad \varepsilon_{\pm}=3C_{\pm}$                                                  |          |            |
| BH3FR_EQ0079 (28)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 109  | ■1.000 | $V_{xy}=e_xe_yf(r)$                                                                                                                          |          |            |
| BH3FR_EQ0081 (29)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 109  | ■1.000 | $4\,\pi^2\,e_{\pm}=\pm 3\sqrt{2\vartheta\hbar/R_-},\quad \vartheta=5\eta+2\sqrt{\eta}+1$                                                     |          |            |
| BH3FR_EQ0082 (29a)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 109  | ■1.000 | $(2\,\pi)^5\alpha'=9\vartheta$                                                                                                               |          |            |
| BH3FR_EQ0083 (30)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 109  | ■1.000 | $eR\sqrt{\tau}=\varepsilon nF,\quad \varepsilon^2\sqrt{F}=s_0=1[m]$                                                                          |          |            |
| BH3FR_EQ0087 (34)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 110  | ■1.000 | $\underline{\eta}=\sqrt[3]{\tau\varepsilon/3}$                                                                                               |          |            |
| BH3FR_EQ0089 (35)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 110  | ■1.000 | $s'^{\prime}=b\tau^{5/6},\quad b=\frac{eE^{-1/3}}{\sqrt[4]{\pi}\sqrt[3]{2}}$                                                                 |          |            |
| BH3FR_EQ0092 (37)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 110  | ■1.000 | $f\sqrt{3/2}\left(\frac{D}{4\sqrt{\tau}}f^3\sqrt{3/2}-1\right)^2=\frac{D}{\sqrt{\tau}}$                                                      |          |            |
| BH3FR_EQ0094 (37b)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 110  | ■1.000 | $aD\approx\tau^{-11/6},\quad 2a=(3\pi2^5)^{-2}E^{-7/3}$                                                                                      |          |            |
| BH3FR_EQ0095 (38)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 111  | ■1.000 | $DA=\dot{D},\quad A\neq 0,\quad A=0$                                                                                                         |          |            |
| BH3FR_EQ0097 (39a)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 111  | ■1.000 | $D\dot{y}=n\dot{D}y$                                                                                                                         |          |            |
| BH3FR_EQ0098 (40)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 111  | ■1.000 | $\begin{aligned} &3W=16\pi\left(\frac{a}{b}\right)^2\sqrt[3]{3\tau^2},\quad a^2N^3\sqrt{3\tau^2}=2\pi \\ &3\alpha'NW=(4\pi)^2 \end{aligned}$ |          |            |
| BH3FR_EQ0100 (41)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 111  | ■1.000 | $DA=\dot{D}>0,\quad \dot{t}<0$                                                                                                               |          |            |
| BH3FR_EQ0101 (41a)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 111  | ■1.000 | $A\ll H,\quad z\approx Bs,\quad v\approx Hs,\quad H=cB$                                                                                      |          |            |
| BH3FR_EQ0102 (42)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 111  | ■1.000 | $cz_D=As,\quad z_S=-\frac{39\lambda}{11c}A,\quad z_a=\frac{16\tau}{9\lambda^2}$                                                              |          |            |
| BH3FR_EQ0103 (42a)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 111  | ■1.000 | $z_k=z_D+z_S+z_a$                                                                                                                            |          |            |
| BH3FR_EQ0104 (43)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 111  | ■1.000 | $D(T)=D_0e^{\psi},\quad \dot{\psi}=A,\quad D_0>0$                                                                                            |          |            |
| BH3FR_EQ0105 (43a)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 112  | ■1.000 | $\frac{d}{dt}\ln m=\frac{36}{11}A>0$                                                                                                         |          |            |
| BH3FR_EQ0106 (44)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 112  | ■1.000 | $\lambda_r^3\approx\frac{c\tau}{2A}$                                                                                                         |          |            |
| BH3FR_EQ0109 (45a)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 112  | ■1.000 | $H\approx\sqrt{\pi e\gamma\sigma}$                                                                                                           |          |            |
| BH3FR_EQ0110 (46)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 112  | ■1.000 | $y\sqrt{\pi e\gamma\sigma}\approx c$                                                                                                         |          |            |
| <p><b>Conf.</b> MathPix confidence: ■<math>\geq 0.9</math> ■<math>0.5\text{--}0.9</math> ■<math>&lt; 0.5</math> — no value</p> <p><b>LaTeX source</b> MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ</p> <p><b>Residual</b> not shown for this document — no residual measurements are paired with these rows.</p> <p><b>[refined: <i>basis</i>]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</p> |      |        |                                                                                                                                              |          |            |



| Identifier          | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Rendered                                                                                                                                                                                                                          | Scan image                                                                                                                                                                                                                         |
|---------------------|------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BH3FR_EQ0222 (100b) | 130  | ■ 1.000 | $q_{\{x\}\left(R_{4}^{+}\right)+q_{\{x\}\left(R_{4}^{-}\right)=C\left(R_{4}^{+}\right)+C\left(R_{4}^{-}\right)=0$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | $q_x\left(R_4^+\right)+q_x\left(R_4^-\right)=C\left(R_4^+\right)+C\left(R_4^-\right)=0$                                                                                                                                           | $q_x(R_4^+)+q_x(R_4^-)=C(R_4^+)+C(R_4^-)=0$                                                                                                                                                                                        |
| BH3FR_EQ0227 (103)  | 131  | ■ 1.000 | $n_{\{j\}}=0,\quad 1\leq\lambda\leq G=k+1,\quad z_{\{\lambda\}}\left(b_{\{\lambda\}}\right)\geq 16$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | $n_j=0,\quad 1\leq\lambda\leq G=k+1,\quad z_\lambda(b_\lambda)\geq 16$                                                                                                                                                            | $n_j=0,\quad 1\leq\lambda\leq G=k+1,\quad z_\lambda(b_\lambda)\geq 16$                                                                                                                                                             |
| BH3FR_EQ0231 (105a) | 131  | ■ 1.000 | $\alpha_{(+)}=\alpha,\quad \alpha_{(-)}=\beta,\quad \beta\approx 137\alpha$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $\alpha_{(+)}=\alpha,\quad \alpha_{(-)}=\beta,\quad \beta\approx 137\alpha$                                                                                                                                                       | $\alpha_{(+)}=\alpha,\quad \alpha_{(-)}=\beta,\quad \beta\approx 137\alpha$                                                                                                                                                        |
| BH3FR_EQ0233 (106a) | 132  | ■ 1.000 | $\begin{array}{l} 2\ B_{\mathrm{H}}\!=\!B\cos\varphi,\ \&\ B_{\mathrm{V}}\!=\!B\sin\varphi,\ \&\ B\!=\!2AD\mu S(t),\\ D\!=\!\sigma R^2w,\ \&\ \mu(x_1,x_2,x_3)>0,\ \&\ 0\leq S(t)\leq +1\\ \left(x_{\mathrm{1}},\ x_{\mathrm{2}},\ x_{\mathrm{3}}\right)>0,\ \&\ \theta\leqq S(t)\leqq +1 \end{array}$                                                                                                                                                                                                                                                                                                          | $2B_H=B\cos\varphi,\quad B_V=B\sin\varphi,\quad B=2AD\mu S(t),\\ D=\sigma R^2w,\quad \mu(x_1,x_2,x_3)>0,\quad 0\leq S(t)\leq +1$                                                                                                  | $2B_H=B\cos\varphi,\quad B_V=B\sin\varphi,\quad B=2AD\mu S(t),\\ D=\sigma R^2w,\quad \mu(x_1,x_2,x_3)>0,\quad 0\leq S(t)\leq +1$                                                                                                   |
| BH3FR_EQ0234 (106b) | 132  | ■ 1.000 | $D>\frac{\Delta}{2\mu A}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | $D>\frac{\Delta}{2\mu A}$                                                                                                                                                                                                         | $D>\frac{A}{2\mu A}$                                                                                                                                                                                                               |
| BH3FR_EQ0240 (109)  | 133  | ■ 1.000 | $w(v\ x)=w_{\{1\}}^{\{2-k\}}+w_{\{2\}}^{\{k-1\}}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | $w(vx)=w_1^{2-k}+w_2^{k-1}$                                                                                                                                                                                                       | $w(vx)=w_1^{2-k}+w_2^{k-1}$                                                                                                                                                                                                        |
| BH3FR_EQ0244 (110)  | 134  | ■ 1.000 | $f(N)=\left(1-Q(1-\kappa)(2-k)\ Y_{23}\right)\left(\frac{aN}{N+2}+b\sqrt{N(N-2)}-ib\delta_{1N}\right)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | $f(N)=(1-Q(1-\kappa)(2-k)Y_{23})\left(\frac{aN}{N+2}+b\sqrt{N(N-2)}-ib\delta_{1N}\right)$                                                                                                                                         | $f(N)=(1-Q(1-\kappa)(2-k)Y_{23})\left(\frac{aN}{N+2}+b\sqrt{N(N-2)}-ib\delta_{1N}\right)$                                                                                                                                          |
| BH3FR_EQ0248 (110d) | 135  | ■ 1.000 | $\hat{A}=\left(A_{i\ k}\right)_{\{6\}}=\hat{A}^{\ast}\neq\hat{A}^{\times}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | $\hat{A}=(A_{ik})_6=\hat{A}^{\ast}\neq\hat{A}^{\times}$                                                                                                                                                                           | $\hat{A}=(A_{ik})_6=\hat{A}^{\ast}\neq\hat{A}^{\times}$                                                                                                                                                                            |
| BH3FR_EQ0249 (111)  | 135  | ■ 1.000 | $\phi=A_{\{v\}}F_{\{1\}}F_{\{q\}}F_{\{\kappa\}}/F_{\{2\}}+B_{\{v\}}(P+Q)+4q\alpha_{-}/\alpha_{+}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | $\phi=A_vF_1F_qF_\kappa/F_2+B_v(P+Q)+4q\alpha_{-}/\alpha_{+}$                                                                                                                                                                     | $\phi=A_vF_1F_qF_\kappa/F_2+B_v(P+Q)+4q\alpha_{-}/\alpha_{+}$                                                                                                                                                                      |
| BH3FR_EQ0253 (112a) | 136  | ■ 1.000 | $\begin{array}{l} \&\ K=N_{\{1\}}\ Q_{\{1\}}^{\{2\}}\left(1+Q_{\{1\}}\right)+\\ \&\ +N_{\{2\}}\ Q_{\{2\}}\left(2\ Q_{\{2\}}^{\{2\}}+3\ Q_{\{2\}}+1\right)+\\ \&\ +N_{\{3\}}\ Q_{\{3\}}\left(1+Q_{\{3\}}\right)+4\ Q_{\{4\}},\quad \&\ F=N_{\{1\}}\ n_{\{1\}}^{\{2\}}\left(1+n_{\{1\}}\right)^{\{2\}}+\\ \&\ +N_{\{2\}}\ n_{\{2\}}\left(2\ n_{\{2\}}^{\{2\}}+3\ n_{\{2\}}+1\right)+N_{\{3\}}\ n_{\{3\}}\left(1+n_{\{3\}}\right)+4\ n_4,\\ \&\ H=2\ n_1\ Q_1\left(1+3\left(n_1+Q_1+n_1\ Q_1\right)+2\left(n_1^2+Q_1^2\right)\right)N_1+\\ \&\ +6\ n_2\ Q_2\left(1+n_2+Q_2\right)N_2+2\ n_3\ Q_3\ N_3 \end{array}$ | $K=N_1Q_1^2(1+Q_1)^2+N_2Q_2(2Q_2^2+3Q_2+1)+\\ +N_3Q_3(1+Q_3)+4Q_4,\quad F=N_1n_1^2(1+n_1)^2+\\ +N_2n_2(2n_2^2+3n_2+1)+N_3n_3(1+n_3)+4n_4\\ H=2n_1Q_1(1+3(n_1+Q_1+n_1Q_1)+2(n_1^2+Q_1^2))N_1+\\ +6n_2Q_2(1+n_2+Q_2)N_2+2n_3Q_3N_3$ | $K=N_1Q_1^2(1+Q_1)^2+N_2Q_2(2Q_2^2+3Q_2+1)+\\ +N_3Q_3(1+Q_3)+4Q_4,\quad F=N_1n_1^2(1+n_1)^2+\\ +N_2n_2(2n_2^2+3n_2+1)+N_3n_3(1+n_3)+4n_4,\\ H=2n_1Q_1(1+3(n_1+Q_1+n_1Q_1)+2(n_1^2+Q_1^2))N_1+\\ +6n_2Q_2(1+n_2+Q_2)N_2+2n_3Q_3N_3$ |
| BH3FR_EQ0254 (112b) | 136  | ■ 1.000 | $N_{\{1\}}=\alpha_{\{1\}},\quad 3\ N_{\{2\}}=2\ \alpha_{\{2\}},\quad N_{\{3\}}=2\ \alpha_{\{3\}}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | $N_1=\alpha_1,\quad 3N_2=2\alpha_2,\quad N_3=2\alpha_3$                                                                                                                                                                           | $N_1=\alpha_1,\quad 3N_2=2\alpha_2,\quad N_3=2\alpha_3$                                                                                                                                                                            |
| BH3FR_EQ0255 (113)  | 136  | ■ 1.000 | $k\ \mu_{\{+\}}\ \alpha_{\{1\}}\left(L_{\{1\}}+Q_{\{1\}}\right)^{\{3\}}=2(k+1)(P+1)^{\{2\}}/k\ M_{\{x\}}\ Y_{\{45\}}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | $k\mu_{+\alpha_1}(L_1+Q_1)^3=2(k+1)(P+1)^{2/k}M_xY_{45}$                                                                                                                                                                          | $k\mu_{+\alpha_1}(L_1+Q_1)^3=2(k+1)(P+1)^{2/k}M_xY_{45}$                                                                                                                                                                           |

**Conf.** MathPix confidence: ■ ≥0.9   ■ 0.5–0.9   ■ <0.5   — no value

**LaTeX source** MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

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| Identifier          | Page | Conf.  | LaTeX source                                                                                                                                                                                                                                                   | Rendered                                                                                                                                                                                                                            | Scan image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------|------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BH3FR_EQ0257 (113b) | 137  | ■1.000 | \begin{aligned} & \alpha_1\left(L_1+Q_1\right)^3+\alpha_2\left(L_2+Q_2\right)^2+\alpha_3\left(L_3+Q_3\right)+\\ & \alpha_4\left(L_4+Q_4\right)+\exp \left(-\frac{2 k-1}{3 Q_4}\left(L_4+Q_4\right)\right)=W(v x)\left(1+f\left(L_N\right)\right) \end{aligned} | $\alpha_1\left(L_1+Q_1\right)^3+\alpha_2\left(L_2+Q_2\right)^2+\alpha_3\left(L_3+Q_3\right)+\exp \left(-\frac{2 k-1}{3 Q_4}\left(L_4+Q_4\right)\right)=W(v x)\left(1+f\left(L_N\right)\right)$                                      | $\alpha_1\left(L_1+Q_1\right)^3+\alpha_2\left(L_2+Q_2\right)^2+\alpha_3\left(L_3+Q_3\right)+\exp \left(-\frac{2 k-1}{3 Q_4}\left(L_4+Q_4\right)\right)=W(v x)\left(1+f\left(L_N\right)\right)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| BH3FR_EQ0258 (113c) | 137  | ■1.000 | M_{\max }\left(L_{\mathrm{j}}\right)_1^4 \geq M_L\left(L_N\right), \quad E\left(\beta_{j>1}>0\right)=\left(M_{\max }-M_L\right) c^2                                                                                                                            | $M_{\max }\left(L_j\right)_1^4 \geq M_L\left(L_N\right), \quad E\left(\beta_{j>1}>0\right)=\left(M_{\max }-M_L\right) c^2$                                                                                                          | $M_{\max }\left(L_j\right)_1^4 \geq M_L\left(L_N\right), \quad E\left(\beta_{j>1}>0\right)=\left(M_{\max }-M_L\right) c^2$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| BH3FR_EQ0261 (116)  | 137  | ■1.000 | \lambda c\left(M\left(N_m\right)-M\left(N_m+1\right)\right)=h                                                                                                                                                                                                  | $\lambda c\left(M\left(N_m\right)-M\left(N_m+1\right)\right)=h$                                                                                                                                                                     | $\lambda c\left(M\left(N_m\right)-M\left(N_m+1\right)\right)=h$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| BH3FR_EQ0264 (118a) | 137  | ■1.000 | m\left(v_{\mathrm{x}}\right)=\mu \alpha_{+} \phi_{q=0} \neq 0                                                                                                                                                                                                  | $m\left(v_x\right)=\mu \alpha_{+} \phi_{q=0} \neq 0$                                                                                                                                                                                | $m\left(v_x\right)=\mu \alpha_{+} \phi_{q=0} \neq 0$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| BH3FR_EQ0266 (118c) | 138  | ■1.000 | v_{\beta} \rightarrowtail v_{\mathrm{P}}+v_{\mathrm{R}}                                                                                                                                                                                                        | $v_{\beta} \rightleftarrows v_P+v_R$                                                                                                                                                                                                | $v_{\beta} \rightleftarrows v_P+v_R$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| BH3FR_EQ0270 (M3)   | 139  | ■1.000 | \partial^k \varphi=\sum_{v=0}^k(-1)^v a_v(k) \varphi(n-v), \quad a_v(k)=\binom{k}{v}                                                                                                                                                                           | $\partial^k \varphi=\sum_{v=0}^k(-1)^v a_v(k) \varphi(n-v), \quad a_v(k)=\binom{k}{v}$                                                                                                                                              | $\partial^k \varphi=\sum_{v=0}^k(-1)^v a_v(k) \varphi(n-v), \quad a_v(k)=\binom{k}{v}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| BH3FR_EQ0278 (M7)   | 140  | ■1.000 | 0<\left \partial_{\varepsilon}\right  \ll 1, \quad \partial_{\varepsilon} \varphi \approx \varphi \partial_{\varepsilon} \ln \varphi, \quad \partial_{\varepsilon} e^{\varphi} \approx e^{\varphi} \partial_{\varepsilon} \varphi                              | $0<\left \partial_{\varepsilon}\right  \ll 1, \quad \partial_{\varepsilon} \varphi \approx \varphi \partial_{\varepsilon} \ln \varphi, \quad \partial_{\varepsilon} e^{\varphi} \approx e^{\varphi} \partial_{\varepsilon} \varphi$ | $0<\left \partial_{\varepsilon}\right  \ll 1, \quad \partial_{\varepsilon} \varphi \approx \varphi \partial_{\varepsilon} \ln \varphi, \quad \partial_{\varepsilon} e^{\varphi} \approx e^{\varphi} \partial_{\varepsilon} \varphi$                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| BH3FR_EQ0292        | 142  | ■1.000 | \partial^2-3 \partial+(\quad)=0, \quad 2 \xi=1+\sqrt{5}                                                                                                                                                                                                        | $\partial^2-3 \partial+(\quad)=O, \quad 2 \xi=1+\sqrt{5}$                                                                                                                                                                           | $\partial^2-3 \partial+(\quad)=O, \quad 2 \xi=1+\sqrt{5}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| BH3FR_EQ0298        | 143  | ■1.000 | {\bar{\gamma}}^2; n=\text{const}, \quad \xi^k=X^k; n, \quad 1 \leq k \leq N                                                                                                                                                                                    | $^2\bar{\gamma}; n=\text{const}, \quad \xi^{\underline{k}}=X^{\underline{k}}; n, \quad 1 \leq k \leq N$                                                                                                                             | $^2\bar{\gamma}; n=\text{const}, \quad \xi^{\underline{k}}=X^{\underline{k}}; n, \quad 1 \leq k \leq N$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| BH3FR_EQ0052 (18)   | 105  | —      | —                                                                                                                                                                                                                                                              | —                                                                                                                                                                                                                                   | $\frac{\partial}{\partial x^k} \rightarrow \frac{\check{\partial}_k}{\alpha_k}=\check{\partial}_k, \quad g_{i k} \rightarrow \gamma_{i k} ; n, \quad {}^2 \overline{\gamma}\left(C_k\right)_1^6=$<br>$=s p\left({}^2 \overline{\kappa} \times {}^2 \overline{\kappa}\right)={}^2 \overline{\gamma}^{\times}, \quad {}^2 \overline{\kappa}={}^2 \overline{\kappa}^{\times}, \quad \widehat{\{\}} \rightarrow \widehat{[\quad]}; n,$<br>$\widehat{[\quad]}=\widehat{[\quad]}^{\times}, \quad I_{(\pm)}^{\left\{s_1\right\}\left\{s_2\right\}} \rightarrow\left(\right)_{(\pm)}^{\left(s_1\right)\left(s_2\right)}, \quad(\varepsilon \leq 3) \triangleq+,$<br>$(\varepsilon>3) \triangleq-$ |

**Conf.** MathPix confidence: ■≥0.9   ■0.5–0.9   ■<0.5   — no value

**LaTeX source** MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

**Residual** not shown for this document — no residual measurements are paired with these rows.

**[refined: *basis*]** beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

### Inline formulas (first occurrence)

| Identifier   | Page | Conf. | LaTeX source                                                              | Rendered                             |
|--------------|------|-------|---------------------------------------------------------------------------|--------------------------------------|
| BH3FR_F00001 | 8    | —     | $Y_{\{j\}}$                                                               | $Y_j$                                |
| BH3FR_F00002 | 8    | —     | $j \geqq 1$                                                               | $j \geq 1$                           |
| BH3FR_F00003 | 8    | —     | $Y_{\{k\}}=1$                                                             | $Y_k = 1$                            |
| BH3FR_F00004 | 8    | —     | $Y_{\{j\}} \neq 1$                                                        | $Y_j \neq 1$                         |
| BH3FR_F00005 | 8    | —     | $Y_{\{k\}}$                                                               | $Y_k$                                |
| BH3FR_F00006 | 9    | —     | $\tau=0$                                                                  | $\tau = 0$                           |
| BH3FR_F00007 | 9    | —     | $C$                                                                       | $C$                                  |
| BH3FR_F00008 | 9    | —     | $\{ \ }^{\wedge}\{\text{text { } \}}$                                     | <i>(not rendered)</i>                |
| BH3FR_F00009 | 9    | —     | $\hat{C}$                                                                 | $\hat{C}$                            |
| BH3FR_F00010 | 9    | —     | $C_{\{i\ k\}}$                                                            | $C_{ik}$                             |
| BH3FR_F00011 | 9    | —     | $C_{\{i\ k\}} \neq C_{\{i\ k\}}^{\wedge\{*\}}$                            | $C_{ik} \neq C_{ik}^*$               |
| BH3FR_F00012 | 9    | —     | $x$                                                                       | $x$                                  |
| BH3FR_F00013 | 9    | —     | $\left(C_{\{i\ k\}}\right)^{\wedge\{\times\}}=C_{\{k\ i\}}^{\wedge\{*\}}$ | $(C_{ik})^{\times} = C_{ki}^*$       |
| BH3FR_F00014 | 9    | —     | $[a,\ b]=a\ b-b\ a$                                                       | $[a,b] = ab - ba$                    |
| BH3FR_F00015 | 9    | —     | $(a\ \times\ b)_{\pm} = a\ b\ \pm\ b\ a$                                  | $(a \times b)_{\pm} = ab \pm ba$     |
| BH3FR_F00016 | 9    | —     | $[a,\ b]$                                                                 | $[a,b]$                              |
| BH3FR_F00017 | 9    | —     | $R_{\{3\}}$                                                               | $R_3$                                |
| BH3FR_F00018 | 9    | —     | $R_{\{n\}}$                                                               | $R_n$                                |
| BH3FR_F00019 | 9    | —     | $n$                                                                       | $n$                                  |
| BH3FR_F00020 | 9    | —     | $\operatorname{div}_{\{n\}}$                                              | $\operatorname{div}_n$               |
| BH3FR_F00021 | 9    | —     | $\operatorname{grad}_{\{n\}}$                                             | $\operatorname{grad}_n$              |
| BH3FR_F00022 | 9    | —     | $\operatorname{rot}_{\{n\}}$                                              | $\operatorname{rot}_n$               |
| BH3FR_F00023 | 9    | —     | $R_{\{\mathrm{n}\}}$                                                      | $R_{\mathrm{n}}$                     |
| BH3FR_F00024 | 9    | —     | $m$                                                                       | $m$                                  |
| BH3FR_F00025 | 9    | —     | $\overline{\operatorname{div}_{\{n\}}}$                                   | $\overline{\operatorname{div}_n}$    |
| BH3FR_F00026 | 9    | —     | $m+1$                                                                     | $m + 1$                              |
| BH3FR_F00027 | 9    | —     | $\widehat{\operatorname{div}_{\{n\}}}$                                    | $\widehat{\operatorname{div}_n}$     |
| BH3FR_F00028 | 9    | —     | $\tau>0$                                                                  | $\tau > 0$                           |
| BH3FR_F00029 | 12   | —     | $E=m\ .\ c^{\wedge\{2\}}$                                                 | $E = m.c^2$                          |
| BH3FR_F00030 | 12   | —     | $2,7^{\wedge\{\circ\}}\ K$                                                | $2,7^{\circ}K$                       |
| BH3FR_F00031 | 16   | —     | $\mathrm{W}^{\wedge\{\ \pm\}}$                                            | $W^{\pm}$                            |
| BH3FR_F00032 | 16   | —     | $\mathrm{Z}^{\wedge\{\circ\}}$                                            | $Z^{\circ}$                          |
| BH3FR_F00033 | 19   | —     | $\tau$                                                                    | $\tau$                               |
| BH3FR_F00034 | 19   | —     | $W^{\{ \ \pm\}}$                                                          | $W^{\pm}$                            |
| BH3FR_F00035 | 19   | —     | $Z^{\{0^{\wedge\{-\}}\}}$                                                 | $Z^{0^-}$                            |
| BH3FR_F00036 | 20   | —     | $\mathrm{W}\ \pm,\ \mathrm{Z}^{\wedge\{\mathrm{o}\}}$                     | $W\pm, Z^{\circ}$                    |
| BH3FR_F00037 | 21   | —     | $S\ U(2)\ \times\ U(1)$                                                   | $SU(2) \times U(1)$                  |
| BH3FR_F00038 | 21   | —     | $Z^{\{o\}}$                                                               | $Z^{\circ}$                          |
| BH3FR_F00039 | 21   | —     | $S\ U(5)$                                                                 | $SU(5)$                              |
| BH3FR_F00040 | 21   | —     | $e,\ \mu$                                                                 | $e, \mu$                             |
| BH3FR_F00041 | 21   | —     | $v_{\{\mathrm{varepsilon}\}},\ v_{\{\mu\}},\ v_{\{\tau\}}$                | $v_{\varepsilon}, v_{\mu}, v_{\tau}$ |
| BH3FR_F00042 | 22   | —     | $S\ U(3)$                                                                 | $SU(3)$                              |
| BH3FR_F00043 | 22   | —     | $S\ O(3)$                                                                 | $SO(3)$                              |
| BH3FR_F00044 | 23   | —     | $10^{\{-33\}}\ c\ m$                                                      | $10^{-33}cm$                         |
| BH3FR_F00045 | 24   | —     | $N$                                                                       | $N$                                  |
| BH3FR_F00046 | 24   | —     | $N=8$                                                                     | $N = 8$                              |
| BH3FR_F00047 | 24   | —     | $s=0,56$                                                                  | $s = 0,56$                           |
| BH3FR_F00048 | 24   | —     | $s=1\ /\ 2,28$                                                            | $s = 1/2, 28$                        |
| BH3FR_F00049 | 24   | —     | $s=1,8$                                                                   | $s = 1, 8$                           |
| BH3FR_F00050 | 24   | —     | $s=3\ /\ 2$                                                               | $s = 3/2$                            |
| BH3FR_F00051 | 24   | —     | $s=2$                                                                     | $s = 2$                              |

**Conf.** MathPix confidence: ■  $\geq 0.9$  ■ 0.5-0.9 ■  $< 0.5$  — no value

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| Identifier   | Page | Conf. | LaTeX source                                                                 | Rendered                                                             |
|--------------|------|-------|------------------------------------------------------------------------------|----------------------------------------------------------------------|
| BH3FR_F00052 | 24   | —     | N>8                                                                          | $N > 8$                                                              |
| BH3FR_F00053 | 24   | —     | c                                                                            | $c$                                                                  |
| BH3FR_F00054 | 24   | —     | \omega=4 / 3 c                                                               | $\omega = 4/3c$                                                      |
| BH3FR_F00055 | 24   | —     | x_{1}, x_{2}, x_{3}, \omega t                                                | $x_1, x_2, x_3, \omega t$                                            |
| BH3FR_F00056 | 24   | —     | R_{+4}                                                                       | $R_{+4}$                                                             |
| BH3FR_F00057 | 25   | —     | R_{-4}                                                                       | $R_{-4}$                                                             |
| BH3FR_F00058 | 25   | —     | x_{1}, x_{2}, x_{3}                                                          | $x_1, x_2, x_3$                                                      |
| BH3FR_F00059 | 25   | —     | R_{6}                                                                        | $R_6$                                                                |
| BH3FR_F00060 | 25   | —     | g_{i k}                                                                      | $g_{ik}$                                                             |
| BH3FR_F00061 | 25   | —     | g_{i k+}                                                                     | $g_{ik+}$                                                            |
| BH3FR_F00062 | 25   | —     | g_{i k-}                                                                     | $g_{ik-}$                                                            |
| BH3FR_F00063 | 25   | —     | \bar{\kappa}^i                                                               | $\bar{\kappa}^i$                                                     |
| BH3FR_F00064 | 25   | —     | i                                                                            | $i$                                                                  |
| BH3FR_F00065 | 25   | —     | k                                                                            | $k$                                                                  |
| BH3FR_F00066 | 25   | —     | n_{i}                                                                        | $n_i$                                                                |
| BH3FR_F00067 | 25   | —     | n_{k}                                                                        | $n_k$                                                                |
| BH3FR_F00068 | 25   | —     | \gamma_{(\mu\nu)}^{ik}                                                       | $\gamma_{(\mu\nu)}^{ik}$                                             |
| BH3FR_F00069 | 26   | —     | \left[\begin{array}{c}\widehat{\kappa\lambda}\\-t\\ \mu\nu\end{array}\right] | $\left[\begin{array}{c}\kappa\lambda\\-t\\ \mu\nu\end{array}\right]$ |
| BH3FR_F00070 | 26   | —     | \mu, \nu=1                                                                   | $\mu, \nu = 1$                                                       |
| BH3FR_F00071 | 26   | —     | R_{4}                                                                        | $R_4$                                                                |
| BH3FR_F00072 | 27   | —     | R_{6}\mathrm{ab}                                                             | $R_6\mathrm{ab}$                                                     |
| BH3FR_F00073 | 27   | —     | x_{5}                                                                        | $x_5$                                                                |
| BH3FR_F00074 | 27   | —     | x_{6}                                                                        | $x_6$                                                                |
| BH3FR_F00075 | 27   | —     | x_{1}                                                                        | $x_1$                                                                |
| BH3FR_F00076 | 27   | —     | x_{4}, x_{5}, x_{6}                                                          | $x_4, x_5, x_6$                                                      |
| BH3FR_F00077 | 27   | —     | x_{5}=i \,\varepsilon                                                        | $x_5 = i\varepsilon$                                                 |
| BH3FR_F00078 | 27   | —     | x_{6}=i \,\eta                                                               | $x_6 = i\eta$                                                        |
| BH3FR_F00079 | 27   | —     | x_{4}                                                                        | $x_4$                                                                |
| BH3FR_F00080 | 27   | —     | \tau \rightarrow 0                                                           | $\tau \rightarrow 0$                                                 |
| BH3FR_F00081 | 29   | —     | D                                                                            | $D$                                                                  |
| BH3FR_F00082 | 29   | —     | D=D(\tau)                                                                    | $D = D(\tau)$                                                        |
| BH3FR_F00083 | 29   | —     | e^{2} / \hbar c                                                              | $e^2/\hbar c$                                                        |
| BH3FR_F00084 | 29   | —     | \gamma, h, \,\varepsilon_o                                                   | $\gamma, h, \varepsilon_o$                                           |
| BH3FR_F00085 | 29   | —     | \mu_o                                                                        | $\mu_o$                                                              |
| BH3FR_F00086 | 32   | —     | n \geqq 4                                                                    | $n \geqslant 4$                                                      |
| BH3FR_F00087 | 32   | —     | m \geqq 1                                                                    | $m \geqslant 1$                                                      |
| BH3FR_F00088 | 32   | —     | n-m>0                                                                        | $n-m>0$                                                              |
| BH3FR_F00089 | 32   | —     | 1 \leqq \mathrm{~}\!^{-m}<n                                                  | $1 \leqslant \mathrm{~}\!^{-m} < n$                                  |
| BH3FR_F00090 | 32   | —     | 4 \leqq n<\infty                                                             | $4 \leqslant n < \infty$                                             |
| BH3FR_F00091 | 34   | —     | \pm \infty                                                                   | $\pm\infty$                                                          |
| BH3FR_F00092 | 35   | —     | 4^3=64                                                                       | $4^3=64$                                                             |
| BH3FR_F00093 | 35   | —     | 64-28=36                                                                     | $64-28=36$                                                           |
| BH3FR_F00094 | 36   | —     | \beta                                                                        | $\beta$                                                              |
| BH3FR_F00095 | 37   | —     | \infty                                                                       | $\infty$                                                             |
| BH3FR_F00096 | 38   | —     | p                                                                            | $p$                                                                  |
| BH3FR_F00097 | 38   | —     | \tau=                                                                        | $\tau=$                                                              |
| BH3FR_F00098 | 39   | —     | a                                                                            | $a$                                                                  |
| BH3FR_F00099 | 40   | —     | b                                                                            | $b$                                                                  |
| BH3FR_F00100 | 40   | —     | d                                                                            | $d$                                                                  |
| BH3FR_F00101 | 43   | —     | m_{L}                                                                        | $m_L$                                                                |
| BH3FR_F00102 | 43   | —     | m_{\mathrm{e}}>m_{\mathrm{L}}                                                | $m_e > m_L$                                                          |
| BH3FR_F00103 | 43   | —     | m_{\mathrm{e}}                                                               | $m_e$                                                                |

**Conf.** MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 ■ — no value

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| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Page | Conf. | LaTeX source              | Rendered                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------|---------------------------|---------------------------|
| BH3FR_F00104                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 43   | —     | $m_{\{B\}}$               | $m_B$                     |
| BH3FR_F00105                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 43   | —     | $1\%$                     | 1%                        |
| BH3FR_F00106                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 46   | —     | $\pi$                     | $\pi$                     |
| BH3FR_F00107                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 46   | —     | $e$                       | $e$                       |
| BH3FR_F00108                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 46   | —     | $A$                       | $A$                       |
| BH3FR_F00109                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 46   | —     | $A \not= 0$               | $A \neq 0$                |
| BH3FR_F00110                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 46   | —     | $A=0$                     | $A = 0$                   |
| BH3FR_F00111                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 47   | —     | $D/2$                     | $D/2$                     |
| BH3FR_F00112                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 50   | —     | $v$                       | $v$                       |
| BH3FR_F00113                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 50   | —     | $\eta$                    | $\eta$                    |
| BH3FR_F00114                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 58   | —     | $(a, b)$                  | $(a, b)$                  |
| BH3FR_F00115                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 58   | —     | $(c, d)$                  | $(c, d)$                  |
| BH3FR_F00116                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 61   | —     | $1/2$                     | $1/2$                     |
| BH3FR_F00117                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 62   | —     | $Q$                       | $Q$                       |
| BH3FR_F00118                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 62   | —     | $P$                       | $P$                       |
| BH3FR_F00119                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 63   | —     | $\pm 1$                   | $\pm 1$                   |
| BH3FR_F00120                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 64   | —     | $k, P, Q$                 | $k, P, Q$                 |
| BH3FR_F00121                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 64   | —     | $q$                       | $q$                       |
| BH3FR_F00122                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 64   | —     | $\beta \approx 1$         | $\beta \approx 1$         |
| BH3FR_F00123                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 65   | —     | $k=1$                     | $k = 1$                   |
| BH3FR_F00124                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 65   | —     | $k=2$                     | $k = 2$                   |
| BH3FR_F00125                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 66   | —     | $\varepsilon$             | $\varepsilon$             |
| BH3FR_F00126                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 66   | —     | $\kappa$                  | $\kappa$                  |
| BH3FR_F00127                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 66   | —     | $q_x$                     | $q_x$                     |
| BH3FR_F00128                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 67   | —     | $P/2$                     | $P/2$                     |
| BH3FR_F00129                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 67   | —     | $Q/2$                     | $Q/2$                     |
| BH3FR_F00130                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 67   | —     | $\Delta$                  | $\Delta$                  |
| BH3FR_F00131                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 67   | —     | $o$                       | $o$                       |
| BH3FR_F00132                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 67   | —     | $o^{--}$                  | $o^{--}$                  |
| BH3FR_F00133                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 67   | —     | $\Xi$                     | $\Xi$                     |
| BH3FR_F00134                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 68   | —     | $G=k+1$                   | $G = k + 1$               |
| BH3FR_F00135                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 69   | —     | $e-$                      | $e-$                      |
| BH3FR_F00136                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 69   | —     | $s$                       | $s$                       |
| BH3FR_F00137                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 69   | —     | $H$                       | $H$                       |
| BH3FR_F00138                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 69   | —     | $e^{-}$                   | $e^{-}$                   |
| BH3FR_F00139                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 70   | —     | $(+2)$                    | $(+2)$                    |
| BH3FR_F00140                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 73   | —     | $\xi$                     | $\xi$                     |
| BH3FR_F00141                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 73   | —     | $k, P, Q, \kappa, C, q_x$ | $k, P, Q, \kappa, C, q_x$ |
| BH3FR_F00142                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 74   | —     | $G$                       | $G$                       |
| BH3FR_F00143                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 79   | —     | $\pi, e$                  | $\pi, e$                  |
| BH3FR_F00144                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 79   | —     | $\zeta$                   | $\zeta$                   |
| BH3FR_F00145                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 79   | —     | $B=k-1$                   | $B = k - 1$               |
| BH3FR_F00146                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 80   | —     | $(2, 21)$                 | $(2, 21)$                 |
| BH3FR_F00147                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 80   | —     | $(2, 24)$                 | $(2, 24)$                 |
| BH3FR_F00148                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 80   | —     | $(2, 83)$                 | $(2, 83)$                 |
| BH3FR_F00149                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 80   | —     | $(2, 220)$                | $(2, 220)$                |
| BH3FR_F00150                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 80   | —     | $(2, 67)$                 | $(2, 67)$                 |
| BH3FR_F00151                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 80   | —     | $(2, 13)$                 | $(2, 13)$                 |
| BH3FR_F00152                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 80   | —     | $(2, 325)$                | $(2, 325)$                |
| BH3FR_F00153                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 81   | —     | $(2, 27)$                 | $(2, 27)$                 |
| BH3FR_F00154                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 81   | —     | $(2, 285)$                | $(2, 285)$                |
| BH3FR_F00155                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 81   | —     | $(2, 283)$                | $(2, 283)$                |
| BH3FR_F00156                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 81   | —     | $(2, 62)$                 | $(2, 62)$                 |
| BH3FR_F00157                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 82   | —     | $(2, 262)$                | $(2, 262)$                |
| <b>Conf.</b> MathPix confidence: <span style="color: green;">■</span> $\geq 0.9$ <span style="color: orange;">■</span> $0.5\text{--}0.9$ <span style="color: red;">■</span> $< 0.5$ <span style="color: gray;">—</span> no value<br><b>LaTeX source</b> MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ<br><b>Residual</b> not shown for this document — no residual measurements are paired with these rows.<br><b>[refined: <i>basis</i>]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here. |      |       |                           |                           |

| Identifier                                                                                                                                                                                                                                                                                  | Page | Conf. | LaTeX source                | Rendered                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------|-----------------------------|-----------------------------|
| BH3FR_F00158                                                                                                                                                                                                                                                                                | 82   | —     | $(2, 82)$                   | $(2, 82)$                   |
| BH3FR_F00159                                                                                                                                                                                                                                                                                | 82   | —     | $(2, 186)$                  | $(2, 186)$                  |
| BH3FR_F00160                                                                                                                                                                                                                                                                                | 82   | —     | $(2, 201)$                  | $(2, 201)$                  |
| BH3FR_F00161                                                                                                                                                                                                                                                                                | 82   | —     | $(2, 197)$                  | $(2, 197)$                  |
| BH3FR_F00162                                                                                                                                                                                                                                                                                | 82   | —     | $(2, 158)$                  | $(2, 158)$                  |
| BH3FR_F00163                                                                                                                                                                                                                                                                                | 82   | —     | $(2, 112)$                  | $(2, 112)$                  |
| BH3FR_F00164                                                                                                                                                                                                                                                                                | 82   | —     | $(2, 115)$                  | $(2, 115)$                  |
| BH3FR_F00165                                                                                                                                                                                                                                                                                | 82   | —     | $(2, 108)$                  | $(2, 108)$                  |
| BH3FR_F00166                                                                                                                                                                                                                                                                                | 83   | —     | $(2, 58)$                   | $(2, 58)$                   |
| BH3FR_F00167                                                                                                                                                                                                                                                                                | 84   | —     | $(2, 261)$                  | $(2, 261)$                  |
| BH3FR_F00168                                                                                                                                                                                                                                                                                | 85   | —     | $(2, 213)$                  | $(2, 213)$                  |
| BH3FR_F00169                                                                                                                                                                                                                                                                                | 85   | —     | $(2, 353)$                  | $(2, 353)$                  |
| BH3FR_F00170                                                                                                                                                                                                                                                                                | 85   | —     | $(2, 122)$                  | $(2, 122)$                  |
| BH3FR_F00171                                                                                                                                                                                                                                                                                | 85   | —     | $(2, 156)$                  | $(2, 156)$                  |
| BH3FR_F00172                                                                                                                                                                                                                                                                                | 86   | —     | $(2, 25)$                   | $(2, 25)$                   |
| BH3FR_F00173                                                                                                                                                                                                                                                                                | 86   | —     | $(2, 123)$                  | $(2, 123)$                  |
| BH3FR_F00174                                                                                                                                                                                                                                                                                | 86   | —     | $(2, 263)$                  | $(2, 263)$                  |
| BH3FR_F00175                                                                                                                                                                                                                                                                                | 86   | —     | $(2, 203)$                  | $(2, 203)$                  |
| BH3FR_F00176                                                                                                                                                                                                                                                                                | 86   | —     | $(2, 196)$                  | $(2, 196)$                  |
| BH3FR_F00177                                                                                                                                                                                                                                                                                | 86   | —     | $(2, 202)$                  | $(2, 202)$                  |
| BH3FR_F00178                                                                                                                                                                                                                                                                                | 86   | —     | $(2, 198)$                  | $(2, 198)$                  |
| BH3FR_F00179                                                                                                                                                                                                                                                                                | 87   | —     | $(2, 91)$                   | $(2, 91)$                   |
| BH3FR_F00180                                                                                                                                                                                                                                                                                | 88   | —     | $(2, 119)$                  | $(2, 119)$                  |
| BH3FR_F00181                                                                                                                                                                                                                                                                                | 88   | —     | $(2, 233)$                  | $(2, 233)$                  |
| BH3FR_F00182                                                                                                                                                                                                                                                                                | 89   | —     | $(2, 40)$                   | $(2, 40)$                   |
| BH3FR_F00183                                                                                                                                                                                                                                                                                | 91   | —     | $(2, 190)$                  | $(2, 190)$                  |
| BH3FR_F00184                                                                                                                                                                                                                                                                                | 91   | —     | $(2, 324)$                  | $(2, 324)$                  |
| BH3FR_F00185                                                                                                                                                                                                                                                                                | 91   | —     | $(2, 195)$                  | $(2, 195)$                  |
| BH3FR_F00186                                                                                                                                                                                                                                                                                | 91   | —     | $(2, 341)$                  | $(2, 341)$                  |
| BH3FR_F00187                                                                                                                                                                                                                                                                                | 91   | —     | $(2, 191)$                  | $(2, 191)$                  |
| BH3FR_F00188                                                                                                                                                                                                                                                                                | 92   | —     | $(2, 23)$                   | $(2, 23)$                   |
| BH3FR_F00189                                                                                                                                                                                                                                                                                | 92   | —     | $(2, 211)$                  | $(2, 211)$                  |
| BH3FR_F00190                                                                                                                                                                                                                                                                                | 92   | —     | $(2, 230)$                  | $(2, 230)$                  |
| BH3FR_F00191                                                                                                                                                                                                                                                                                | 92   | —     | $(2, 337)$                  | $(2, 337)$                  |
| BH3FR_F00192                                                                                                                                                                                                                                                                                | 92   | —     | $(2, 329)$                  | $(2, 329)$                  |
| BH3FR_F00193                                                                                                                                                                                                                                                                                | 93   | —     | $(2, 187)$                  | $(2, 187)$                  |
| BH3FR_F00194                                                                                                                                                                                                                                                                                | 93   | —     | $(2, 232)$                  | $(2, 232)$                  |
| BH3FR_F00195                                                                                                                                                                                                                                                                                | 94   | —     | $(2, 231)$                  | $(2, 231)$                  |
| BH3FR_F00196                                                                                                                                                                                                                                                                                | 94   | —     | $(2, 216)$                  | $(2, 216)$                  |
| BH3FR_F00197                                                                                                                                                                                                                                                                                | 94   | —     | $(2, 188)$                  | $(2, 188)$                  |
| BH3FR_F00198                                                                                                                                                                                                                                                                                | 94   | —     | $(2, 348)$                  | $(2, 348)$                  |
| BH3FR_F00199                                                                                                                                                                                                                                                                                | 94   | —     | $(2, 209)$                  | $(2, 209)$                  |
| BH3FR_F00200                                                                                                                                                                                                                                                                                | 94   | —     | $(2, 80)$                   | $(2, 80)$                   |
| BH3FR_F00201                                                                                                                                                                                                                                                                                | 95   | —     | $(2, 75)$                   | $(2, 75)$                   |
| BH3FR_F00202                                                                                                                                                                                                                                                                                | 96   | —     | $(2, 215)$                  | $(2, 215)$                  |
| BH3FR_F00203                                                                                                                                                                                                                                                                                | 96   | —     | $(2, 243)$                  | $(2, 243)$                  |
| BH3FR_F00204                                                                                                                                                                                                                                                                                | 97   | —     | $(2, 189)$                  | $(2, 189)$                  |
| BH3FR_F00205                                                                                                                                                                                                                                                                                | 99   | —     | $(2, 225)$                  | $(2, 225)$                  |
| BH3FR_F00206                                                                                                                                                                                                                                                                                | 130  | —     | $0, -1$                     | $0, -1$                     |
| BH3FR_F00207                                                                                                                                                                                                                                                                                | 130  | —     | $0(-1)$                     | $0(-1)$                     |
| BH3FR_F00208                                                                                                                                                                                                                                                                                | 130  | —     | $(1200) \ 0(+1, 0, -1)$     | $(1200)0(+1, 0, -1)$        |
| BH3FR_F00209                                                                                                                                                                                                                                                                                | 130  | —     | $(2111) - 2(0, -1)$         | $(2111) - 2(0, -1)$         |
| BH3FR_F00210                                                                                                                                                                                                                                                                                | 130  | —     | $(2310) - 2(+1, 0, -1, -2)$ | $(2310) - 2(+1, 0, -1, -2)$ |
| BH3FR_F00211                                                                                                                                                                                                                                                                                | 130  | —     | $(2330) \ 0(+2, +1, 0, -1)$ | $(2330)0(+2, +1, 0, -1)$    |
| <b>Conf.</b> MathPix confidence: <span>■</span> $\geq 0.9$ <span>■</span> $0.5-0.9$ <span>■</span> $< 0.5$ <span>—</span> no value                                                                                                                                                          |      |       |                             |                             |
| <b>LaTeX source</b> MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ                           |      |       |                             |                             |
| <b>Residual</b> not shown for this document — no residual measurements are paired with these rows.                                                                                                                                                                                          |      |       |                             |                             |
| <b>[refined: <i>basis</i>]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here. |      |       |                             |                             |

| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Page | Conf. | LaTeX source        | Rendered            |
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| BH3FR_F00212                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 130  | —     | $(4)(1101)+1(+1,0)$ | $(4)(1101)+1(+1,0)$ |
| BH3FR_F00213                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 130  | —     | $(6)(2010)-1(0)$    | $(6)(2010)-1(0)$    |
| <p><b>Conf.</b> MathPix confidence: <span>■</span> <math>\geq 0.9</math> <span>■</span> 0.5-0.9 <span>■</span> <math>&lt; 0.5</math> — no value</p> <p><b>LaTeX source</b> MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ</p> <p><b>Residual</b> not shown for this document — no residual measurements are paired with these rows.</p> <p><b>[refined: <i>basis</i>]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</p> |      |       |                     |                     |